

# **Removing Payment Friction: Evidence-Based Churn Prevention for PT Asuransi Nusantara Sejahtera**

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Final Stage 1 — Data Analytics Competition

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## **Executive Summary**

PT Asuransi Nusantara Sejahtera loses 21.1% of its customers to lapse, surrender or non-renewal, which puts about IDR 9.9B of annual premium (20.8% of the premium base) at risk in the analysed sample of 8,000 customers — roughly IDR 1,490.5B across the full portfolio of 1,200,000. Using records of 8,000 policyholders, we combined statistical driver analysis, survival analysis, interpretable regression, a cross-validated machine-learning ensemble and doubly robust causal inference to answer three questions: who leaves, when they leave, and which actions are worth funding.

Churn is most strongly associated with premium change at last renewal (%), payment method and auto-renewal; customers in the riskiest group of premium change at last renewal (%) ('13.2–30.6') churn at 33.5%, 1.6 times the average. Risk is concentrated in time: the churn hazard peaks in tenure interval [18, 24) months (0.99% per month, about 1.4 times the average). The final model reaches an out-of-fold ROC-AUC 0.760 (95% CI 0.748–0.772); the top 20% of scored customers contain 46% of all churners (2.3× random targeting). A retention campaign aimed at the top 56% of customers by expected loss is estimated to return IDR 807.7M net (ROI 1.2×), whereas random targeting of the same size yields IDR 514.8M. Causal analysis indicates that moving customers to 'premium change at last renewal (%) ≤ median (4.1)' would lower churn by about 11.7 percentage points, a result that survives placebo and sensitivity tests.

We recommend (1) Smarter renewal pricing; (2) Shift manual payers to automatic payment; (3) Make renewal effortless (default auto-renewal). Each action should start as a randomised pilot; detecting a reduction of 11.7 pp requires about 278 customers per arm.

For ANS, whose renewal premiums are 58% of premium income and whose lost policies each leave about Rp1.1 million of acquisition cost unrecovered, the programme protects revenue and recovers acquisition spend at the same time.

## **1. Introduction**

PT Asuransi Nusantara Sejahtera (ANS) is a multi-line insurer that sells life, health, motor and property policies through agents, bancassurance partners, brokers and an online channel. Policy non-renewal and lapse rose from 15.2% in 2022 to 20.6% in 2024, and the Board has asked for an explanation of who is leaving, when and why, together with a retention programme for 2026 within a budget of Rp1.5 billion.

ANS manages around 1.2 million active policies and booked gross written premium of Rp4.8 trillion in 2024; renewal premiums represent 58% of its premium income. Management estimates that every lost policyholder leaves about Rp1.1 million of

acquisition cost unrecovered. Renewal premiums accounted for Rp77.07 trillion, about 42% of Indonesian life insurance premiums, in 2024 ("AAJI catat premi," 2025). Surrender claims paid by Indonesian life insurers still amounted to Rp62.72 trillion in 2025, 19% lower than in 2024 ("AAJI catat nilai klaim surrender," 2026).

For an individual insurer, churn (lapse, surrender or non-renewal) affects several parts of the business at once. It erodes the renewal premium base, leaves acquisition costs unrecovered when policies exit early, distorts the persistency assumptions used in pricing (Eling & Kochanski, 2013) and is often a signal of unmet needs or poor service (Brockett et al., 2008). The economics of retention are well documented: Reichheld and Sasser (1990) reported that reducing defections by 5% raised profits by 25% to 85% across several service businesses. Not every exit is equally avoidable, however: Gottlieb and Smetters (2021) found that forgetting to pay explained 37.8% of recent life-insurance lapses, so part of churn can be addressed through process design rather than price.

This study answers 4 research questions (RQ): Management's objective is to maximise the profit of retention spending.

- RQ1. What are the main drivers of churn, and how large is each effect?
- RQ2. When during the customer lifecycle is churn risk highest?
- RQ3. How accurately can churn be predicted for the customers in the test set?
- RQ4. Which retention programme within the Rp1.5 billion budget maximises the financial return?

## **2. Data and Methodology**

### **2.1 Data and preparation**

The training data contain 8,000 customers (plus 2,000 customers to be scored) and 27 original variables covering January 2022 – June 2025 (snapshot 30 June 2025). The target is 'churn', defined as a policy that was not renewed or lapsed within the observation window (January 2022 – June 2025) (churn = Yes); the churn rate is 21.1%, so models are evaluated with ROC-AUC, precision–recall and profit rather than accuracy. 1 related table(s) were aggregated to one row per customer (counts, sums, recency).

Preparation was automated and audited: 6 type corrections and harmonisations were applied (for example numbers or ranges stored as text, date parsing and inconsistent category labels), 15 duplicate records were removed and impossible values were set to missing. Most importantly, cancellation reason is recorded only around or after the churn event; such post-event variables leak the outcome, so they and all derived features

were removed before any analysis (single-feature and missingness tests flagged them automatically). We engineered 18 insurance-specific features, such as tenure (years), tenure band, new customer (<12 months), age band and premium-to-income ratio.

## 2.2 Analytical framework

The analysis follows five layers (Figure B.1). (1) Univariate tests ( $\chi^2$  and Mann–Whitney U) with effect sizes, Information Value and Benjamini–Hochberg correction screen candidate drivers (Benjamini & Hochberg, 1995). (2) Because churn is a time-to-event outcome with censored observations, Kaplan–Meier curves, log-rank tests, a Cox model, an accelerated failure time model and restricted mean survival time describe when customers leave (Kaplan & Meier, 1958; Cox, 1972). (3) A logistic regression gives adjusted odds ratios that managers can read directly. (4) 9 algorithms — CatBoost, Explainable boosting machine (EBM), Extra trees, Hist. gradient boosting, LightGBM, Logistic regression, Neural network (MLP), Random forest and XGBoost — were trained with stratified 5-fold cross-validation, tuned with Bayesian optimisation and combined by ensemble selection; tree ensembles remain the strongest family on tabular data (Grinsztajn et al., 2022), and SHAP values explain each prediction (Lundberg & Lee, 2017). (5) Doubly robust estimators (AIPW and the DR-learner) estimate the average and heterogeneous effects of controllable levers (Chernozhukov et al., 2018; Kennedy, 2023). Results are translated into money with customer lifetime value, the expected maximum profit measure (Verbraken et al., 2013) and a campaign simulation.

## 2.3 Validation

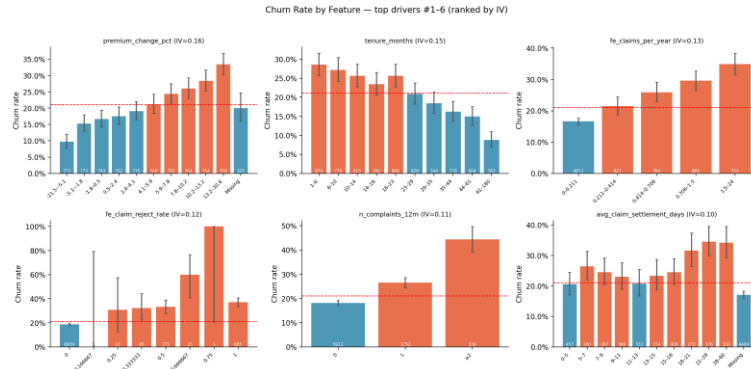
All reported performance figures are out-of-fold estimates; target-dependent encodings and hyperparameter search were fitted inside training folds only. Models are compared with DeLong tests for correlated AUCs (DeLong et al., 1988) with Holm correction, uncertainty is reported with bootstrap and DeLong confidence intervals, and calibration is checked with the Brier score and expected calibration error. Sanity checks support the validity of the pipeline: a model trained on shuffled labels scores an AUC of 0.519, and an adversarial classifier separates training from scoring data with AUC 0.50. Appendix B documents the full validation design.

## 3. Findings

### 3.1 Who churns? (RQ1)

The strongest correlates of churn are premium change at last renewal (%) (IV 0.16): '13.2–30.6' churns at 33.5%, 1.6× the average; payment method (IV 0.10): 'Bank Transfer' churns at 25.5%, 1.2× the average; auto-renewal (IV 0.08): 'No' churns at 26.4%, 1.3× the average (Table 1). Several variables — age, claims status Approved

count, sales channel and policy start date month — are statistically detectable but practically negligible ( $IV < 0.02$ ), a reminder that with 8,000 records significance alone is not evidence of importance.



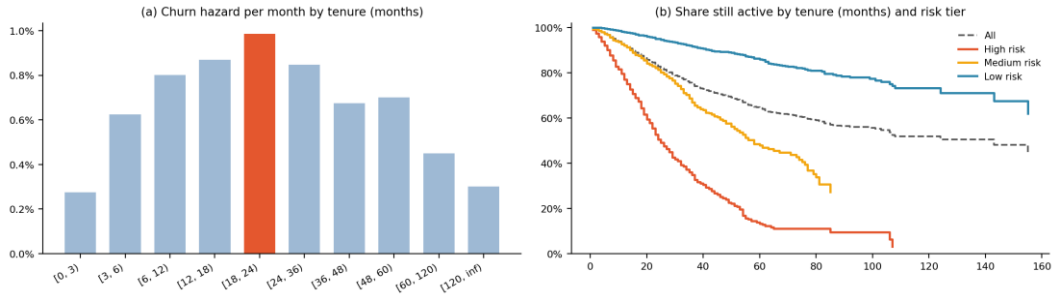
**Figure 1.** Churn rate by the strongest drivers (bars: churn rate with 95% Wilson CI; dashed line: overall churn rate).

**Table 1.** Evidence for churn drivers across methods (overall churn rate 21.1%; 'methods agreeing' counts significant univariate test, logistic regression, Cox model and top-10 SHAP).

| Driver                             | Riskiest group | Churn rate | Lift  | IV   | Odds ratio | Hazard ratio | SHAP rank | Methods agreeing |
|------------------------------------|----------------|------------|-------|------|------------|--------------|-----------|------------------|
| premium change at last renewal (%) | 13.2–30.6      | 33.5%      | 1.59× | 0.16 | 1.64       | 1.39         | 1         | 4/4              |
| payment method                     | Bank Transfer  | 25.5%      | 1.21× | 0.10 | 2.33       | 1.83         | 3         | 4/4              |
| auto-renewal                       | No             | 26.4%      | 1.25× | 0.08 | 0.51       | 0.61         | 4         | 4/4              |
| complaints in the last 12 months   | ≥2             | 44.3%      | 2.10× | 0.11 | 1.33       | 1.23         | 5         | 4/4              |
| tenure (months)                    | 1–6            | 28.6%      | 1.35× | 0.15 | 0.63       | —            | 2         | 3/4              |
| number of ANS products             | 1              | 23.8%      | 1.13× | 0.05 | 0.76       | —            | 6         | 3/4              |
| claim rejection rate               | 1              | 37.1%      | 1.76× | 0.12 | 1.26       | 1.11         | 11        | 3/4              |

### 3.2 When do customers churn? (RQ2)

Survival analysis treats active customers as censored observations. The Kaplan–Meier estimates show 96.8% after 6 months; 92.2% after 12 months; 82.8% after 24 months. The churn hazard peaks in tenure interval [18, 24) months at 0.99% per month, against an average of 0.72% (Figure 2).



**Figure 2.** (a) Churn hazard per month by tenure interval; (b) Kaplan–Meier survival by predicted risk tier (out-of-sample validation of the score over time).

### 3.3 Who is most at risk? Interpretable segments

A shallow decision tree turns the drivers into segments operations teams can act on (Table F.1). The riskiest segment — claims per year of tenure > 0.41 AND premium change at last renewal (%) > 3.85 AND payment method  $\neq$  Auto-debit AND claim rejection rate > 0.12 — has 266 customers (3.3% of the base) churning at 55.3% (2.6 $\times$ ) and contains 8.7% of all churners, whereas the most loyal segment churns at only 6.1%.

### 3.4 How well can churn be predicted? (RQ3)

**Table 2.** Out-of-fold performance (5-fold stratified CV; random baseline ROC-AUC 0.500, PR-AUC 0.211). EMPC = expected maximum profit per customer (Verbraken et al., 2013); p-values Holm-corrected.

| Model                      | ROC-AUC (95% CI)    | PR-AUC | Brier | EMPC (IDR) | DeLong p vs final |
|----------------------------|---------------------|--------|-------|------------|-------------------|
| Final ensemble: hill climb | 0.760 (0.747–0.773) | 0.475  | 0.140 | 115.01K    | —                 |
| xgb_fs                     | 0.758 (0.745–0.771) | 0.469  | 0.141 | 114.75K    | 0.025             |
| Logistic regression        | 0.757 (0.744–0.769) | 0.470  | 0.141 | 114.50K    | 0.013             |
| xgb_multiseed              | 0.755 (0.742–0.768) | 0.465  | 0.142 | 114.29K    | <0.001            |
| catboost_multiseed         | 0.755 (0.742–0.768) | 0.469  | 0.141 | 114.18K    | 0.002             |

The final model reaches an out-of-fold ROC-AUC 0.760 (95% CI 0.748–0.772) and a PR-AUC of 0.475, 2.3 times the random baseline (Table 2). The interpretable logistic regression reaches 0.757; the ensemble is significantly better ( $\Delta$ AUC +0.004, DeLong p 0.013), so the added complexity buys real accuracy. A glass-box Explainable Boosting Machine reaches 0.748. Ranking by expected maximum profit agrees with ranking by AUC. Concentration is strong: the highest-risk decile churns at 55.6% and the top 20% contain 46% of churners.

### **3.5 Why is a customer at risk? Explaining the model**

SHAP values attribute each prediction to the features (Lundberg & Lee, 2017). The model relies mostly on premium change at last renewal (%) (13%), tenure (months) (10%) and payment method (10%) of total attribution (Figure G.4), consistent with the statistical evidence, so the model behaves in a way managers can audit.

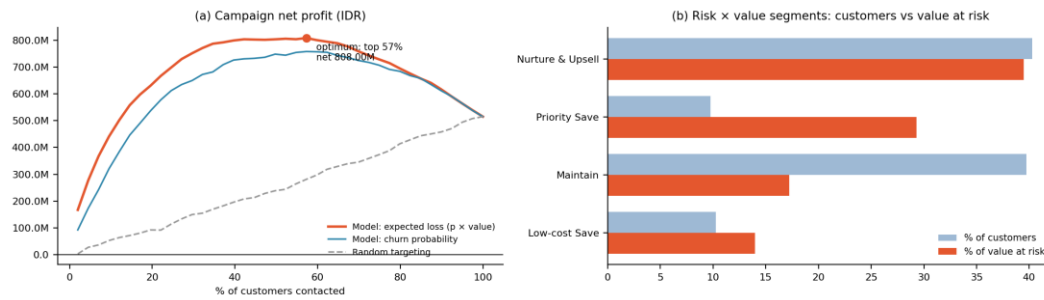
## **4. Business Impact (RQ4)**

### **4.1 Value at risk and prioritisation**

Weighting each customer's churn probability by annual premium, about IDR 9.9B of annual premium (20.8% of the premium base) is expected to be lost in the analysed sample, or roughly IDR 1,490.5B when scaled to the portfolio of 1,200,000 customers. Crossing risk with value (Figure 3b) isolates a 'Priority Save' group — 9.7% of customers with high risk and above-median premium — that holds 29% of the value at risk and warrants personal contact; high-risk, low-value customers can be served through automated digital touchpoints.

### **4.2 Retention campaign economics**

We simulated a retention campaign on the out-of-fold predictions, using the parameters given in the case: a contact-and-incentive cost of IDR 150K per customer, that 30% of contacted would-be churners are retained and a 20% margin over a 5-year horizon (discount rate 10%). Contacting the top 56% by expected loss maximises net benefit at IDR 807.7M (ROI 1.2×; Figure 3a). The conclusion is robust: the campaign stays profitable in 9 of 9 combinations of success rate ( $\pm 50\%$ ) and cost ( $\times 0.5$ – $\times 2$ ) in the sensitivity analysis. Scaled to the full portfolio of 1,200,000 customers, the same rule would contact about 672,000 customers at a cost of IDR 100.8B for a net benefit of about IDR 121.2B. The budget of IDR 1.5B is the binding constraint: it pays for about 10,000 contacts (0.8% of the portfolio). Spending it on the customers with the highest expected loss — among whom 68.7% would otherwise churn — is estimated to return IDR 12.2B net (ROI 8.1×), so the budget should be concentrated on the top of the ranking rather than spread thinly.



**Figure 3.** (a) Net profit of a retention campaign by share of customers contacted under three targeting rules; (b) share of customers vs share of value at risk by risk  $\times$  value segment.

### 4.3 What works? Causal evidence and uplift

Prediction identifies who is at risk, not which action changes behaviour. For each controllable lever we estimated the average effect of the favourable condition on churn with a cross-fitted doubly robust (AIPW) estimator that adjusts for observed confounders (Chernozhukov et al., 2018). The largest effects are 'premium change at last renewal (%)  $\leq$  median (4.1)' (-11.7 pp, 95% CI -13.6 to -9.9; naive difference -11.1 pp), 'auto-renewal = 1' (-10.0 pp, 95% CI -11.8 to -8.1; naive difference -9.7 pp) and 'payment method = Auto-debit' (-9.9 pp, 95% CI -11.9 to -8.0; naive difference -10.4 pp). 4 of 5 effects pass all robustness checks (placebo treatment  $\approx$  0, adequate overlap and E-values above 1.25; VanderWeele & Ding, 2017). We also tested whether some customers respond more than others (DR-learner; Kennedy, 2023). For 'premium change at last renewal (%)  $\leq$  median (4.1)' the effect is broadly homogeneous (most vs least responsive fifth: 12.0 vs 11.7 pp,  $p$  0.940), and targeting the top 20% by churn risk avoids 68.2 churners per 1,000 eligible customers versus 30.6 with uplift-based targeting, so risk-based targeting is appropriate for this lever; the pilots should still re-test responsiveness (Ascarza, 2018).

### 4.4 Churn personas

## 5. Recommendations and Implementation

**Table 3.** Recommended retention actions, ranked by evidence and expected value.

| # | Action                                         | Key evidence                                        | Expected impact                 | KPI                                       | When        |
|---|------------------------------------------------|-----------------------------------------------------|---------------------------------|-------------------------------------------|-------------|
| 1 | Smarter renewal pricing                        | '13.2–30.6' churns at 33.5% (1.6x avg); IV 0.16     | -2.99 pp churn at full adoption | Churn rate among customers with increases | Medium term |
| 2 | Shift manual payers to automatic payment       | 'Bank Transfer' churns at 25.5% (1.2x avg); IV 0.10 | measured in pilot               | % of policies on auto-pay                 | Quick win   |
| 3 | Make renewal effortless (default auto-renewal) | '0' churns at 26.4% (1.3x avg); IV 0.08             | -3.43 pp churn at full adoption | Auto-renew adoption                       | Quick win   |
| 4 | Complaint recovery programme                   | ' $\geq$ 2' churns at 44.3% (2.1x avg); IV 0.11     | -1.66 pp churn at full adoption | Complaint resolution time                 | Quick win   |

| # | Action                                         | Key evidence                            | Expected impact   | KPI                   | When      |
|---|------------------------------------------------|-----------------------------------------|-------------------|-----------------------|-----------|
| 5 | Bundling & cross-sell to raise switching costs | 'I' churns at 23.8% (1.1x avg); IV 0.05 | measured in pilot | Products per customer | Long term |

1. Smarter renewal pricing. Cap or phase in premium increases for at-risk loyal customers, explain the value behind increases, and offer coverage/deductible adjustments instead of lapse.

2. Shift manual payers to automatic payment. Offer a small incentive (e.g. premium discount/cashback) and one-click enrolment for auto-debit/recurring card payment; send pre-due reminders to remaining manual payers.

Implementation follows a pilot → measure → scale logic (Figure J.4). Every action keeps a randomised control group: with a baseline churn of 48.4% in the High-risk tier, detecting a 11.7 pp reduction requires 278 customers per arm ( $\alpha = 0.05$ , power 80%). Pilot results then train uplift models so that future campaigns target customers by expected response  $\times$  value rather than by risk alone (Ascarza, 2018). The model is deployed as a monthly early-warning score with drift and calibration monitoring and quarterly retraining (Appendix J).

Two company initiatives make these actions cheaper to deliver: the self-service app planned for Q1 2026 can carry onboarding and payment reminders, and the auto-debit partnership under negotiation removes a major source of payment friction. We therefore recommend running the first-quarter pilots through these two channels, with agents reserved for Priority Save customers.

## 6. Discussion and Conclusion

In answer to the research questions — RQ1: churn is driven mainly by premium change at last renewal (%), payment method and auto-renewal; RQ2: risk peaks in tenure interval [18, 24) months; RQ3: churn can be predicted with ROC-AUC 0.760 (95% CI 0.748–0.772), concentrating 46% of churners in the top 20%; RQ4: a value-targeted campaign returns about IDR 807.7M net in the analysed sample (about IDR 121.2B at portfolio scale); within the IDR 1.5B budget, contacting the 10,000 highest-loss customers returns about IDR 12.2B net, and the most promising lever is 'premium change at last renewal (%)  $\leq$  median (4.1)'. These results are consistent with prior evidence. Using more than one million contracts, Eling and Kiesenbauer (2014) identified contract age, product type and policyholder age as key drivers of lapse. In a survey of recent lapsers, forgetting to pay explained 37.8% of life-insurance lapses (Gottlieb & Smetters, 2021). In the U.S. property and casualty market, around half of customers satisfied with their claims experience intend to renew, compared with about a third of dissatisfied customers (Fallis et al., 2022).

Limitations. The data are a single snapshot from one insurer, so results may not generalise and may shift with economic conditions. Drivers identified by statistical tests, regression and SHAP are associations; the causal estimates rely on the assumption of no unmeasured confounding, which we probe with E-values but cannot rule out. The campaign economics use the parameters given in the case, which we stress-test in a sensitivity analysis. Variables such as competitor prices, service interactions and household income shocks are not observed. A randomised pilot of the recommended actions is the next step to measure their true effect.

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Appendices

The appendices are not counted in the page limit. They document every analysis step so that each result in the main text can be traced and reproduced; all figures and tables were generated by the accompanying notebook.

Appendix A. Data Overview and Quality

This appendix documents the raw data, data quality checks and the transformations applied before analysis.

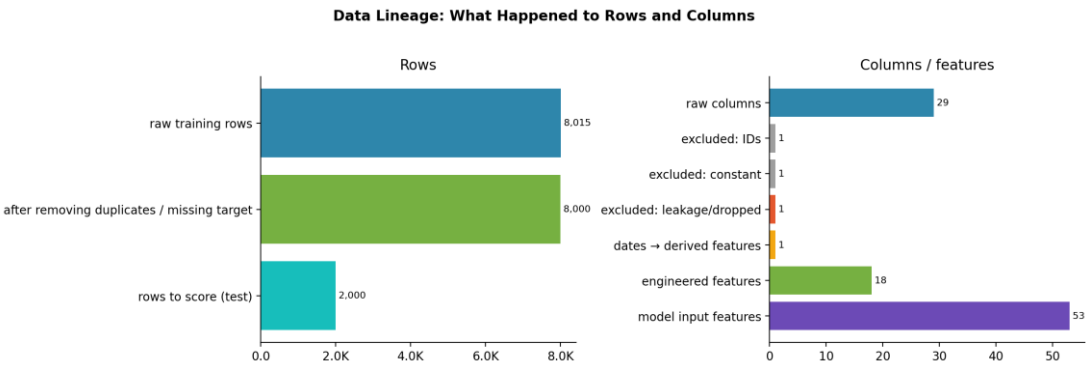


Figure A.1. Data Lineage: What Happened to Rows and Columns

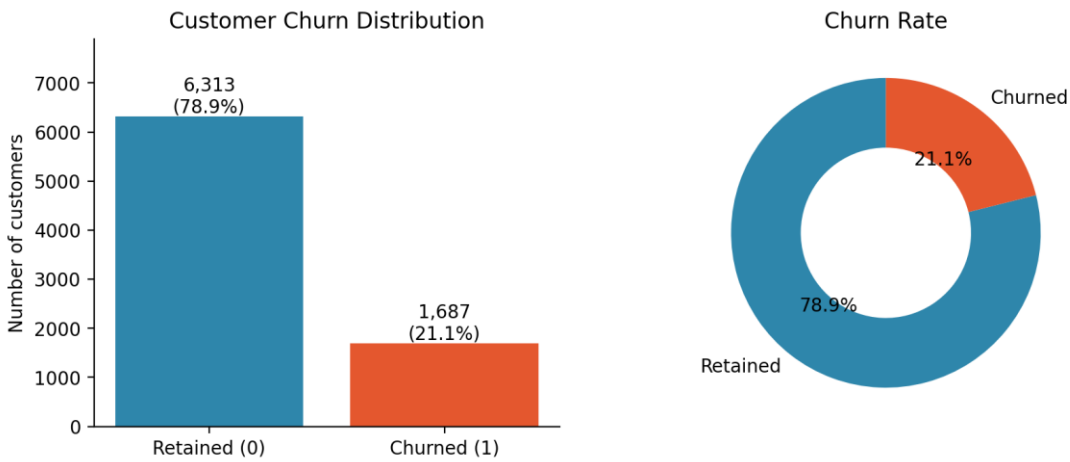
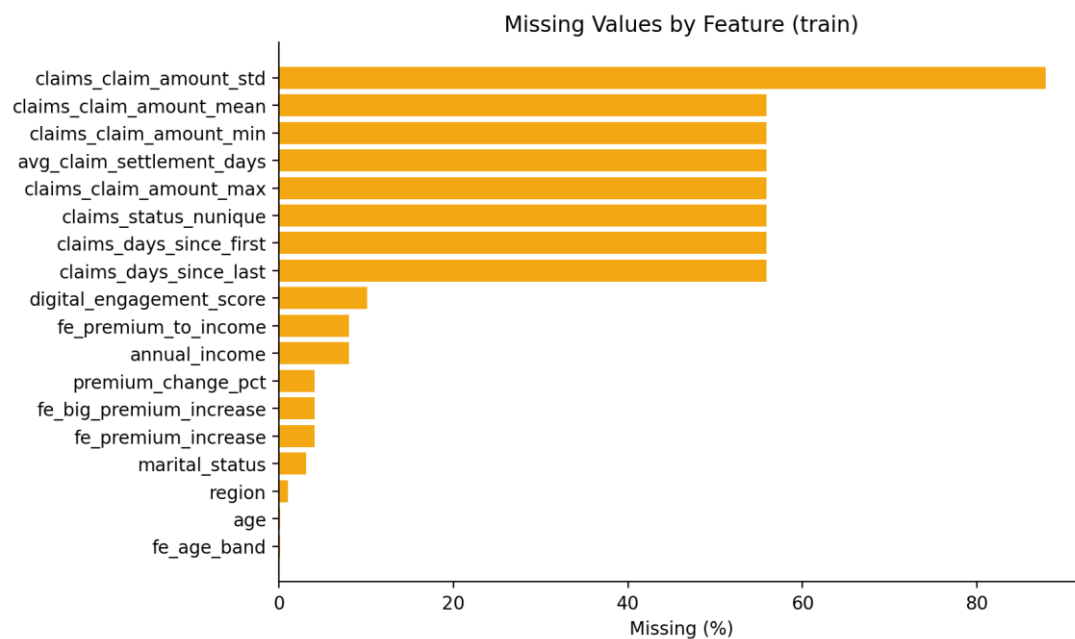
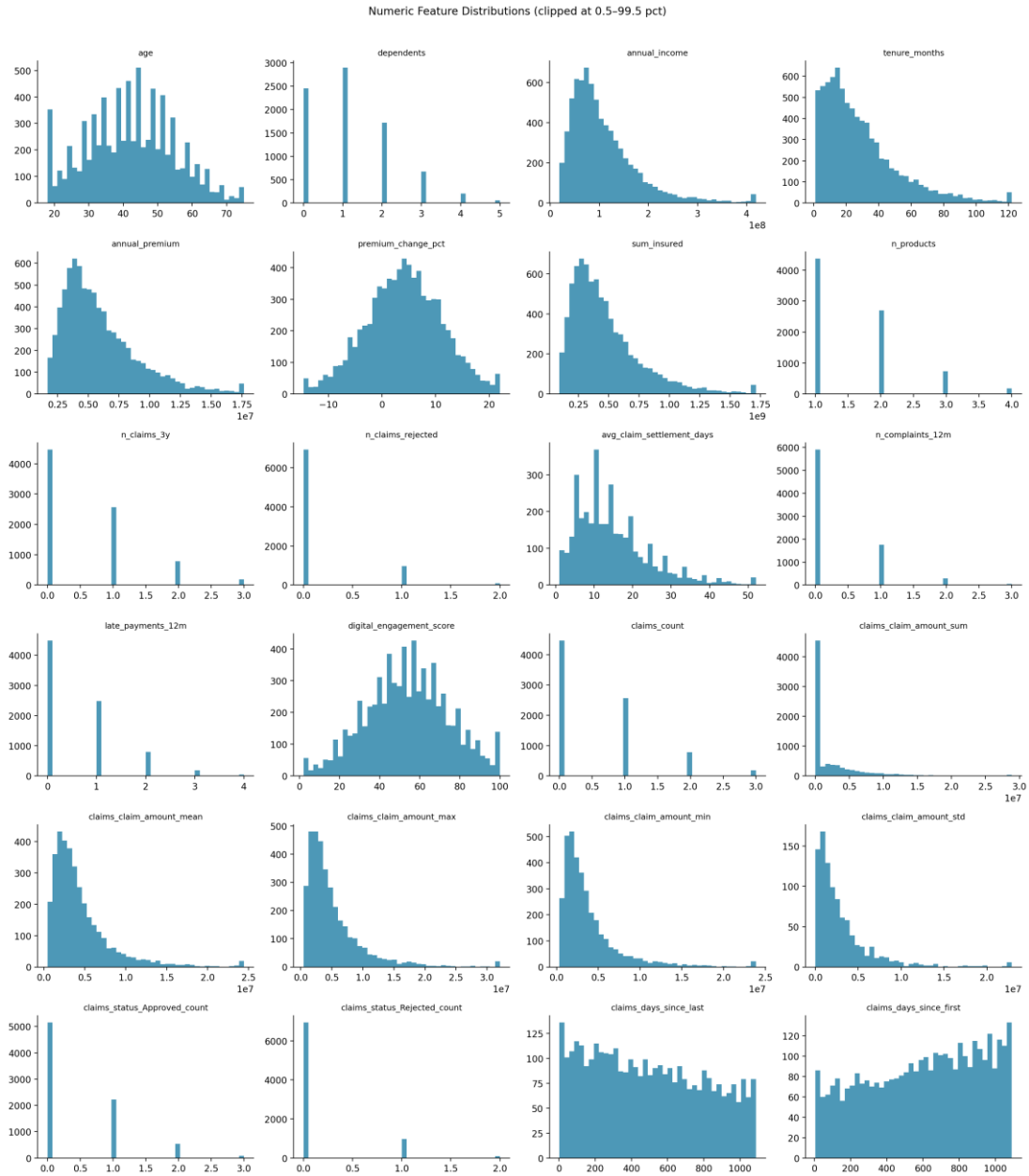


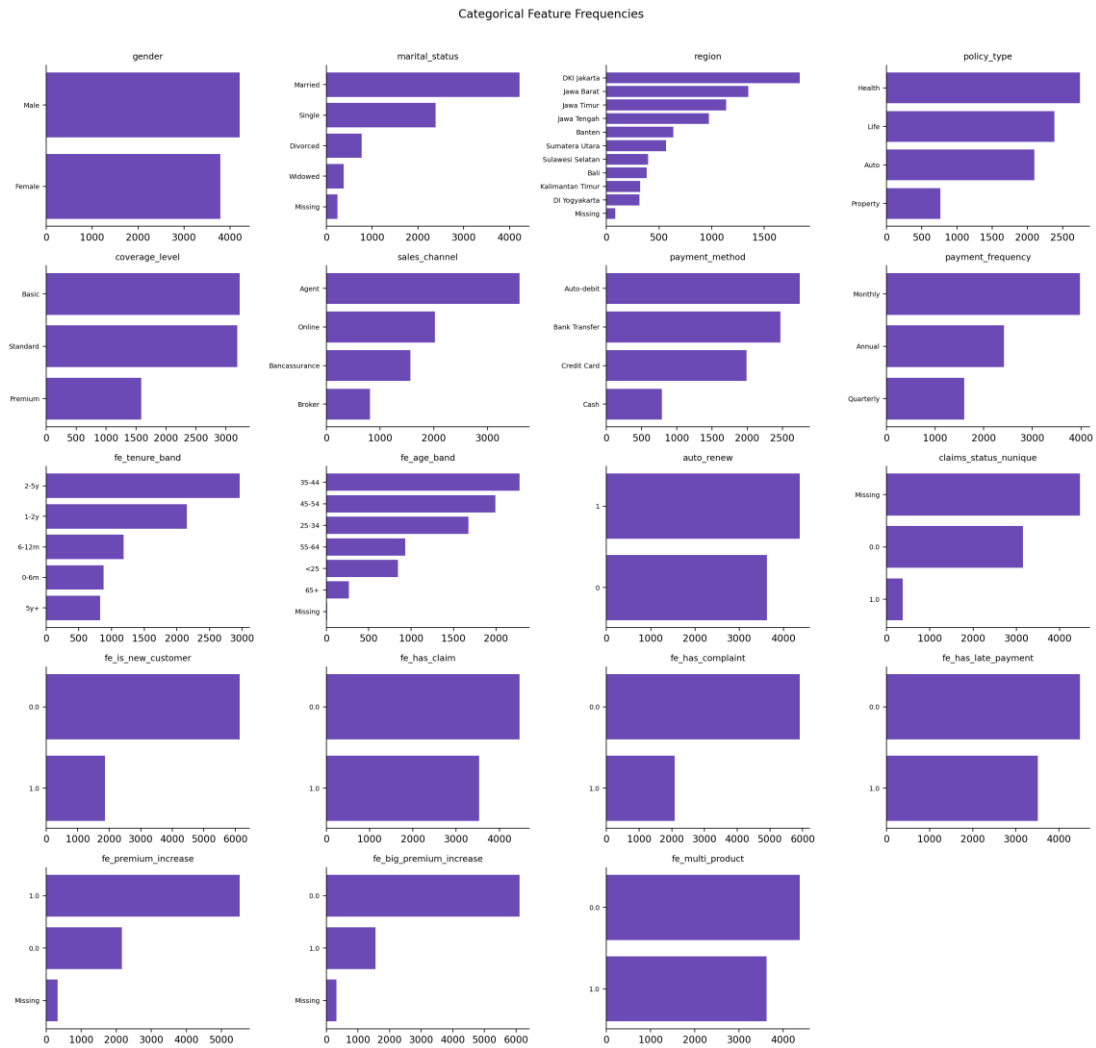
Figure A.2. Customer Churn Distribution



**Figure A.3.** Missing Values by Feature (train)



**Figure A.4.** Numeric Feature Distributions (clipped at 0.5–99.5 pct)



**Figure A.5.** Categorical Feature Frequencies

**Table A.1.** Data dictionary and quality report (inferred type, missing values, cardinality).

| Column            | Inferred type | Role              | Missing % | N unique | Top value   | Top freq % |
|-------------------|---------------|-------------------|-----------|----------|-------------|------------|
| customer id       | id            |                   | 0         | 8,000    | CUST000001  | 0.010      |
| age               | numeric       | age               | 0.100     | 68       | 18.0        | 3.670      |
| gender            | binary_cat    | gender            | 0         | 2        | Male        | 52.6       |
| marital status    | categorical   |                   | 3.090     | 4        | Married     | 54.4       |
| dependents        | numeric       | dependents        | 0         | 8        | 1           | 36.2       |
| region            | categorical   | region            | 1.040     | 10       | DKI Jakarta | 23.2       |
| annual income     | numeric       | income            | 8.040     | 2,210    | 71.40M      | 0.200      |
| policy type       | categorical   | policy type       | 0         | 4        | Health      | 34.3       |
| coverage level    | categorical   |                   | 0         | 3        | Basic       | 40.3       |
| sales channel     | categorical   | channel           | 0         | 4        | Agent       | 45.0       |
| agent id          | high_card     | agent             | 0         | 400      | NONE        | 55.0       |
| payment method    | categorical   | payment method    | 0         | 4        | Auto-debit  | 34.4       |
| payment frequency | categorical   | payment frequency | 0         | 3        | Monthly     | 49.8       |
| policy start date | date          |                   | 0         | 146      | 2024-04-30  | 2.740      |

| Column                               | Inferred type | Role           | Missing % | N unique | Top value | Top freq % |
|--------------------------------------|---------------|----------------|-----------|----------|-----------|------------|
|                                      |               |                |           |          | 00:00:00  |            |
| tenure (months)                      | numeric       | tenure         | 0         | 146      | 14.0      | 2.740      |
| annual premium                       | numeric       | premium        | 0         | 5,321    | 5.79M     | 0.090      |
| premium change at last renewal (%)   | numeric       | premium_change | 4.060     | 416      | 3.900     | 0.790      |
| sum insured                          | numeric       | sum insured    | 0         | 5,084    | 287.50M   | 0.090      |
| number of ANS products               | numeric       | products       | 0         | 7        | 1         | 54.7       |
| claims in the last 3 years           | numeric       | claims         | 0         | 7        | 0         | 55.9       |
| rejected claims                      | numeric       |                | 0         | 4        | 0         | 86.7       |
| average claim settlement time (days) | numeric       |                | 55.9      | 65       | 10.0      | 5.690      |
| complaints in the last 12 months     | numeric       | complaints     | 0         | 7        | 0         | 73.9       |
| late payments in the last 12 months  | numeric       | late_payment   | 0         | 7        | 0         | 56.1       |
| digital engagement score             | numeric       | engagement     | 10.1      | 101      | 48.0      | 2.320      |
| auto-renewal                         | binary        | auto-renewal   | 0         | 2        | 1         | 54.6       |
| currency                             | constant      |                | 0         | 1        | IDR       | 100.0      |
| cancellation reason                  | categorical   |                | 78.9      | 5        | Moved     | 20.7       |
| claims count                         | numeric       |                | 0         | 7        | 0         | 55.9       |
| claims claim amount sum              | numeric       |                | 0         | 3,033    | 0         | 55.9       |
| claims claim amount mean             | numeric       |                | 55.9      | 3,028    | 3.45M     | 0.110      |
| claims claim amount max              | numeric       |                | 55.9      | 2,948    | 2.12M     | 0.110      |
| claims claim amount min              | numeric       |                | 55.9      | 2,753    | 2.61M     | 0.140      |
| claims claim amount std              | numeric       |                | 87.9      | 927      | 1.40M     | 0.310      |
| claims status Approved count         | numeric       |                | 0         | 6        | 0         | 64.4       |
| claims status Rejected count         | numeric       |                | 0         | 4        | 0         | 86.7       |
| claims status nunique                | binary        |                | 55.9      | 2        | 0         | 89.3       |
| claims (days since) last             | numeric       |                | 55.9      | 1,039    | 1.000     | 0.340      |
| claims (days since) first            | numeric       |                | 55.9      | 1,032    | 885.0     | 0.340      |

**Table A.2.** Leakage screen: name-based, missingness-based and single-feature AUC checks.

| Feature                            | Name keyword | Missingness auc | Single feature auc | Statistical evidence | Flagged |
|------------------------------------|--------------|-----------------|--------------------|----------------------|---------|
| cancellation reason                | yes          | 1.000           | 1.000              | yes                  | yes     |
| premium change at last renewal (%) | no           | 0.501           | 0.613              | no                   | no      |
| tenure (years)                     | no           |                 | 0.601              | no                   | no      |
| policy start date (days since)     | no           |                 | 0.601              | no                   | no      |

| Feature                          | Name keyword | Missingness auc | Single feature auc | Statistical evidence | Flagged |
|----------------------------------|--------------|-----------------|--------------------|----------------------|---------|
| tenure (months)                  | no           |                 | 0.601              | no                   | no      |
| policy start date year           | no           |                 | 0.595              | no                   | no      |
| claims per year of tenure        | no           |                 | 0.590              | no                   | no      |
| tenure band                      | no           |                 | 0.588              | no                   | no      |
| claims claim amount sum          | no           |                 | 0.572              | no                   | no      |
| claims in the last 3 years       | no           |                 | 0.570              | no                   | no      |
| claims count                     | no           |                 | 0.570              | no                   | no      |
| complaints in the last 12 months | no           |                 | 0.570              | no                   | no      |
| auto-renewal                     | no           |                 | 0.570              | no                   | no      |
| claims status nunique            | no           | 0.567           | 0.567              | no                   | no      |
| payment method                   | no           |                 | 0.566              | no                   | no      |

Automated cleaning log: gender: 2 variasi ejaan/case disatukan (4  $\rightarrow$  2 level); annual\_income: string  $\rightarrow$  numeric (100.0% berhasil); payment\_method: 4 variasi ejaan/case disatukan (8  $\rightarrow$  4 level); policy\_start\_date: parsed sebagai tanggal (100.0%); auto\_renew: boolean-like  $\rightarrow$  1/0; claims\_status\_nunique: binary numeric [np.float64(1.0), np.float64(2.0)]  $\rightarrow$  0/1

## Appendix B. Methodology and Validation Design

Architecture of the analytics pipeline, the cross-validation design and the predictive model, plus tuning results.

Statistical screening. For categorical variables we use Pearson's  $\chi^2$  test with Cramér's V; for numeric variables the Mann–Whitney U test with the rank-biserial correlation, so that every p-value is paired with an effect size. Because dozens of variables are tested, p-values are converted to Benjamini–Hochberg q-values. Weight of evidence and Information Value, standard in credit and insurance scoring, summarise each binned variable's predictive strength (Siddiqi, 2006); proportions are reported with Wilson 95% intervals.

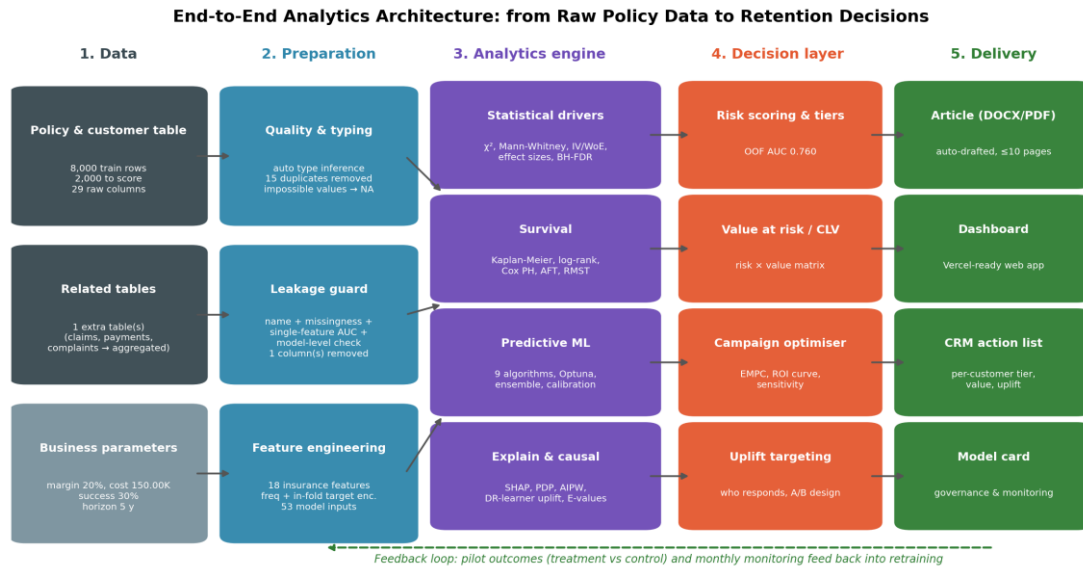
Survival layer. Tenure is the time axis and active customers are right-censored. Kaplan–Meier curves and log-rank tests describe retention; discrete hazards per tenure interval locate the critical window; the Cox model gives adjusted hazard ratios, its proportional-hazards assumption is checked with Schoenfeld residuals (Schoenfeld, 1982), and a Weibull accelerated failure time model provides time ratios that remain interpretable when the assumption fails (Wei, 1992). Restricted mean survival time expresses differences in months or days retained (Royston & Parmar, 2013).

Predictive layer. Logistic regression, random forest, extra trees, histogram gradient boosting, LightGBM, XGBoost and CatBoost (Ke et al., 2017; Chen & Guestrin, 2016; Prokhorenkova et al., 2018), a glass-box Explainable Boosting Machine (Lou et al., 2013) and a neural network are trained on identical folds. Hyperparameters are tuned with the TPE sampler and median pruning (Akiba et al., 2019) on inner folds; the best families are re-trained with several seeds; features with non-positive permutation importance are tested for removal; and six ensembling strategies, including greedy ensemble selection (Caruana et al., 2004) and stacking, are compared, the simplest within 0.0005 of the best being preferred. Probabilities are calibrated by isotonic or Platt scaling when this lowers cross-validated log-loss, and calibration is summarised by the Brier score and expected calibration error (Guo et al., 2017).

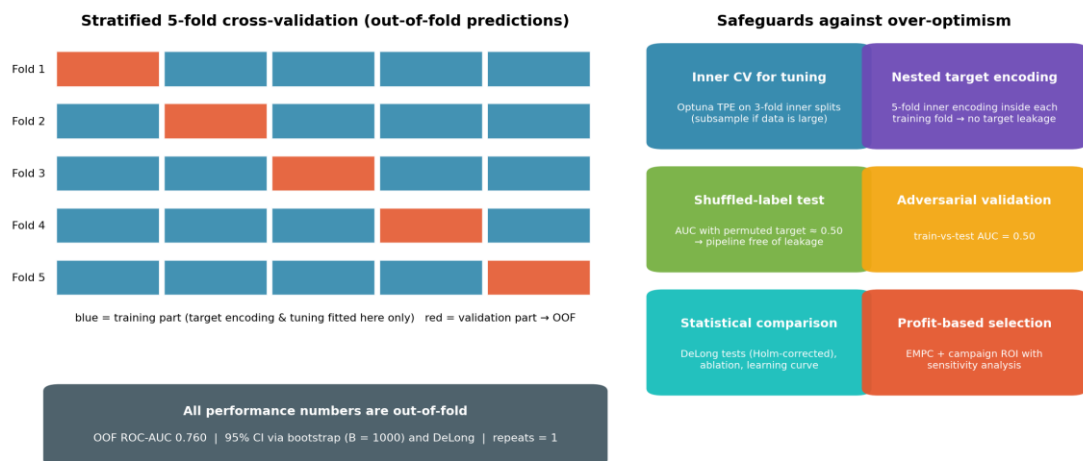
Model comparison. Correlated AUCs are compared with the DeLong test (DeLong et al., 1988) using the fast algorithm, with Holm correction across comparisons; bootstrap intervals complement the analytic ones. Because AUC is not aligned with campaign profit (Verbeke et al., 2012; Höppner et al., 2020), models are also ranked by the expected maximum profit measure for churn, which integrates campaign profit over a Beta-distributed acceptance rate (Verbraken et al., 2013). The risk score is further validated on time-to-churn with Harrell's C-index (Harrell et al., 1982).

Why not only an interpretable model? Interpretable models should be preferred when they perform comparably (Rudin, 2019); we therefore report the logistic regression and the EBM next to the ensemble and test whether the accuracy gain is significant.

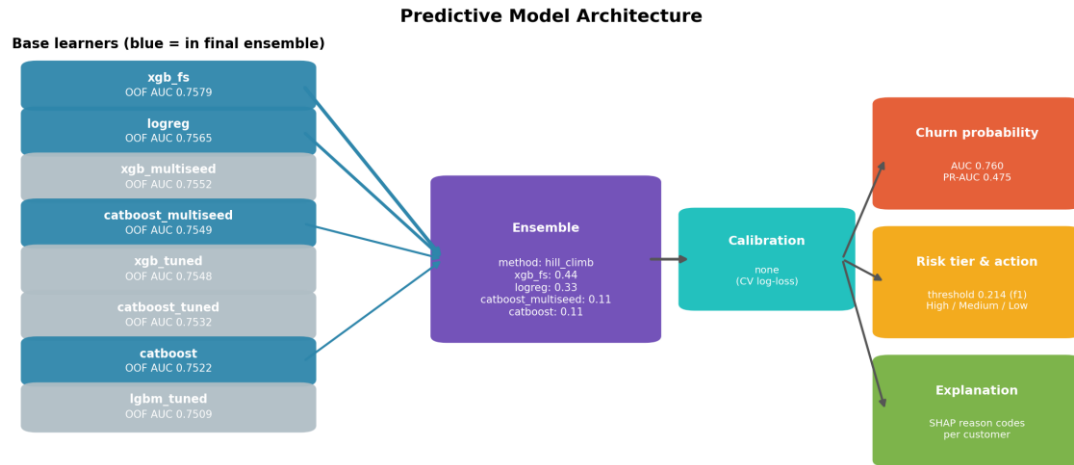
Further methodological support: Uplift models can yield more profitable retention campaigns than churn-risk models (Devriendt et al., 2021).



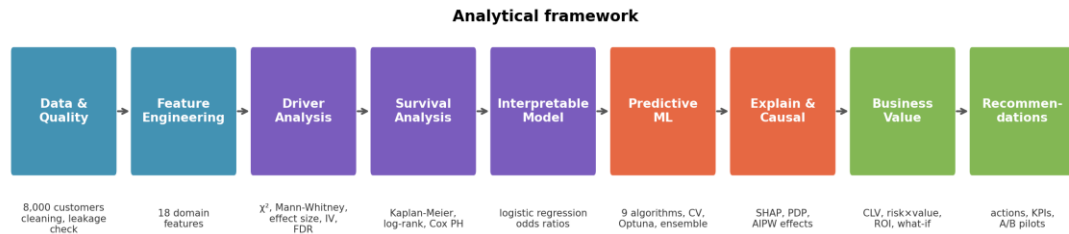
**Figure B.1.** End-to-end analytics architecture: from raw policy data to retention decisions, with a feedback loop from pilots.



**Figure B.2.** Validation design: stratified k-fold cross-validation producing out-of-fold predictions for model comparison and evaluation.



**Figure B.3.** Predictive Model Architecture



**Figure B.4.** Analytical framework

**Table B.1.** Key analysis settings (reproducibility).

| Setting                | Value                                                                  |
|------------------------|------------------------------------------------------------------------|
| RUN MODE               | full                                                                   |
| SEED                   | 42                                                                     |
| N FOLDS                | 5                                                                      |
| N REPEATS              | 1                                                                      |
| METRIC                 | roc auc                                                                |
| MODELS                 | ['logreg', 'rf', 'et', 'hgb', 'lgbm', 'xgb', 'catboost', 'ebm', 'mlp'] |
| USE OPTUNA             | True                                                                   |
| OPTUNA TRIALS          | 40                                                                     |
| FINAL SEEDS            | [42, 2024, 7]                                                          |
| CALIBRATION            | auto                                                                   |
| PROFIT MARGIN          | 0.2                                                                    |
| RETENTION COST         | 150000                                                                 |
| RETENTION SUCCESS RATE | 0.3                                                                    |
| DISCOUNT RATE          | 0.1                                                                    |
| CLV HORIZON YEARS      | 5                                                                      |
| RISK TIER CUTS         | (0.5, 0.8)                                                             |
| WHAT IF ADOPTION       | 0.3                                                                    |
| PILOT POWER            | 0.8                                                                    |
| PILOT ALPHA            | 0.05                                                                   |

**Table B.2.** Bayesian hyperparameter optimisation (Optuna) results.

| Model    | Trials completed | Best tuning score | Default cv auc | Best params                                                                                                                                                                                                                                                                                                  | Tuned cv auc | Gain auc |
|----------|------------------|-------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|
| lgbm     | 39               | 0.753             | 0.745          | {"learning_rate": 0.01422387663504771, "num_leaves": 10, "min_child_samples": 38, "subsample": 0.7191667789934336, "colsample_bytree": 0.3516920077513463, "reg_alpha": 0.0007345564742179169, "reg_lambda": 4.2576268745423745e-08, "min_split_gain": 0.5496283858967218, "cat_smooth": 35.775710006609955} | 0.751        | 0.006    |
| catboost | 10               | 0.755             | 0.752          | {"learning_rate": 0.07599369873560621, "depth": 4, "l2_leaf_reg": 5.864129169696525, "random_strength": 1.640130838029839, "bagging_temperature": 0.1848544555252704}                                                                                                                                        | 0.753        | 0.001    |
| xgb      | 40               | 0.757             | 0.747          | {"learning_rate": 0.048170091670992836, "max_depth": 2, "min_child_weight": 1.385786662745958, "subsample": 0.7058311788274935, "colsample_bytree": 0.9183604973777273, "reg_alpha": 6.03638384257227e-06, "reg_lambda": 0.9796272550751544, "gamma": 4.598639806625283}                                     | 0.755        | 0.008    |

## Appendix C. Driver Analysis (Detailed)

Churn rates for every analysed variable with 95% confidence intervals, predictive strength (Information Value), correlations and interactions.

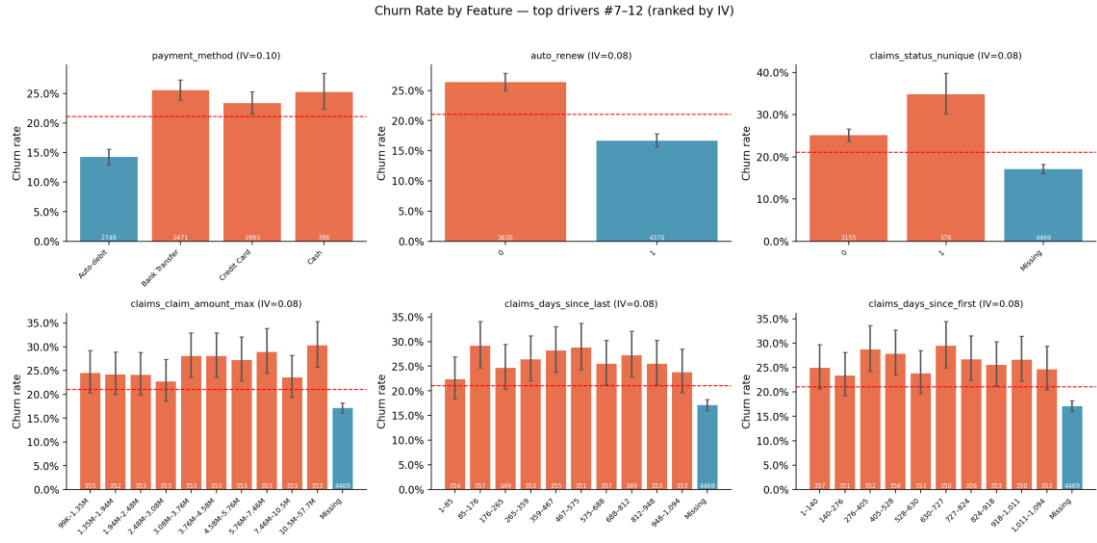


Figure C.1. Churn Rate by Feature — top drivers #7-12 (ranked by IV)

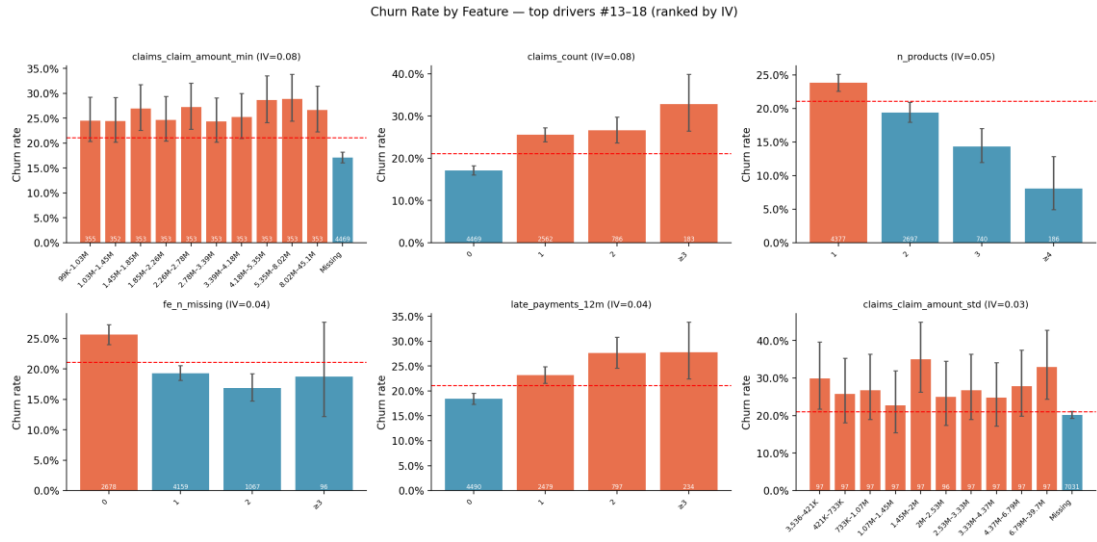
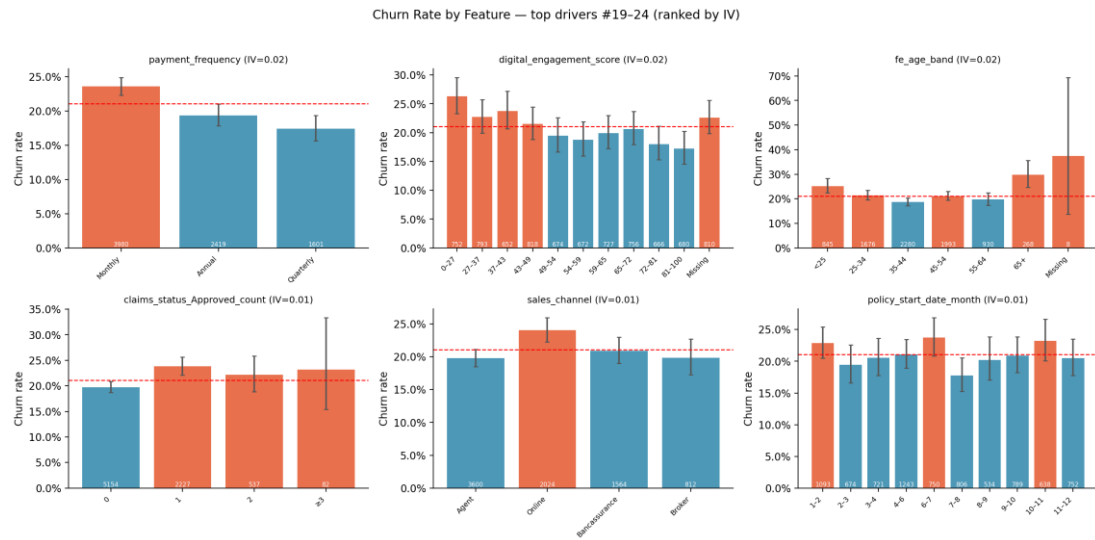
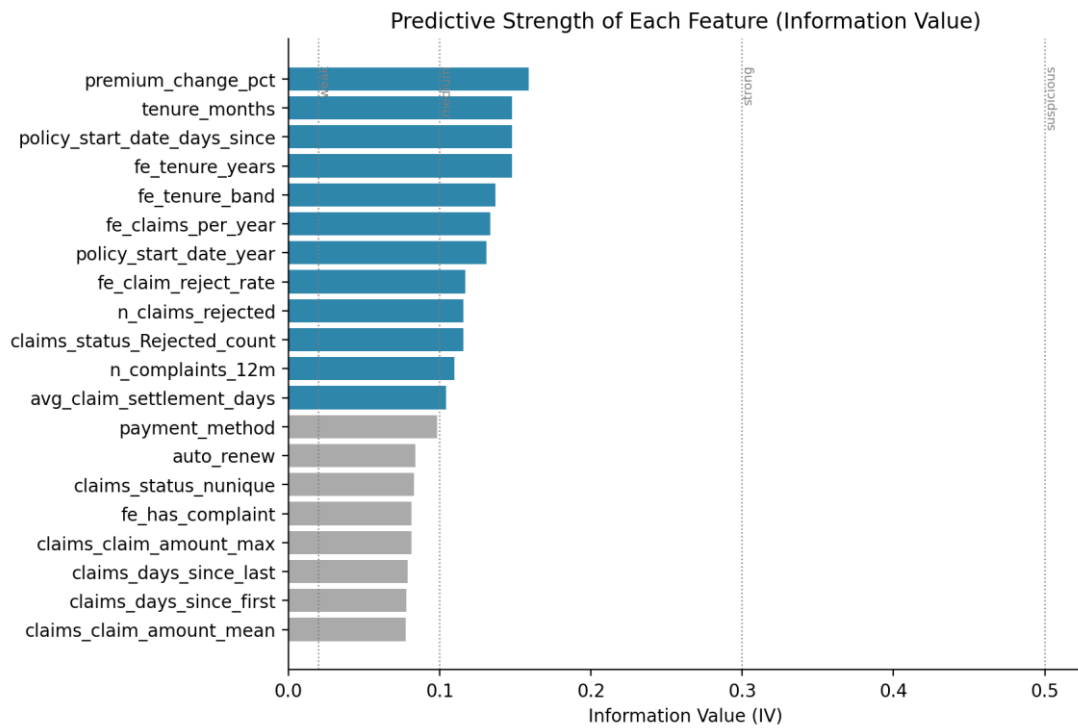


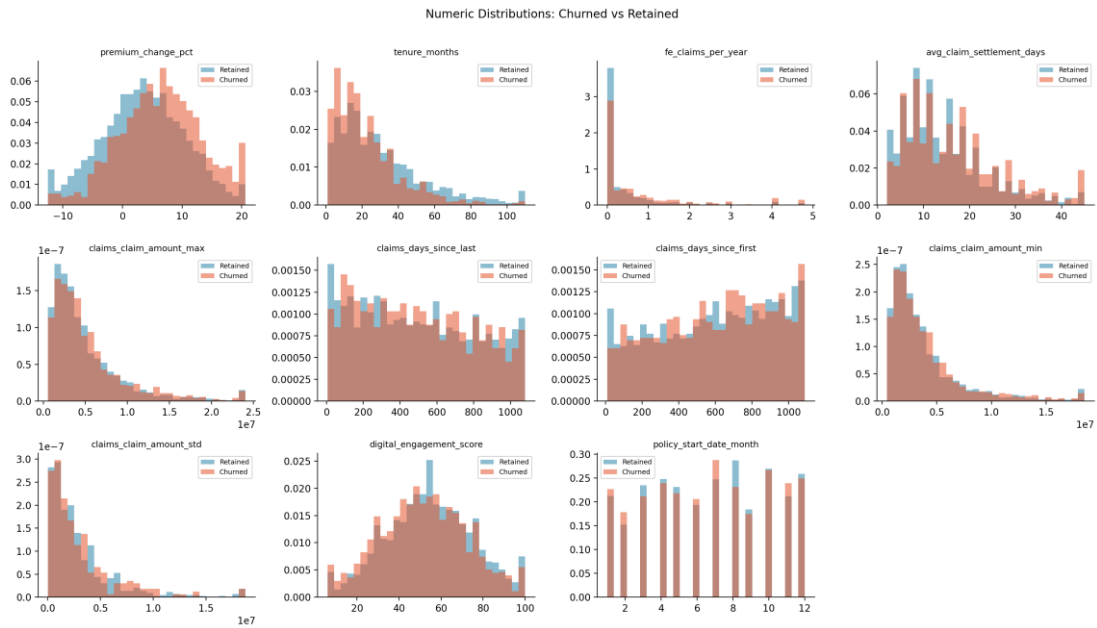
Figure C.2. Churn Rate by Feature — top drivers #13-18 (ranked by IV)



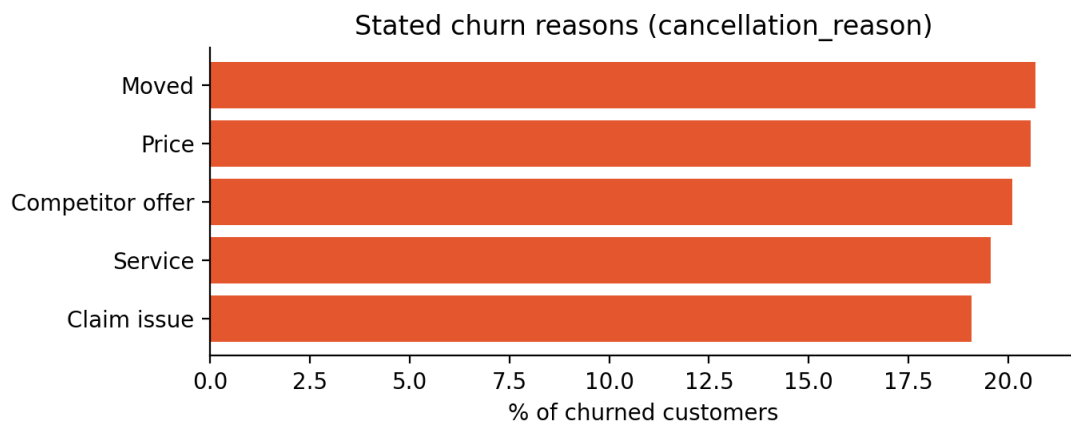
**Figure C.3.** Churn Rate by Feature — top drivers #19–24 (ranked by IV)



**Figure C.4.** Predictive Strength of Each Feature (Information Value)



**Figure C.5.** Numeric Distributions: Churned vs Retained



**Figure C.6.** Stated churn reasons (cancellation reason)

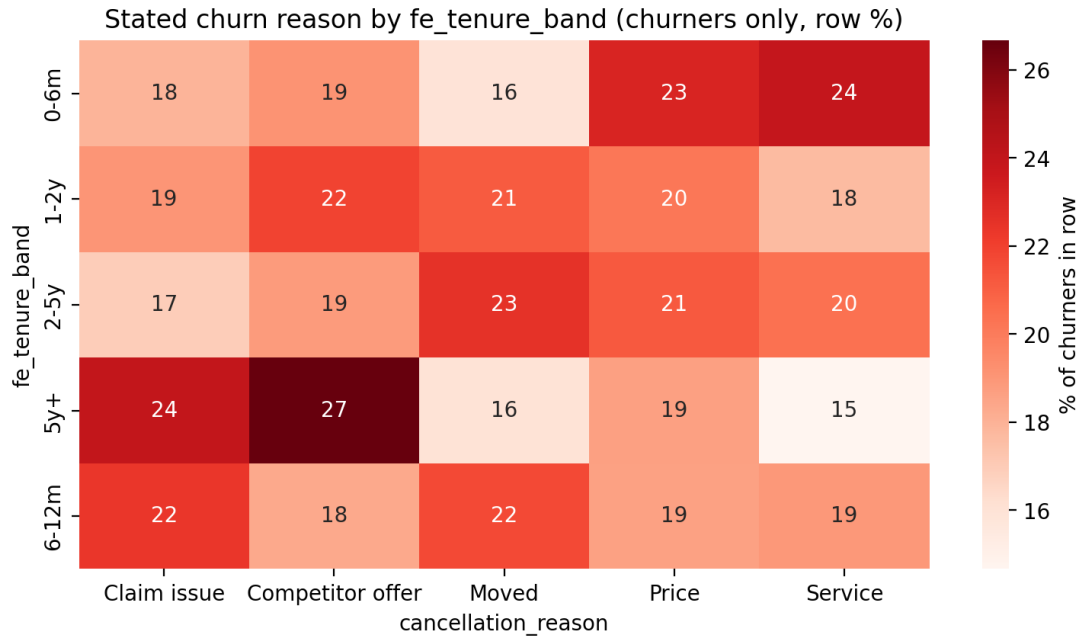


Figure C.7. Stated churn reason by tenure band (churners only, row %)

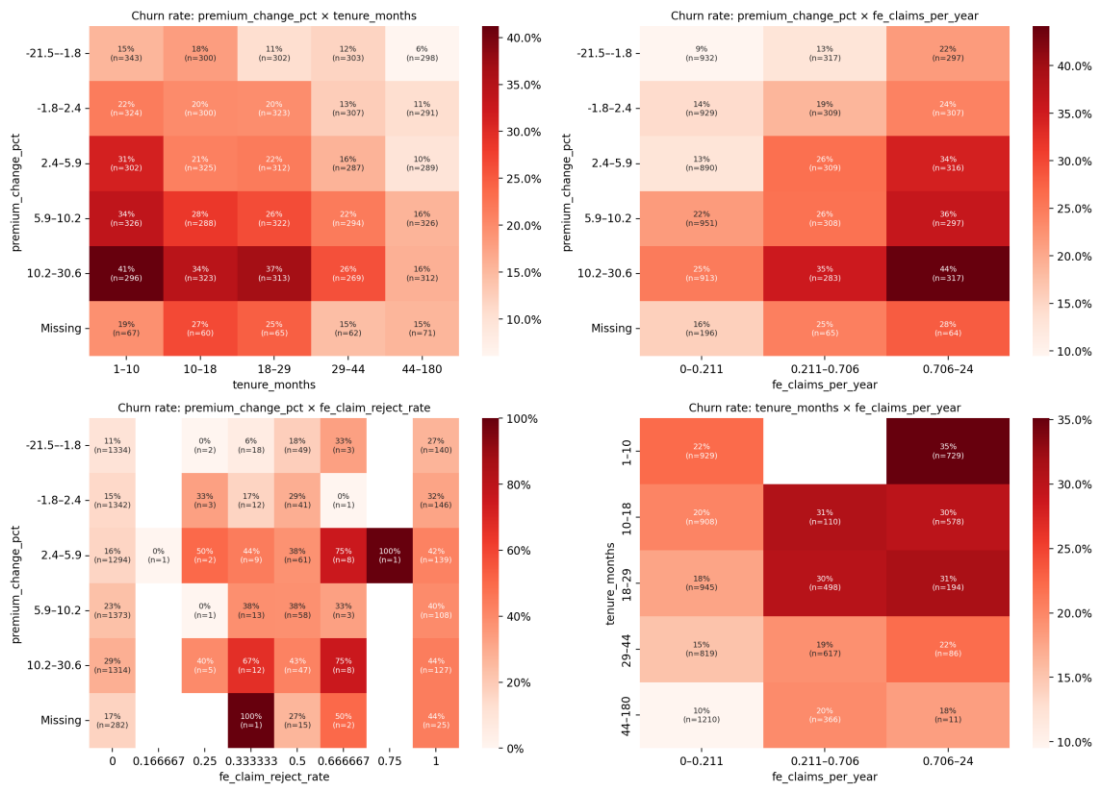


Figure C.8. Churn rate: premium change at last renewal (%) × tenure (months)

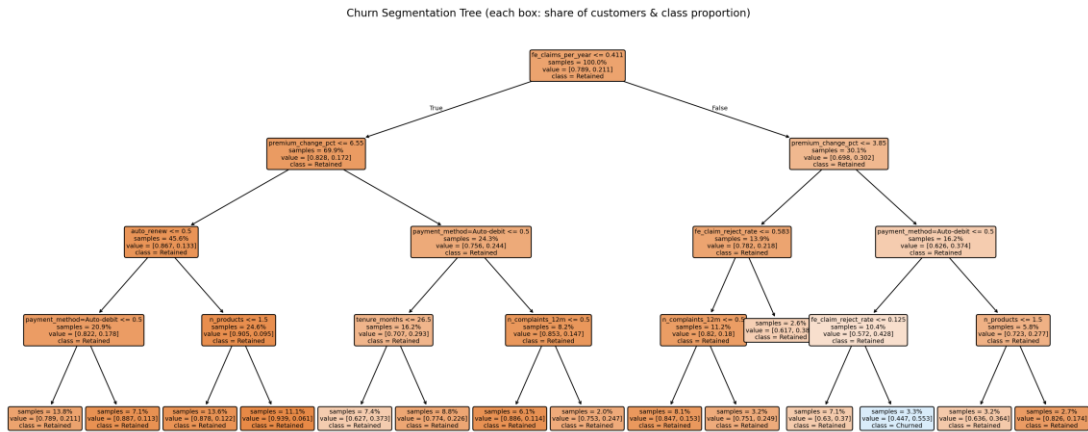


Figure C.9. Churn Segmentation Tree (each box: share of customers & class proportion)

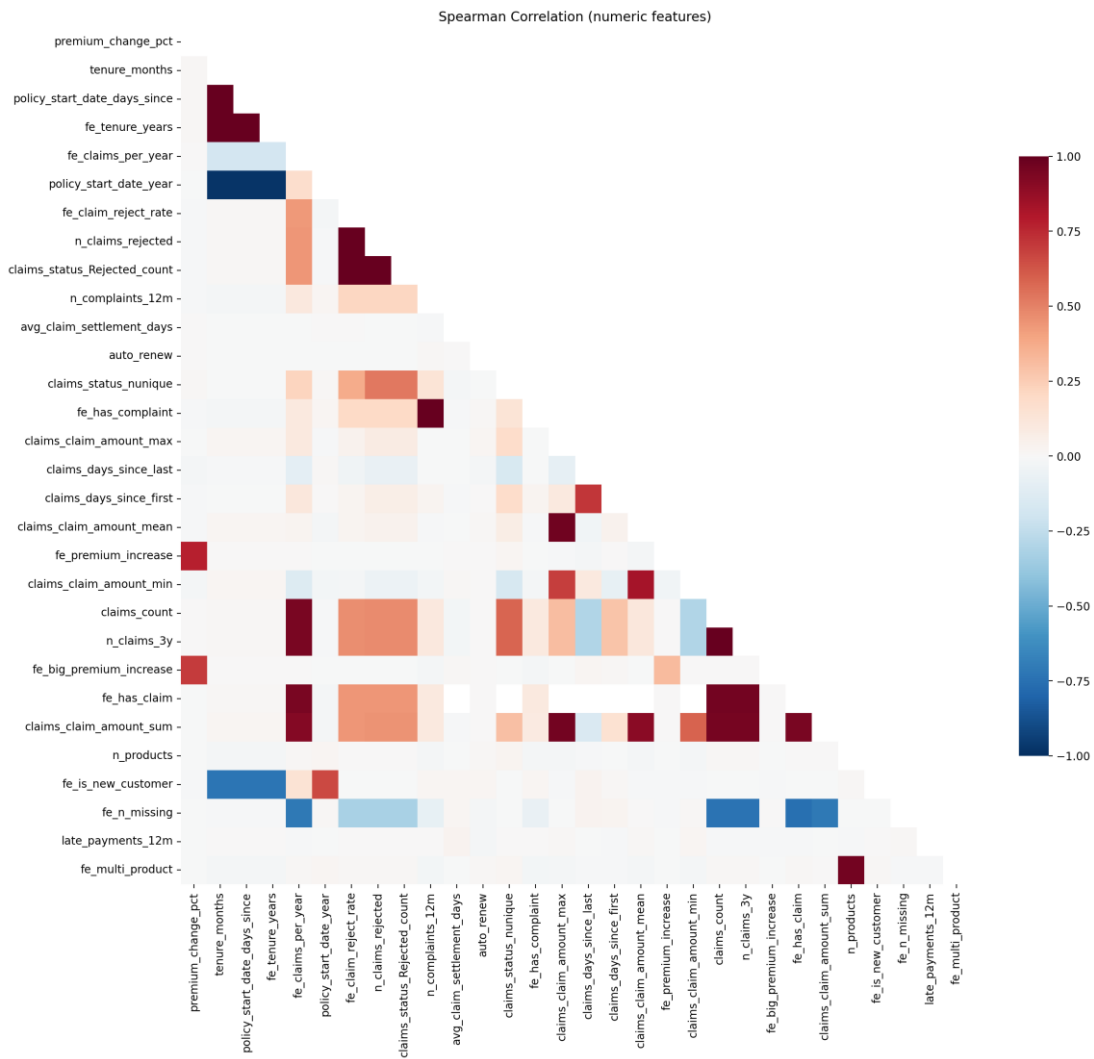
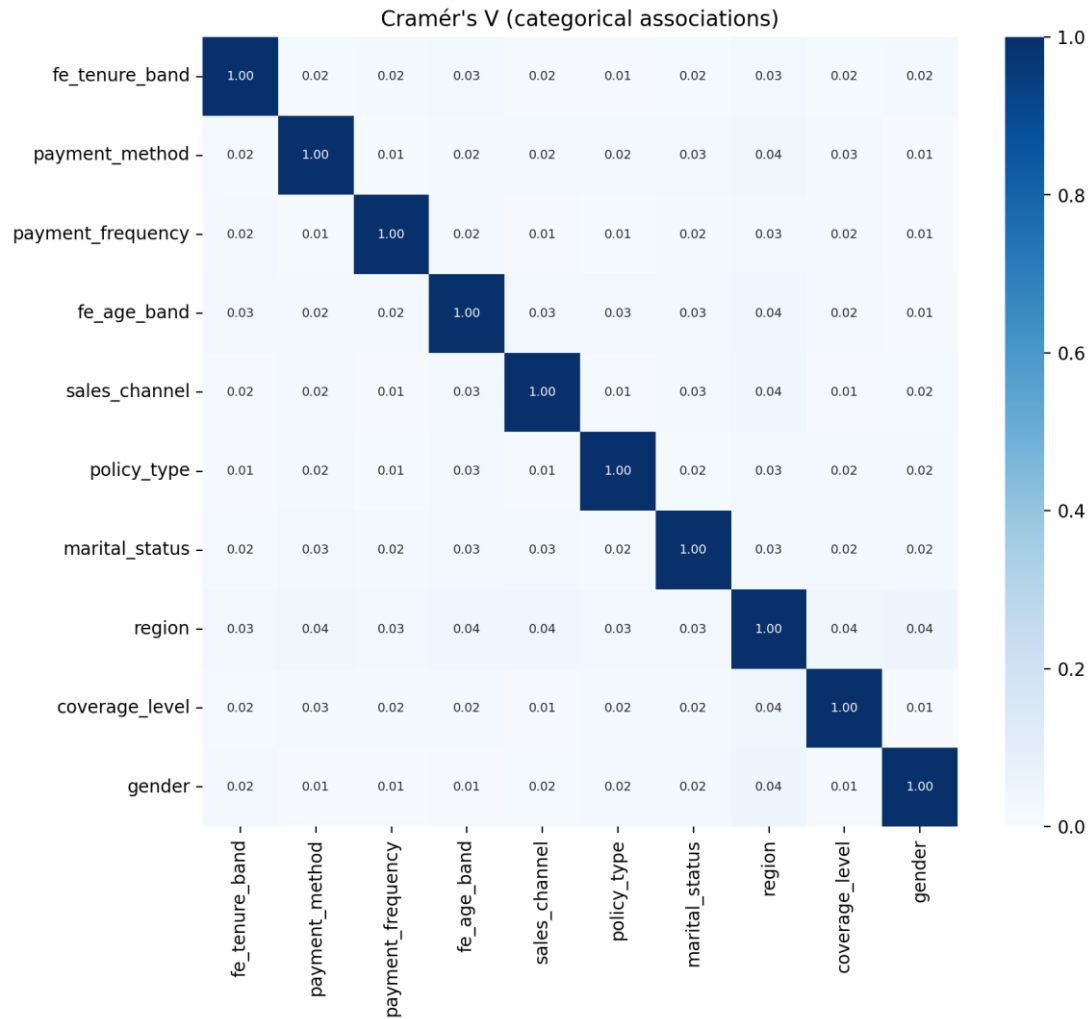


Figure C.10. Spearman Correlation (numeric features)



**Figure C.11.** Cramér's V (categorical associations)

**Table C.1.** Univariate driver statistics (top 25 by Information Value; q-values Benjamini–Hochberg).

| Feature                            | Type        | Iv    | Iv strength | Effect metric   | Effect size | Q value BH | Univariate auc | Highest risk level | Highest risk rate |
|------------------------------------|-------------|-------|-------------|-----------------|-------------|------------|----------------|--------------------|-------------------|
| premium change at last renewal (%) | numeric     | 0.159 | medium      | rank-biserial r | 0.231       | 1.3e-44    | 0.615          | 13.2–30.6          | 0.335             |
| tenure (months)                    | numeric     | 0.148 | medium      | rank-biserial r | -0.201      | 2.9e-36    | 0.601          | 1–6                | 0.286             |
| policy start date (days since)     | numeric     | 0.148 | medium      | rank-biserial r | -0.201      | 2.9e-36    | 0.601          | 31–183             | 0.286             |
| tenure (years)                     | numeric     | 0.148 | medium      | rank-biserial r | -0.201      | 2.9e-36    | 0.601          | 0.0833–0.5         | 0.286             |
| tenure band                        | categorical | 0.137 | medium      | Cramér's V      | 0.142       | 3.7e-33    | 0.588          | 0–6m               | 0.286             |
| claims per year of tenure          | numeric     | 0.134 | medium      | rank-biserial r | 0.180       | 3.6e-35    | 0.590          | 1.5–24             | 0.349             |
| policy start                       | numeric     | 0.131 | medium      | rank-           | 0.191       | 3.4e-34    | 0.595          | 2,024–             | 0.291             |

| Feature                              | Type        | Iv    | Iv strength | Effect metric   | Effect size | Q value BH | Univariate auc | Highest risk level | Highest risk rate |
|--------------------------------------|-------------|-------|-------------|-----------------|-------------|------------|----------------|--------------------|-------------------|
| date year                            |             |       |             | biserial r      |             |            |                | 2,025              |                   |
| claim rejection rate                 | numeric     | 0.117 | medium      | rank-biserial r | 0.122       | 5.1e-38    | 0.561          | 1                  | 0.371             |
| rejected claims                      | numeric     | 0.116 | medium      | rank-biserial r | 0.122       | 2.5e-38    | 0.561          | $\geq 2$           | 0.505             |
| claims status Rejected count         | numeric     | 0.116 | medium      | rank-biserial r | 0.122       | 2.5e-38    | 0.561          | $\geq 2$           | 0.505             |
| complaints in the last 12 months     | numeric     | 0.110 | medium      | rank-biserial r | 0.140       | 3.1e-30    | 0.570          | $\geq 2$           | 0.443             |
| average claim settlement time (days) | numeric     | 0.105 | medium      | rank-biserial r | 0.118       | 1.8e-07    | 0.559          | 21–28              | 0.346             |
| payment method                       | categorical | 0.098 | weak        | Cramér's V      | 0.123       | 1.3e-25    | 0.566          | Bank Transfer      | 0.255             |
| auto-renewal                         | categorical | 0.084 | weak        | Cramér's V      | 0.118       | 1.2e-25    | 0.570          | 0                  | 0.264             |
| claims status nunique                | categorical | 0.083 | weak        | Cramér's V      | 0.120       | 2.1e-25    | 0.567          | 1                  | 0.348             |
| has complained                       | categorical | 0.081 | weak        | Cramér's V      | 0.120       | 1.9e-26    | 0.559          | 1                  | 0.294             |
| claims claim amount max              | numeric     | 0.081 | weak        | rank-biserial r | 0.043       | 0.078      | 0.521          | 10.5M–57.7M        | 0.303             |
| claims (days since) last             | numeric     | 0.079 | weak        | rank-biserial r | -1.4e-04    | 0.995      | 0.500          | 85–176             | 0.291             |
| claims (days since) first            | numeric     | 0.078 | weak        | rank-biserial r | 0.005       | 0.865      | 0.503          | 630–727            | 0.294             |
| claims claim amount mean             | numeric     | 0.078 | weak        | rank-biserial r | 0.042       | 0.078      | 0.521          | 4.99M–6.3M         | 0.286             |
| premium increased at renewal         | categorical | 0.077 | weak        | Cramér's V      | 0.108       | 1.3e-20    | 0.557          | 1                  | 0.239             |
| claims claim amount min              | numeric     | 0.077 | weak        | rank-biserial r | 0.029       | 0.242      | 0.515          | 5.35M–8.02M        | 0.289             |
| claims count                         | numeric     | 0.076 | weak        | rank-biserial r | 0.140       | 7.3e-23    | 0.570          | $\geq 3$           | 0.328             |
| claims in the last 3 years           | numeric     | 0.076 | weak        | rank-biserial r | 0.140       | 7.3e-23    | 0.570          | $\geq 3$           | 0.328             |
| premium increase above 10%           | categorical | 0.074 | weak        | Cramér's V      | 0.117       | 7.9e-24    | 0.550          | 1                  | 0.307             |

**Table C.2.** All decision-tree segments.

| Segment rule                                                              | N customers | Share % | Churn rate | Lift  | Churners captured % |
|---------------------------------------------------------------------------|-------------|---------|------------|-------|---------------------|
| claims per year of tenure > 0.41 AND premium change at last renewal (%) > | 266         | 3.325   | 0.553      | 2.621 | 8.714               |

| Segment rule                                                                                                                                                             | N customers | Share % | Churn rate | Lift  | Churners captured % |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------------|-------|---------------------|
| 3.85 AND payment method $\neq$ Auto-debit AND claim rejection rate $> 0.12$                                                                                              |             |         |            |       |                     |
| claims per year of tenure $> 0.41$ AND premium change at last renewal (%) $\leq 3.85$ AND claim rejection rate $> 0.58$                                                  | 209         | 2.612   | 0.383      | 1.815 | 4.742               |
| claims per year of tenure $\leq 0.41$ AND premium change at last renewal (%) $> 6.55$ AND payment method $\neq$ Auto-debit AND tenure (months) $\leq 26.50$              | 590         | 7.375   | 0.373      | 1.768 | 13.0                |
| claims per year of tenure $> 0.41$ AND premium change at last renewal (%) $> 3.85$ AND payment method $\neq$ Auto-debit AND claim rejection rate $\leq 0.12$             | 568         | 7.100   | 0.370      | 1.753 | 12.4                |
| claims per year of tenure $> 0.41$ AND premium change at last renewal (%) $> 3.85$ AND payment method = Auto-debit AND number of ANS products $\leq 1.50$                | 253         | 3.163   | 0.364      | 1.724 | 5.453               |
| claims per year of tenure $> 0.41$ AND premium change at last renewal (%) $\leq 3.85$ AND claim rejection rate $\leq 0.58$ AND complaints in the last 12 months $> 0.50$ | 253         | 3.163   | 0.249      | 1.181 | 3.734               |
| claims per year of tenure $\leq 0.41$ AND premium                                                                                                                        | 162         | 2.025   | 0.247      | 1.171 | 2.371               |

| Segment rule                                                                                                                                                              | N customers | Share % | Churn rate | Lift  | Churners captured % |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------------|-------|---------------------|
| change at last renewal (%) > 6.55 AND payment method = Auto-debit AND complaints in the last 12 months > 0.50                                                             |             |         |            |       |                     |
| claims per year of tenure $\leq 0.41$ AND premium change at last renewal (%) > 6.55 AND payment method $\neq$ Auto-debit AND tenure (months) > 26.50                      | 705         | 8.812   | 0.226      | 1.070 | 9.425               |
| claims per year of tenure $\leq 0.41$ AND premium change at last renewal (%) $\leq 6.55$ AND auto-renewal = 0 AND payment method $\neq$ Auto-debit                        | 1,105       | 13.8    | 0.211      | 1.000 | 13.8                |
| claims per year of tenure > 0.41 AND premium change at last renewal (%) > 3.85 AND payment method = Auto-debit AND number of ANS products > 1.50                          | 213         | 2.663   | 0.174      | 0.824 | 2.193               |
| claims per year of tenure > 0.41 AND premium change at last renewal (%) $\leq 3.85$ AND claim rejection rate $\leq 0.58$ AND complaints in the last 12 months $\leq 0.50$ | 647         | 8.088   | 0.153      | 0.726 | 5.868               |
| claims per year of tenure $\leq 0.41$ AND premium change at last renewal (%) $\leq 6.55$ AND auto-renewal = 1 AND number of ANS products $\leq 1.50$                      | 1,087       | 13.6    | 0.122      | 0.580 | 7.884               |

| Segment rule                                                                                                                                                                 | N customers | Share % | Churn rate | Lift  | Churners captured % |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|------------|-------|---------------------|
| claims per year of tenure $\leq 0.41$<br>AND premium change at last renewal (%) $> 6.55$ AND payment method = Auto-debit<br>AND complaints in the last 12 months $\leq 0.50$ | 490         | 6.125   | 0.114      | 0.542 | 3.320               |
| claims per year of tenure $\leq 0.41$<br>AND premium change at last renewal (%) $\leq 6.55$ AND auto-renewal = 0<br>AND payment method = Auto-debit                          | 567         | 7.087   | 0.113      | 0.535 | 3.794               |
| claims per year of tenure $\leq 0.41$<br>AND premium change at last renewal (%) $\leq 6.55$ AND auto-renewal = 1<br>AND number of ANS products $> 1.50$                      | 885         | 11.1    | 0.061      | 0.289 | 3.201               |

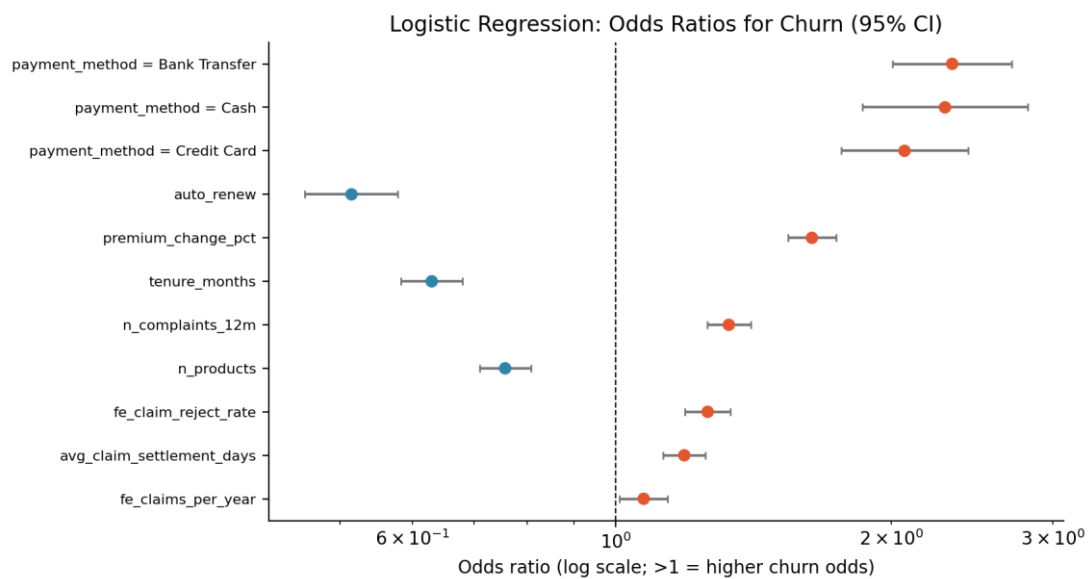
**Table C.3.** Variance inflation factors (numeric features).

| Feature                            | VIF           |
|------------------------------------|---------------|
| tenure (months)                    | 1,000,799.92B |
| tenure (years)                     | 1,000,799.92B |
| claims status Rejected count       | 1,000,799.92B |
| rejected claims                    | 1,000,799.92B |
| claims in the last 3 years         | 1,000,799.92B |
| claims count                       | 1,000,799.92B |
| policy start date (days since)     | 6.57M         |
| claims claim amount mean           | 262.4         |
| claims claim amount max            | 100.1         |
| claims claim amount min            | 73.5          |
| policy start date year             | 47.1          |
| claims claim amount sum            | 34.4          |
| has made a claim                   | 14.1          |
| claim rejection rate               | 12.8          |
| complaints in the last 12 months   | 5.996         |
| has complained                     | 5.874         |
| claims (days since) last           | 5.579         |
| claims (days since) first          | 5.540         |
| premium change at last renewal (%) | 5.055         |
| holds multiple products            | 3.635         |
| number of ANS products             | 3.633         |
| premium increased at renewal       | 2.869         |
| claims status nunique              | 2.593         |

| Feature                              | VIF   |
|--------------------------------------|-------|
| premium increase above 10%           | 2.463 |
| claims per year of tenure            | 2.119 |
| number of missing fields             | 2.053 |
| new customer (<12 months)            | 1.630 |
| average claim settlement time (days) | 1.026 |
| auto-renewal                         | 1.005 |
| late payments in the last 12 months  | 1.002 |

## Appendix D. Interpretable Logistic Regression

Adjusted odds ratios with 95% confidence intervals; numeric variables are standardised (per 1 SD), categories are compared with the most common level.



**Figure D.1.** Logistic Regression: Odds Ratios for Churn (95% CI)

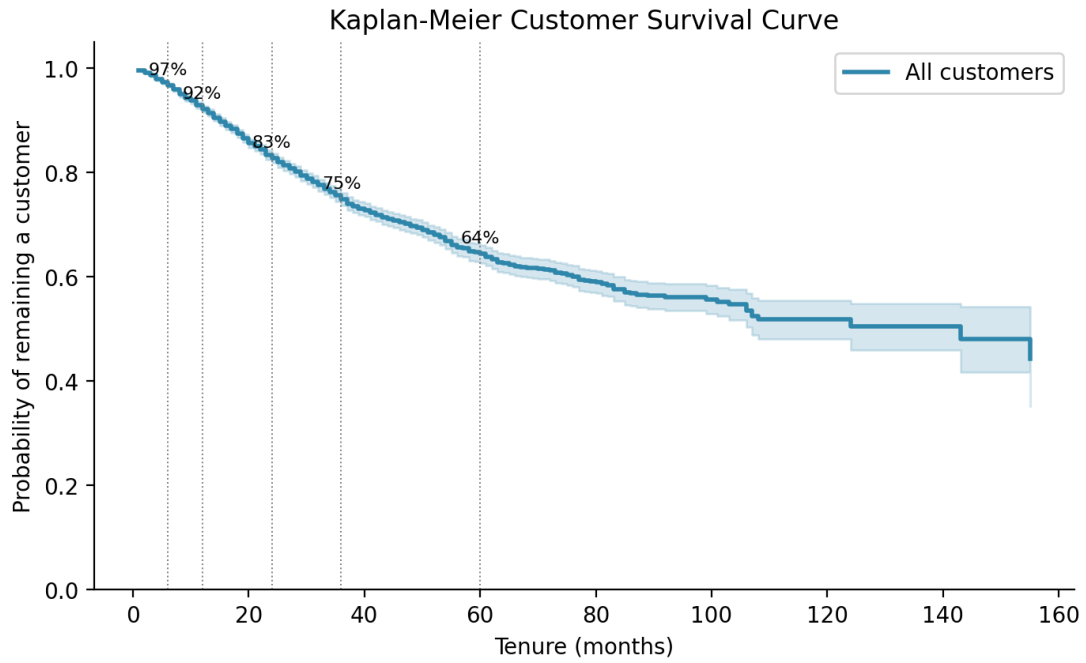
**Table D.1.** Logistic regression: odds ratios, 95% CI, p-values and average marginal effects.

| Term                               | Coef   | Odds ratio | Or ci low | Or ci high | P value | Scale             | Avg marginal effect pp | Interpretation                         |
|------------------------------------|--------|------------|-----------|------------|---------|-------------------|------------------------|----------------------------------------|
| payment method = Bank Transfer     | 0.846  | 2.330      | 2.005     | 2.709      | 3.3e-28 | vs 'Auto-debit'   | 12.1                   | +133% odds of churn (vs 'Auto-debit')  |
| payment method = Cash              | 0.828  | 2.289      | 1.860     | 2.817      | 5.5e-15 | vs 'Auto-debit'   | 11.8                   | +129% odds of churn (vs 'Auto-debit')  |
| payment method = Credit Card       | 0.726  | 2.067      | 1.762     | 2.425      | 4.7e-19 | vs 'Auto-debit'   | 10.4                   | +107% odds of churn (vs 'Auto-debit')  |
| auto-renewal                       | -0.665 | 0.514      | 0.458     | 0.578      | 7.6e-29 | binary (0→1)      | -9.486                 | -49% odds of churn (binary (0→1))      |
| premium change at last renewal (%) | 0.493  | 1.638      | 1.541     | 1.741      | 9.5e-57 | per 1 SD (= 6.99) | 7.040                  | +64% odds of churn (per 1 SD (= 6.99)) |

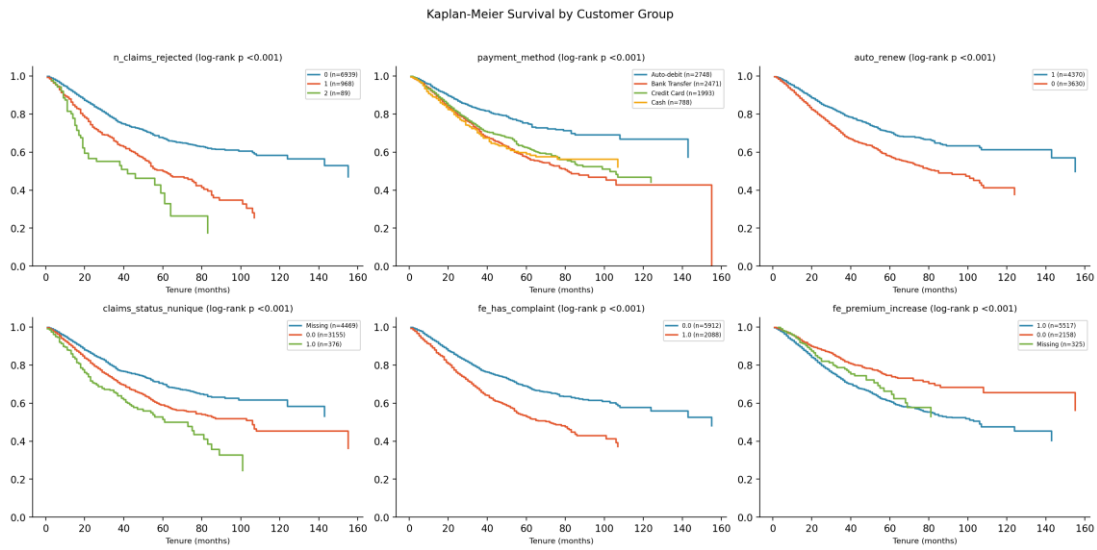
| Term                                 | Coef   | Odds ratio | Or ci low | Or ci high | P value | Scale                 | Avg marginal effect pp | Interpretation                            |
|--------------------------------------|--------|------------|-----------|------------|---------|-----------------------|------------------------|-------------------------------------------|
| tenure (months)                      | -0.463 | 0.629      | 0.582     | 0.680      | 1.1e-31 | per 1 SD (= 23.6)     | -6.606                 | -37% odds of churn (per 1 SD (= 23.6))    |
| complaints in the last 12 months     | 0.284  | 1.329      | 1.258     | 1.403      | 2.2e-24 | per 1 SD (= 0.573)    | 4.056                  | +33% odds of churn (per 1 SD (= 0.573))   |
| number of ANS products               | -0.278 | 0.757      | 0.710     | 0.808      | 3.1e-17 | per 1 SD (= 0.77)     | -3.972                 | -24% odds of churn (per 1 SD (= 0.77))    |
| claims status nunique                | 0.236  | 1.266      | 0.937     | 1.710      | 0.125   | binary (0→1)          | 3.363                  | +27% odds of churn (binary (0→1))         |
| claim rejection rate                 | 0.231  | 1.260      | 1.189     | 1.335      | 3.9e-15 | per 1 SD (= 0.291)    | 3.297                  | +26% odds of churn (per 1 SD (= 0.291))   |
| average claim settlement time (days) | 0.172  | 1.188      | 1.126     | 1.253      | 2.3e-10 | per 1 SD (= 6.49)     | 2.456                  | +19% odds of churn (per 1 SD (= 6.49))    |
| claims per year of tenure            | 0.070  | 1.073      | 1.009     | 1.140      | 0.024   | per 1 SD (= 1.11)     | 1.000                  | +7% odds of churn (per 1 SD (= 1.11))     |
| number of missing fields             | -0.043 | 0.958      | 0.896     | 1.024      | 0.211   | per 1 SD (= 0.698)    | -0.608                 | -4% odds of churn (per 1 SD (= 0.698))    |
| claims claim amount min              | 0.037  | 1.038      | 0.951     | 1.133      | 0.403   | per 1 SD (= 2.64e+06) | 0.534                  | +4% odds of churn (per 1 SD (= 2.64e+06)) |
| claims (days since) last             | 0.026  | 1.026      | 0.930     | 1.133      | 0.606   | per 1 SD (= 208)      | 0.371                  | +3% odds of churn (per 1 SD (= 208))      |
| claims (days since) first            | -0.015 | 0.985      | 0.892     | 1.088      | 0.770   | per 1 SD (= 208)      | -0.211                 | -1% odds of churn (per 1 SD (= 208))      |
| claims claim amount max              | 0.012  | 1.012      | 0.924     | 1.107      | 0.802   | per 1 SD (= 3.37e+06) | 0.165                  | +1% odds of churn (per 1 SD (= 3.37e+06)) |

## Appendix E. Survival Analysis

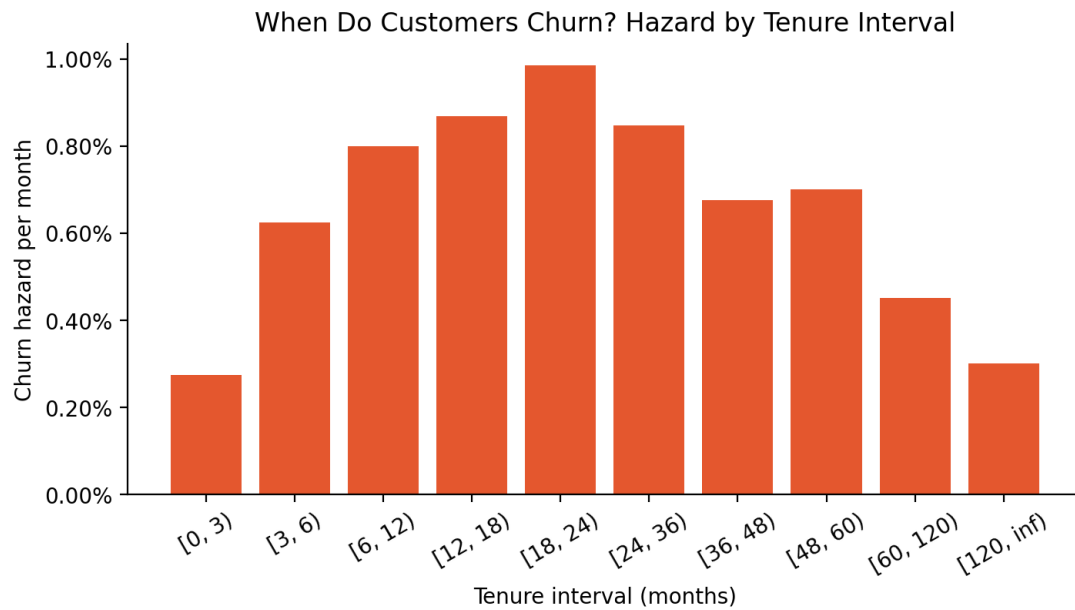
Kaplan–Meier estimates, hazards by tenure, Cox and Weibull AFT models, proportional-hazards diagnostics and validation of risk tiers over time.



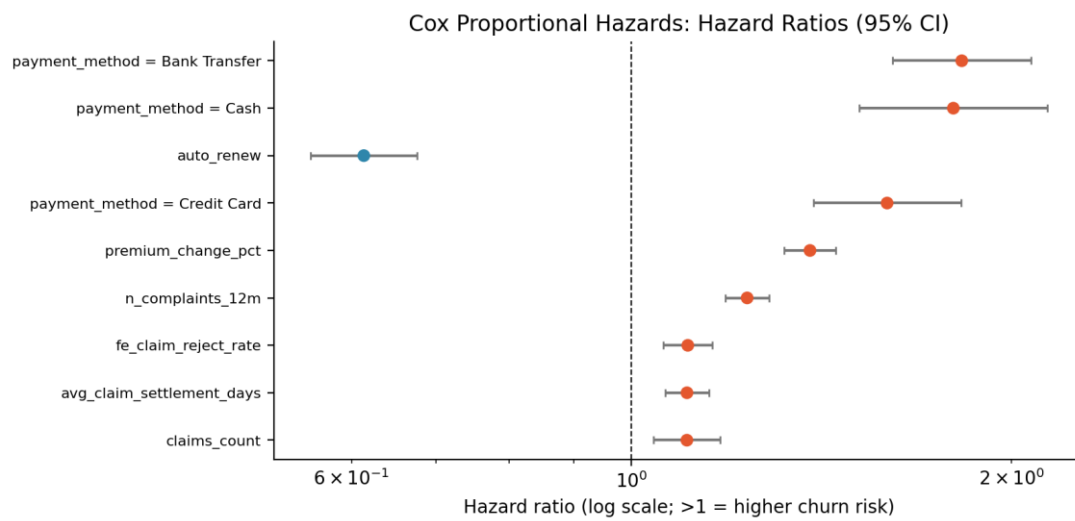
**Figure E.1.** Kaplan-Meier Customer Survival Curve



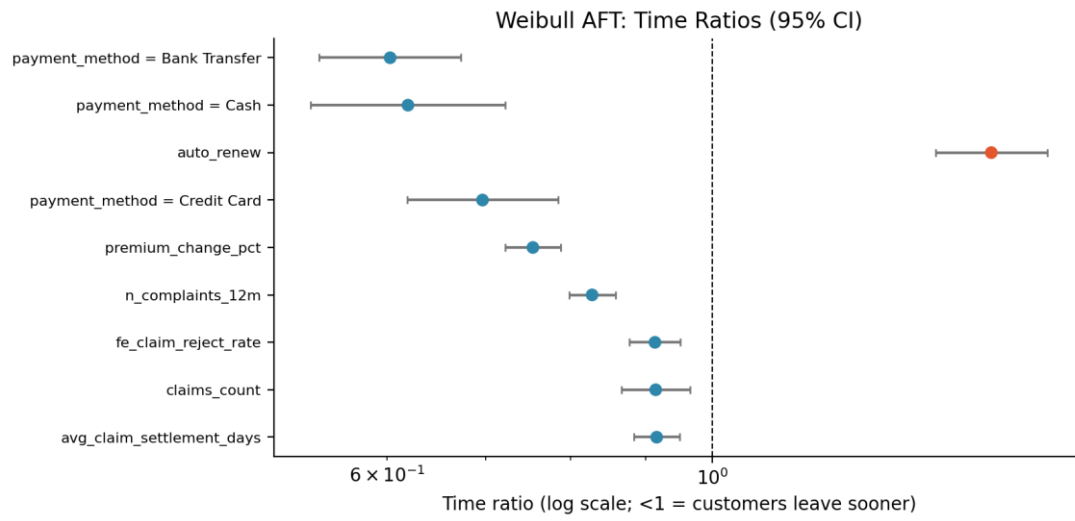
**Figure E.2.** Kaplan-Meier Survival by Customer Group



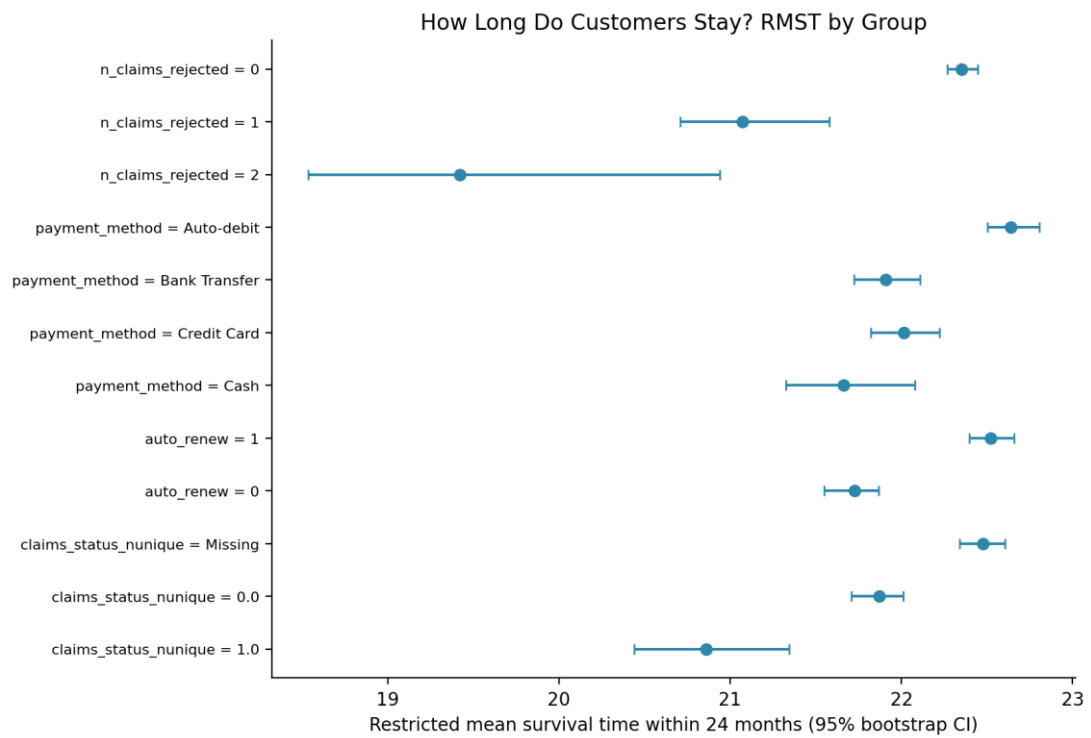
**Figure E.3.** When Do Customers Churn? Hazard by Tenure Interval



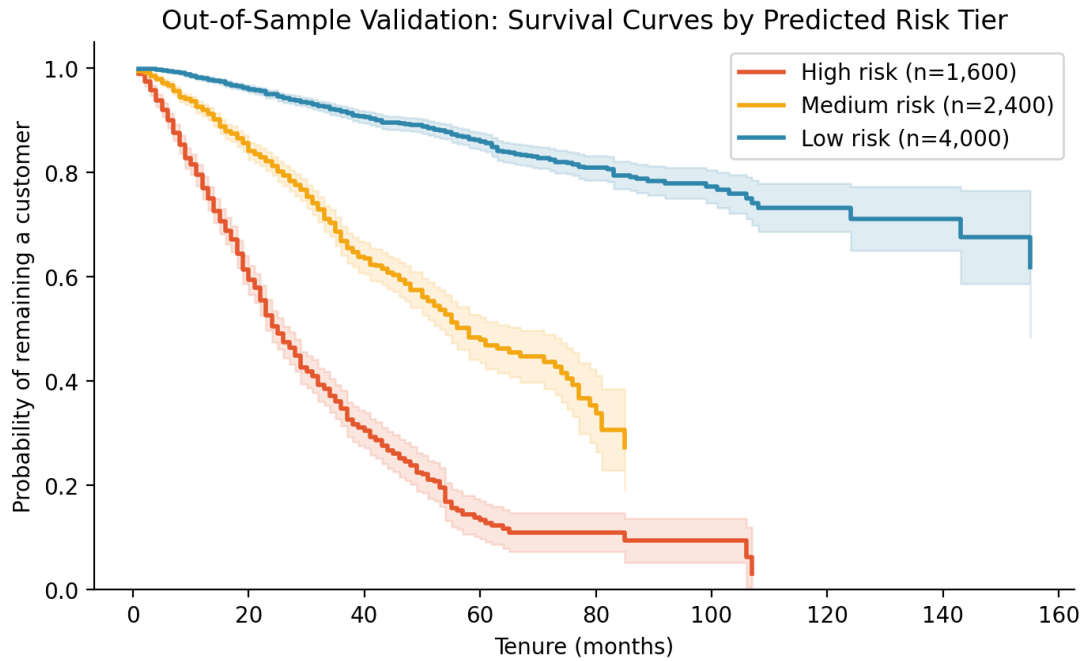
**Figure E.4.** Cox Proportional Hazards: Hazard Ratios (95% CI)



**Figure E.5.** Weibull AFT: Time Ratios (95% CI)



**Figure E.6.** How Long Do Customers Stay? RMST by Group



**Figure E.7.** Out-of-Sample Validation: Survival Curves by Predicted Risk Tier

**Table E.1.** Hazard by tenure interval.

| Interval   | At risk start | Churn events | Person time | Hazard |
|------------|---------------|--------------|-------------|--------|
| [0, 3)     | 8,000         | 65           | 23,681      | 0.003  |
| [3, 6)     | 7,781         | 140          | 22,408      | 0.006  |
| [6, 12)    | 7,296         | 318          | 39,736      | 0.008  |
| [12, 18)   | 6,134         | 283          | 32,571      | 0.009  |
| [18, 24)   | 4,940         | 256          | 25,987      | 0.010  |
| [24, 36)   | 3,938         | 310          | 36,575      | 0.008  |
| [36, 48)   | 2,365         | 145          | 21,463      | 0.007  |
| [48, 60)   | 1,394         | 91           | 12,989      | 0.007  |
| [60, 120)  | 848           | 76           | 16,858      | 0.005  |
| [120, inf) | 47            | 3            | 997.0       | 0.003  |

**Table E.2.** Log-rank tests by group.

| Feature                      | Logrank chi2 | P value |
|------------------------------|--------------|---------|
| rejected claims              | 130.1        | 0       |
| payment method               | 94.8         | 0       |
| auto-renewal                 | 90.7         | 0       |
| claims status nunique        | 83.5         | 0       |
| has complained               | 107.4        | 0       |
| premium increased at renewal | 72.2         | 2.2e-16 |

**Table E.3.** Cox proportional hazards model.

| Term                           | Coef  | Hazard ratio | Hr ci low | Hr ci high | P value | Scale           | Interpretation                      |
|--------------------------------|-------|--------------|-----------|------------|---------|-----------------|-------------------------------------|
| payment method = Bank Transfer | 0.603 | 1.827        | 1.610     | 2.073      | 9.6e-21 | vs 'Auto-debit' | +83% churn hazard (vs 'Auto-debit') |

| Term                                 | Coef    | Hazard ratio | Hr ci low | Hr ci high | P value | Scale                 | Interpretation                           |
|--------------------------------------|---------|--------------|-----------|------------|---------|-----------------------|------------------------------------------|
| payment method = Cash                | 0.587   | 1.799        | 1.516     | 2.135      | 1.8e-11 | vs 'Auto-debit'       | +80% churn hazard (vs 'Auto-debit')      |
| auto-renewal                         | -0.488  | 0.614        | 0.557     | 0.676      | 4.0e-23 | binary (0→1)          | -39% churn hazard (binary (0→1))         |
| payment method = Credit Card         | 0.466   | 1.594        | 1.393     | 1.824      | 1.1e-11 | vs 'Auto-debit'       | +59% churn hazard (vs 'Auto-debit')      |
| premium change at last renewal (%)   | 0.326   | 1.385        | 1.321     | 1.452      | 3.6e-41 | per 1 SD (= 6.99)     | +39% churn hazard (per 1 SD (= 6.99))    |
| complaints in the last 12 months     | 0.211   | 1.235        | 1.187     | 1.285      | 1.9e-25 | per 1 SD (= 0.573)    | +23% churn hazard (per 1 SD (= 0.573))   |
| claim rejection rate                 | 0.103   | 1.108        | 1.060     | 1.159      | 5.7e-06 | per 1 SD (= 0.291)    | +11% churn hazard (per 1 SD (= 0.291))   |
| average claim settlement time (days) | 0.101   | 1.107        | 1.063     | 1.151      | 5.9e-07 | per 1 SD (= 6.49)     | +11% churn hazard (per 1 SD (= 6.49))    |
| claims count                         | 0.101   | 1.106        | 1.041     | 1.176      | 0.001   | per 1 SD (= 0.777)    | +11% churn hazard (per 1 SD (= 0.777))   |
| claims (days since) last             | 0.042   | 1.043        | 0.966     | 1.126      | 0.278   | per 1 SD (= 208)      | +4% churn hazard (per 1 SD (= 208))      |
| claims (days since) first            | -0.041  | 0.960        | 0.889     | 1.036      | 0.296   | per 1 SD (= 208)      | -4% churn hazard (per 1 SD (= 208))      |
| claims status nunique                | 0.034   | 1.035        | 0.817     | 1.311      | 0.778   | binary (0→1)          | +3% churn hazard (binary (0→1))          |
| claims claim amount min              | 0.016   | 1.016        | 0.953     | 1.083      | 0.628   | per 1 SD (= 2.64e+06) | +2% churn hazard (per 1 SD (= 2.64e+06)) |
| claims claim amount max              | 1.7e-04 | 1.000        | 0.938     | 1.066      | 0.996   | per 1 SD (= 3.37e+06) | +0% churn hazard (per 1 SD (= 3.37e+06)) |

**Table E.4.** Schoenfeld test of the proportional-hazards assumption.

| Term                                 | Test statistic | P     | Ph violated |
|--------------------------------------|----------------|-------|-------------|
| auto-renewal                         | 0.370          | 0.543 | no          |
| average claim settlement time (days) | 0.061          | 0.804 | no          |
| claims claim amount max              | 1.976          | 0.160 | no          |
| claims claim amount min              | 1.375          | 0.241 | no          |
| claims count                         | 0.026          | 0.871 | no          |
| claims (days since) first            | 1.184          | 0.277 | no          |
| claims (days since) last             | 1.272          | 0.259 | no          |
| claims status nunique                | 0.498          | 0.480 | no          |
| claim rejection rate                 | 0.038          | 0.846 | no          |
| complaints in the last 12 months     | 9.8e-04        | 0.975 | no          |
| payment method = Bank Transfer       | 5.992          | 0.014 | no          |
| payment method = Cash                | 0.141          | 0.707 | no          |
| payment method = Credit Card         | 0.730          | 0.393 | no          |
| premium change at last renewal (%)   | 0.436          | 0.509 | no          |

**Table E.5.** Weibull accelerated failure time model (time ratios).

| Term                                 | Time ratio | Tr ci low | Tr ci high | P value | Scale                 |
|--------------------------------------|------------|-----------|------------|---------|-----------------------|
| payment method = Bank Transfer       | 0.603      | 0.539     | 0.674      | 4.5e-19 | vs 'Auto-debit'       |
| payment method = Cash                | 0.620      | 0.532     | 0.722      | 7.9e-10 | vs 'Auto-debit'       |
| auto-renewal                         | 1.549      | 1.419     | 1.691      | 1.3e-22 | binary (0→1)          |
| payment method = Credit Card         | 0.697      | 0.619     | 0.784      | 2.2e-09 | vs 'Auto-debit'       |
| premium change at last renewal (%)   | 0.754      | 0.722     | 0.788      | 2.1e-36 | per 1 SD (= 6.99)     |
| complaints in the last 12 months     | 0.828      | 0.798     | 0.859      | 2.8e-24 | per 1 SD (= 0.573)    |
| claim rejection rate                 | 0.914      | 0.878     | 0.951      | 9.4e-06 | per 1 SD (= 0.291)    |
| claims count                         | 0.915      | 0.867     | 0.965      | 0.001   | per 1 SD (= 0.777)    |
| average claim settlement time (days) | 0.916      | 0.883     | 0.950      | 2.2e-06 | per 1 SD (= 6.49)     |
| claims status nunique                | 0.961      | 0.779     | 1.185      | 0.709   | binary (0→1)          |
| claims (days since) first            | 1.031      | 0.966     | 1.099      | 0.360   | per 1 SD (= 208)      |
| claims (days since) last             | 0.972      | 0.912     | 1.037      | 0.387   | per 1 SD (= 208)      |
| claims claim amount min              | 0.983      | 0.931     | 1.038      | 0.539   | per 1 SD (= 2.64e+06) |
| claims claim amount max              | 1.003      | 0.951     | 1.058      | 0.920   | per 1 SD (= 3.37e+06) |

**Table E.6.** Restricted mean survival time by group.

| Feature               | Level         | N     | Rmst | Rmst ci low | Rmst ci high | Diff vs best | Reference level | Tau  |
|-----------------------|---------------|-------|------|-------------|--------------|--------------|-----------------|------|
| rejected claims       | 0             | 6,939 | 22.4 | 22.3        | 22.4         | 0            | 0               | 24.0 |
| rejected claims       | 1             | 968   | 21.1 | 20.7        | 21.6         | -1.281       | 0               | 24.0 |
| rejected claims       | 2             | 89    | 19.4 | 18.5        | 20.9         | -2.930       | 0               | 24.0 |
| payment method        | Auto-debit    | 2,748 | 22.6 | 22.5        | 22.8         | 0            | Auto-debit      | 24.0 |
| payment method        | Bank Transfer | 2,471 | 21.9 | 21.7        | 22.1         | -0.730       | Auto-debit      | 24.0 |
| payment method        | Credit Card   | 1,993 | 22.0 | 21.8        | 22.2         | -0.624       | Auto-debit      | 24.0 |
| payment method        | Cash          | 788   | 21.7 | 21.3        | 22.1         | -0.976       | Auto-debit      | 24.0 |
| auto-renewal          | 1             | 4,370 | 22.5 | 22.4        | 22.7         | 0            | 1               | 24.0 |
| auto-renewal          | 0             | 3,630 | 21.7 | 21.5        | 21.9         | -0.795       | 1               | 24.0 |
| claims status nunique | Missing       | 4,469 | 22.5 | 22.3        | 22.6         | 0            | Missing         | 24.0 |
| claims status nunique | 0.0           | 3,155 | 21.9 | 21.7        | 22.0         | -0.604       | Missing         | 24.0 |

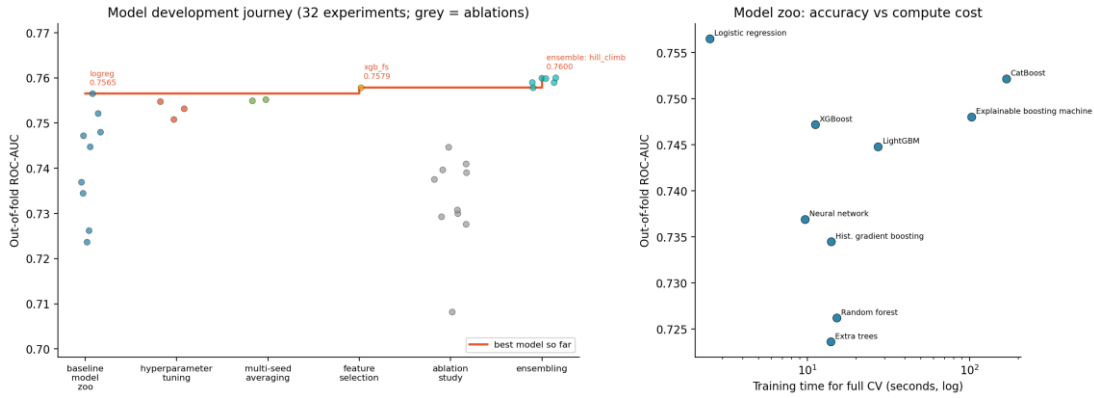
| Feature                     | Level | N   | Rmst | Rmst ci low | Rmst ci high | Diff vs best | Reference level | Tau  |
|-----------------------------|-------|-----|------|-------------|--------------|--------------|-----------------|------|
| claims<br>status<br>nunique | 1.0   | 376 | 20.9 | 20.4        | 21.3         | -1.614       | Missing         | 24.0 |

**Table E.7.** Survival probability by predicted risk tier.

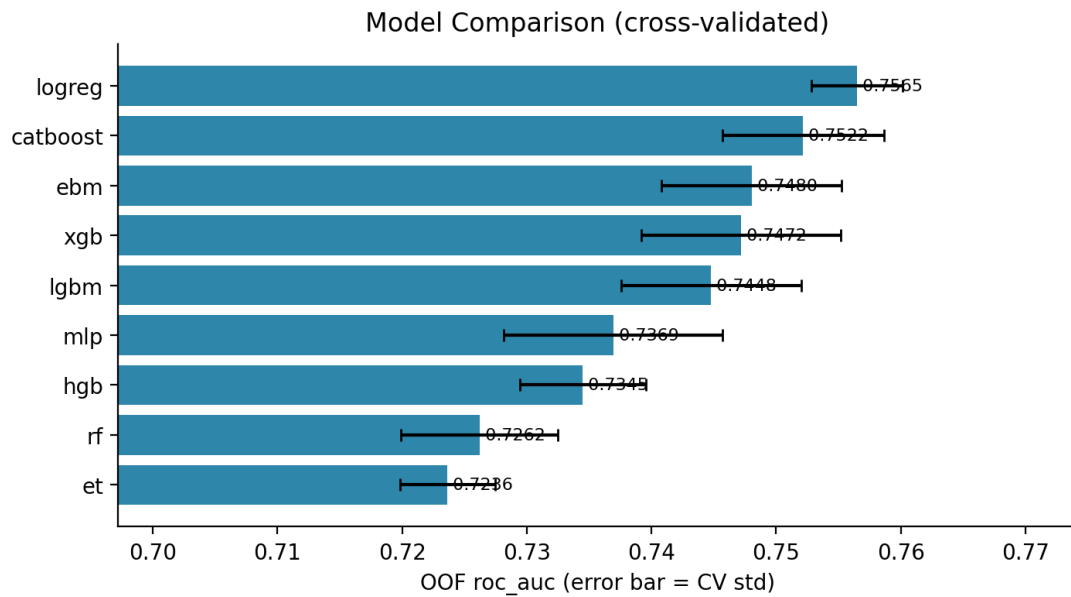
| Tier   | N     | S(6)  | S(12) | S(24) | S(36) | S(60) |
|--------|-------|-------|-------|-------|-------|-------|
| High   | 1,600 | 0.900 | 0.771 | 0.505 | 0.348 | 0.134 |
| Medium | 2,400 | 0.968 | 0.920 | 0.813 | 0.669 | 0.479 |
| Low    | 4,000 | 0.994 | 0.980 | 0.950 | 0.917 | 0.860 |

## Appendix F. Model Development, Comparison and Validation

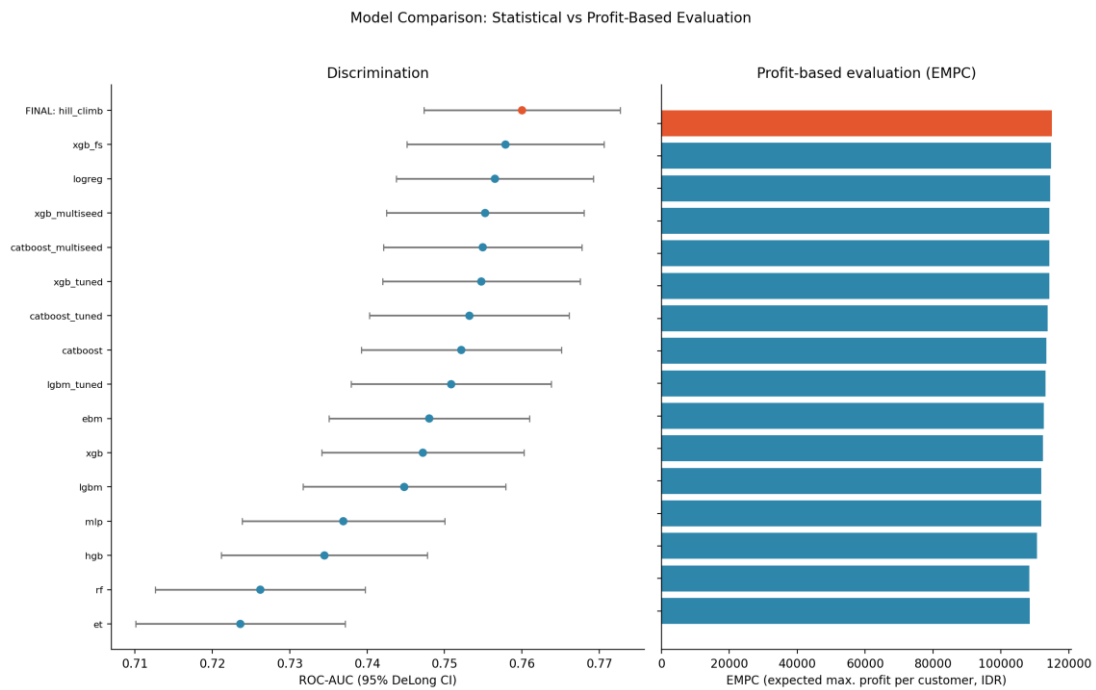
All logged experiments (trial-and-error), statistical comparison of models, ablation study and diagnostic checks.



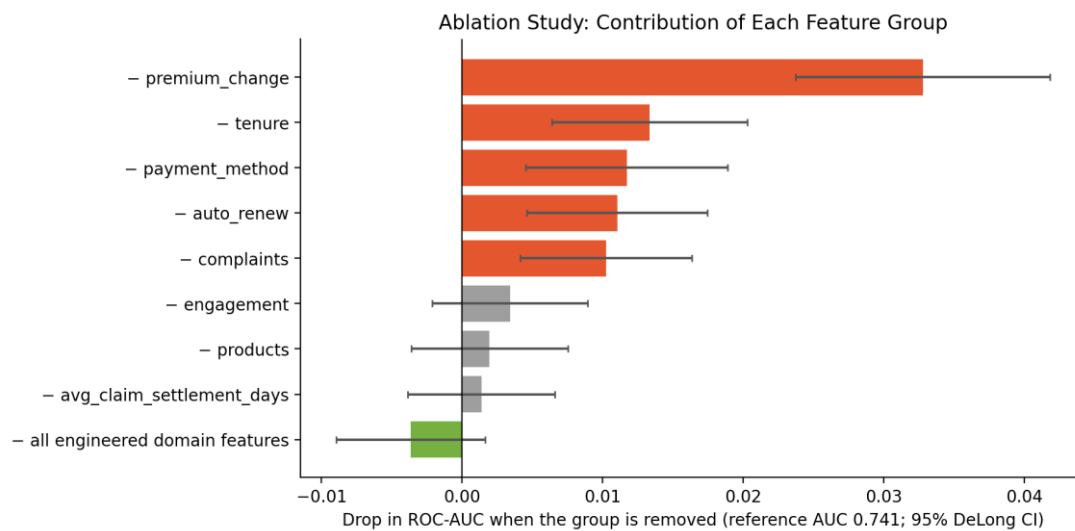
**Figure F.1.** Model development journey (32 experiments; grey = ablations)



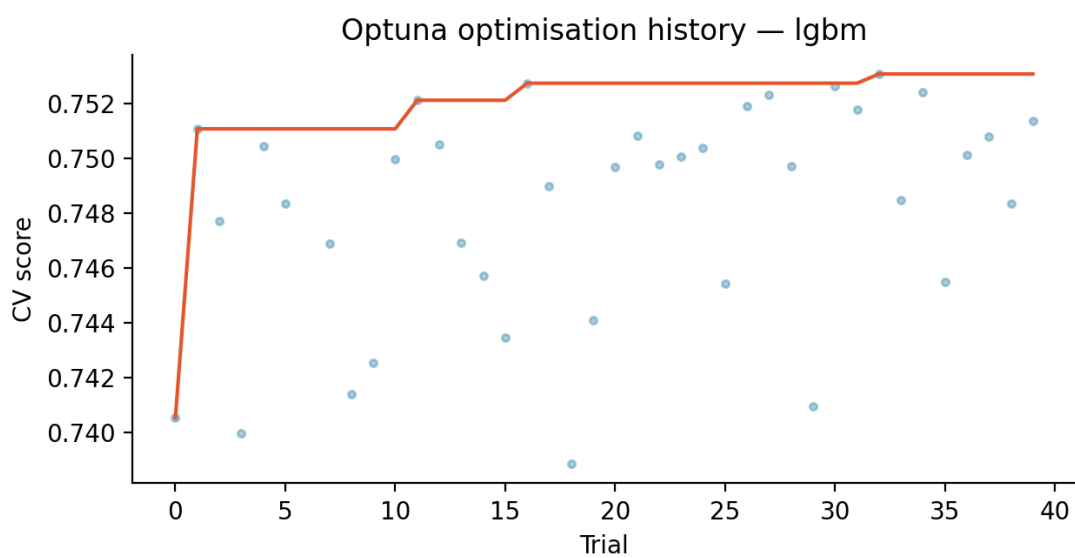
**Figure F.2.** Model Comparison (cross-validated)



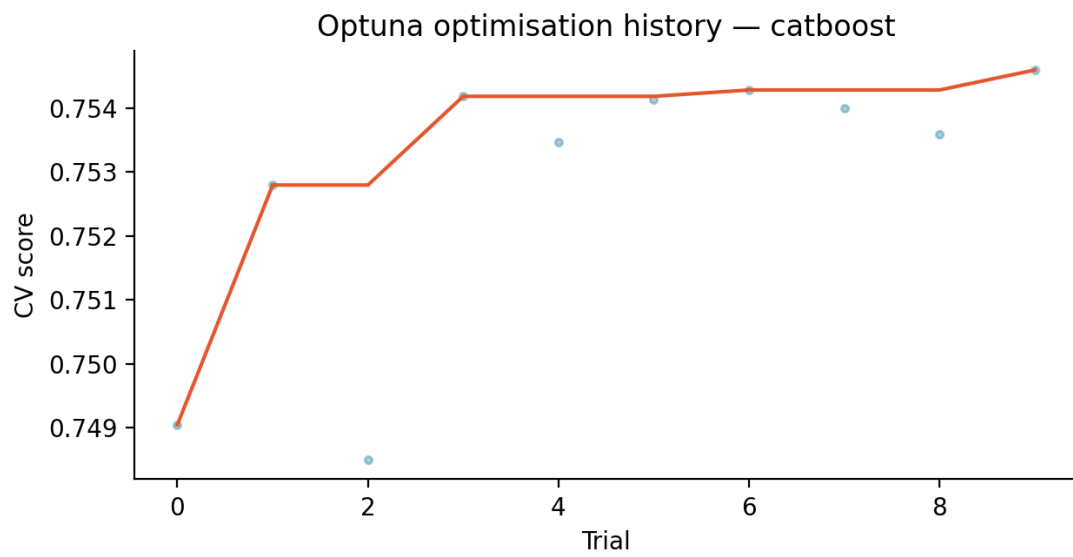
**Figure F.3.** Model Comparison: Statistical vs Profit-Based Evaluation



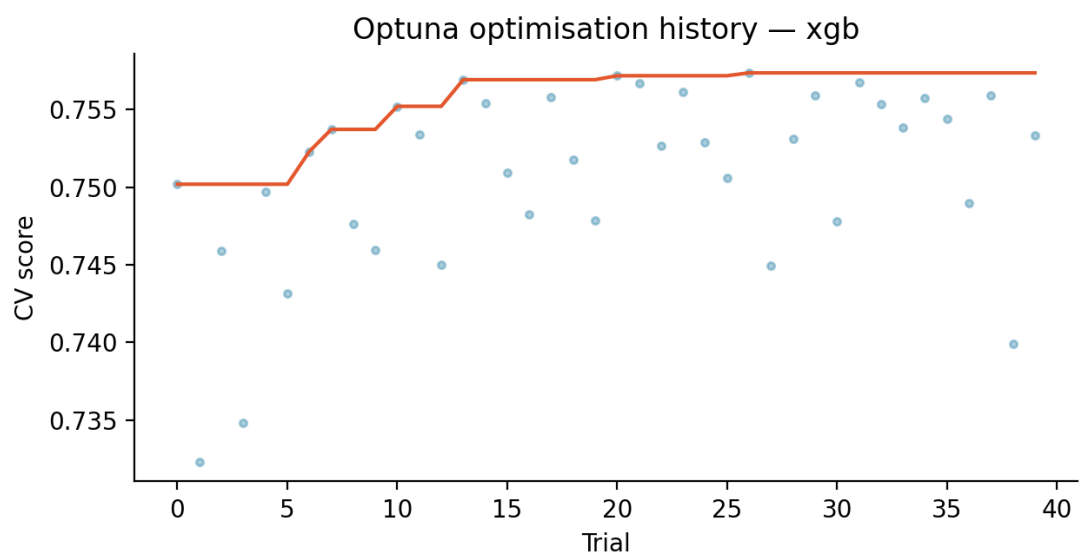
**Figure F.4.** Ablation Study: Contribution of Each Feature Group



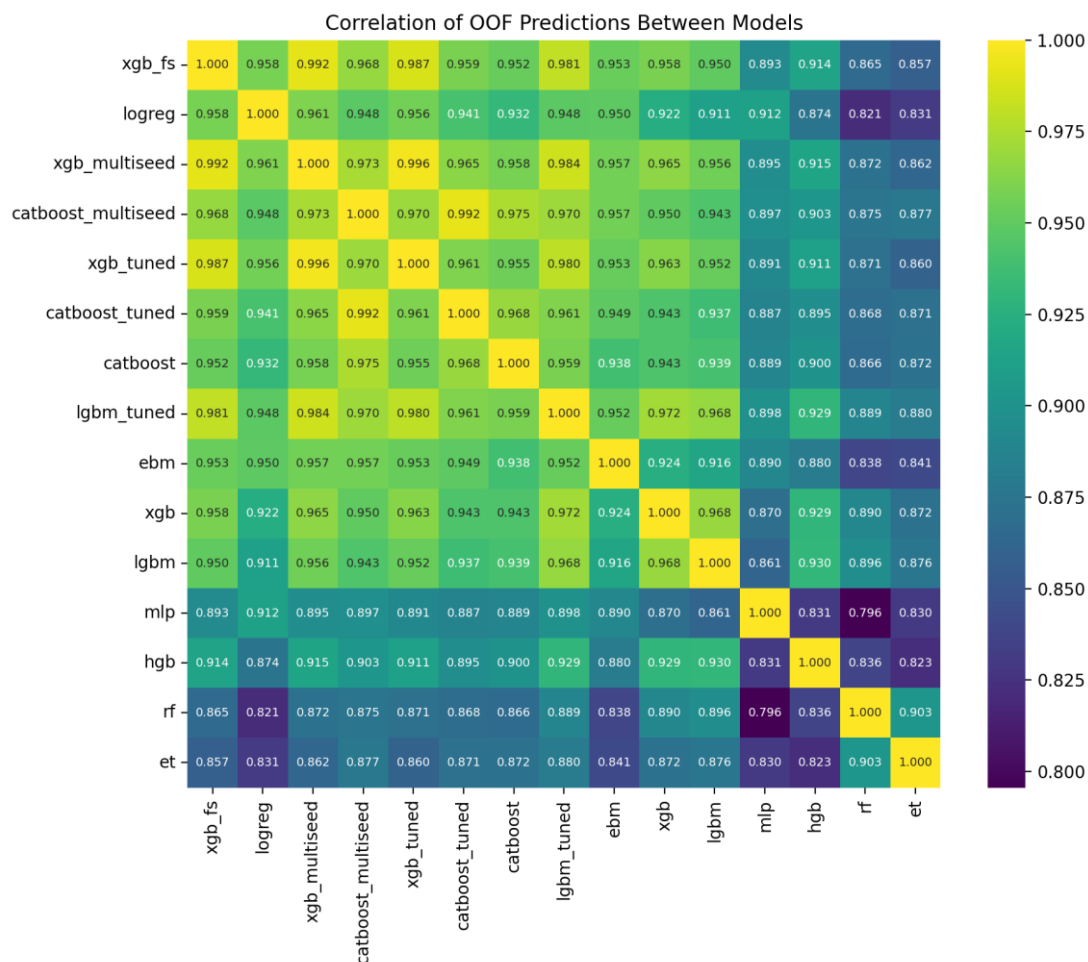
**Figure F.5.** Optuna optimisation history — lgbm



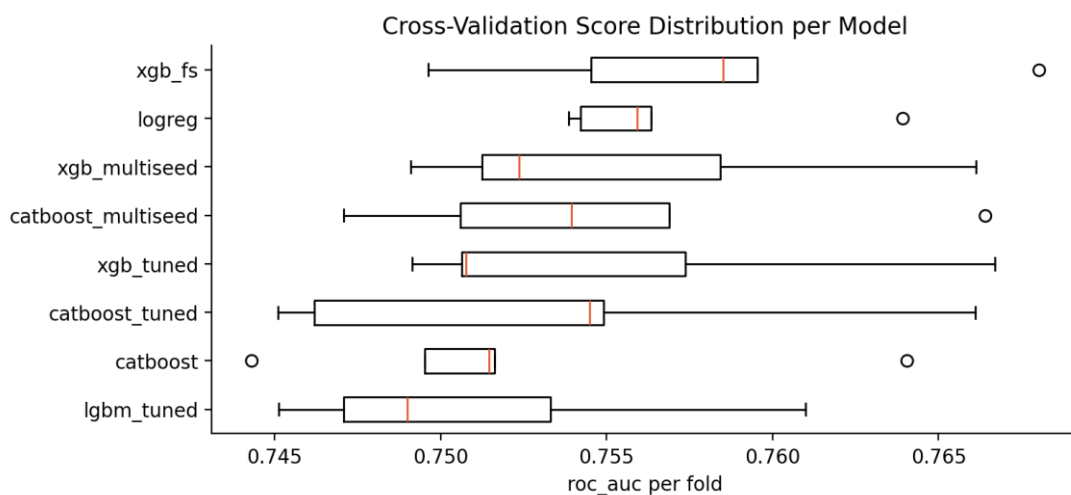
**Figure F.6.** Optuna optimisation history — catboost



**Figure F.7.** Optuna optimisation history — xgb



**Figure F.8.** Correlation of OOF Predictions Between Models



**Figure F.9.** Cross-Validation Score Distribution per Model

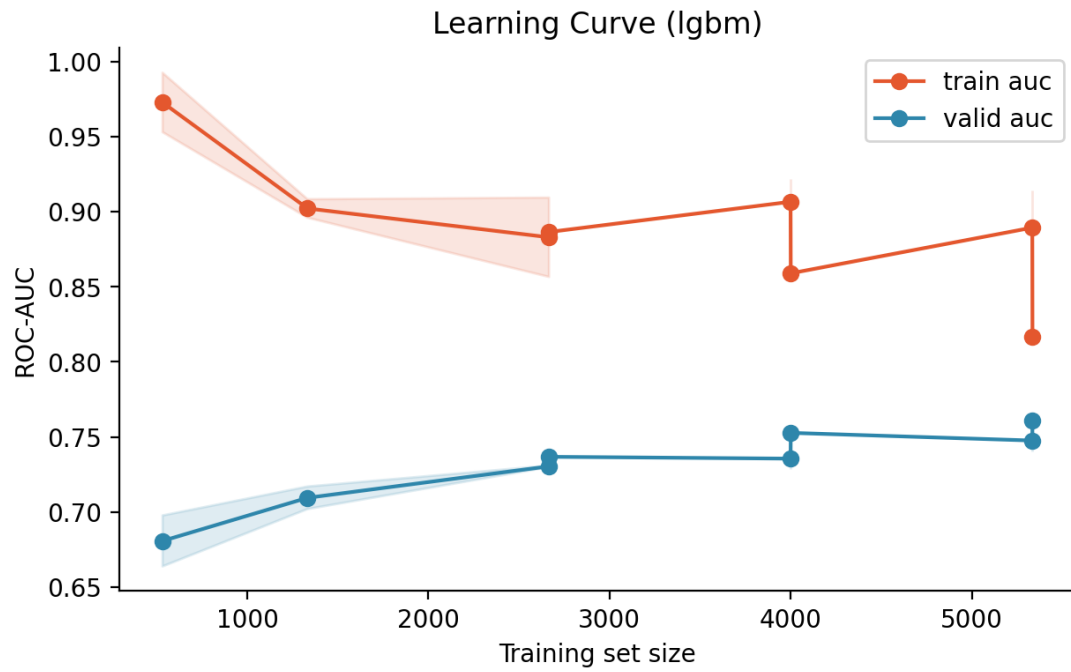


Figure F.10. Learning Curve (lgbm)

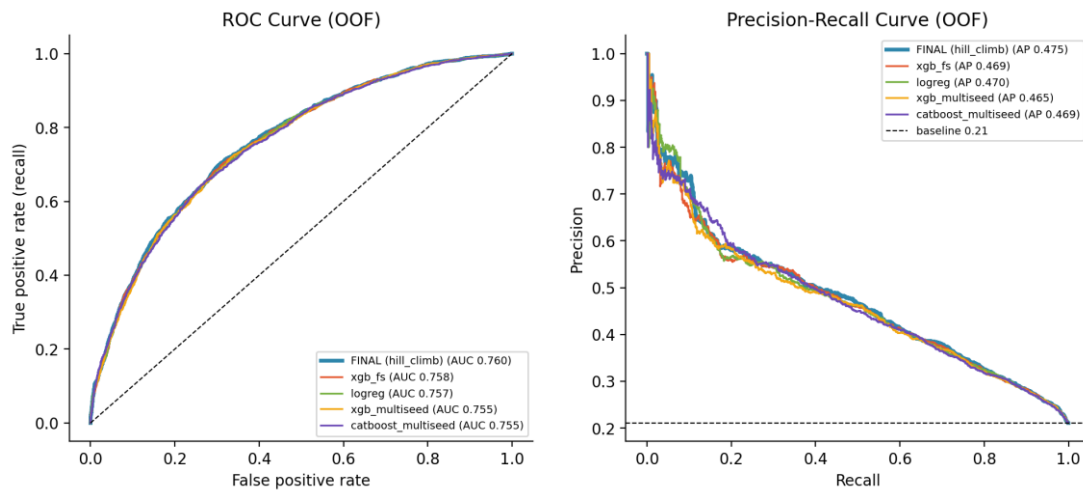
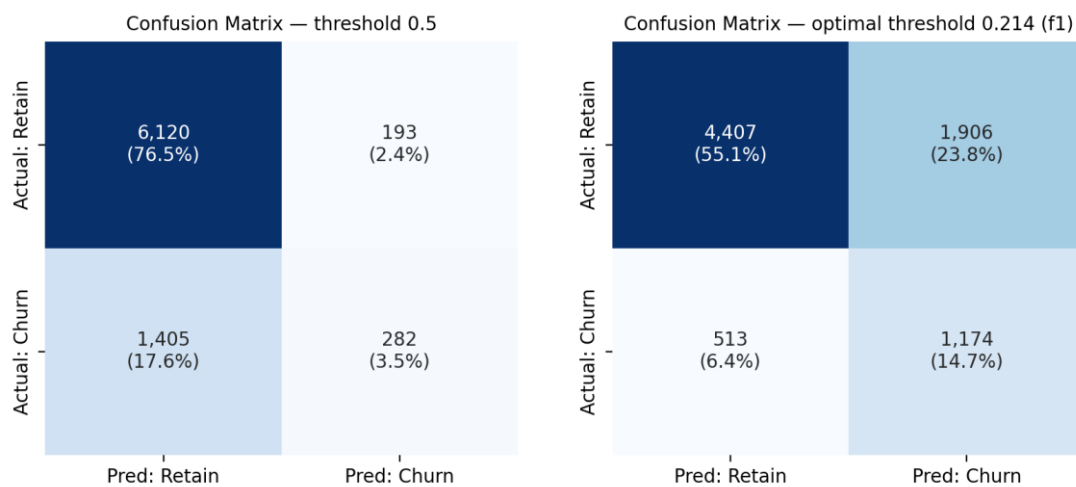
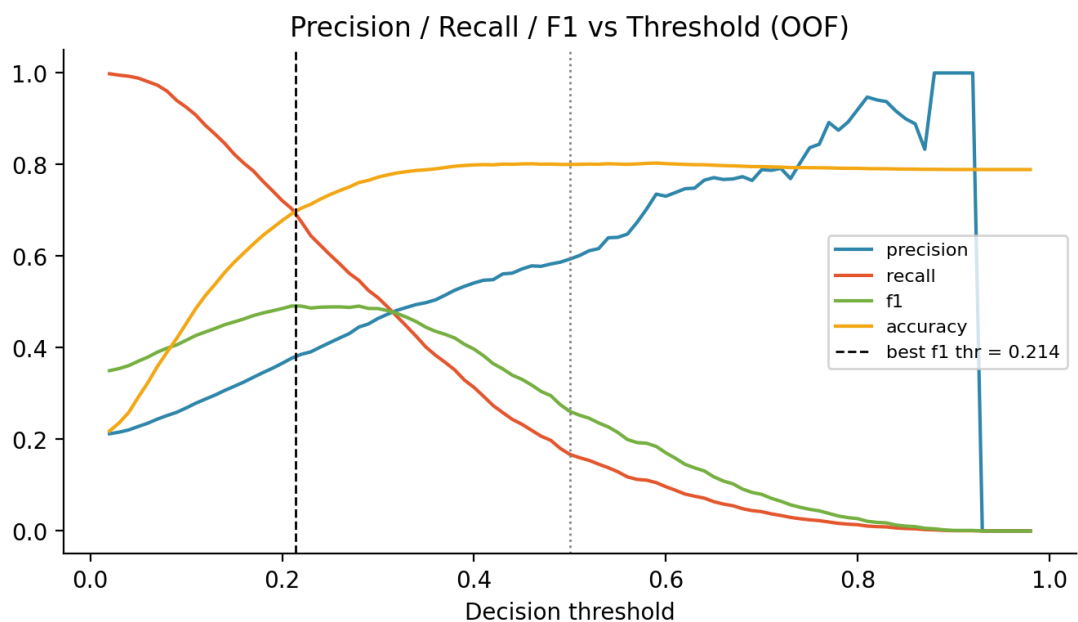


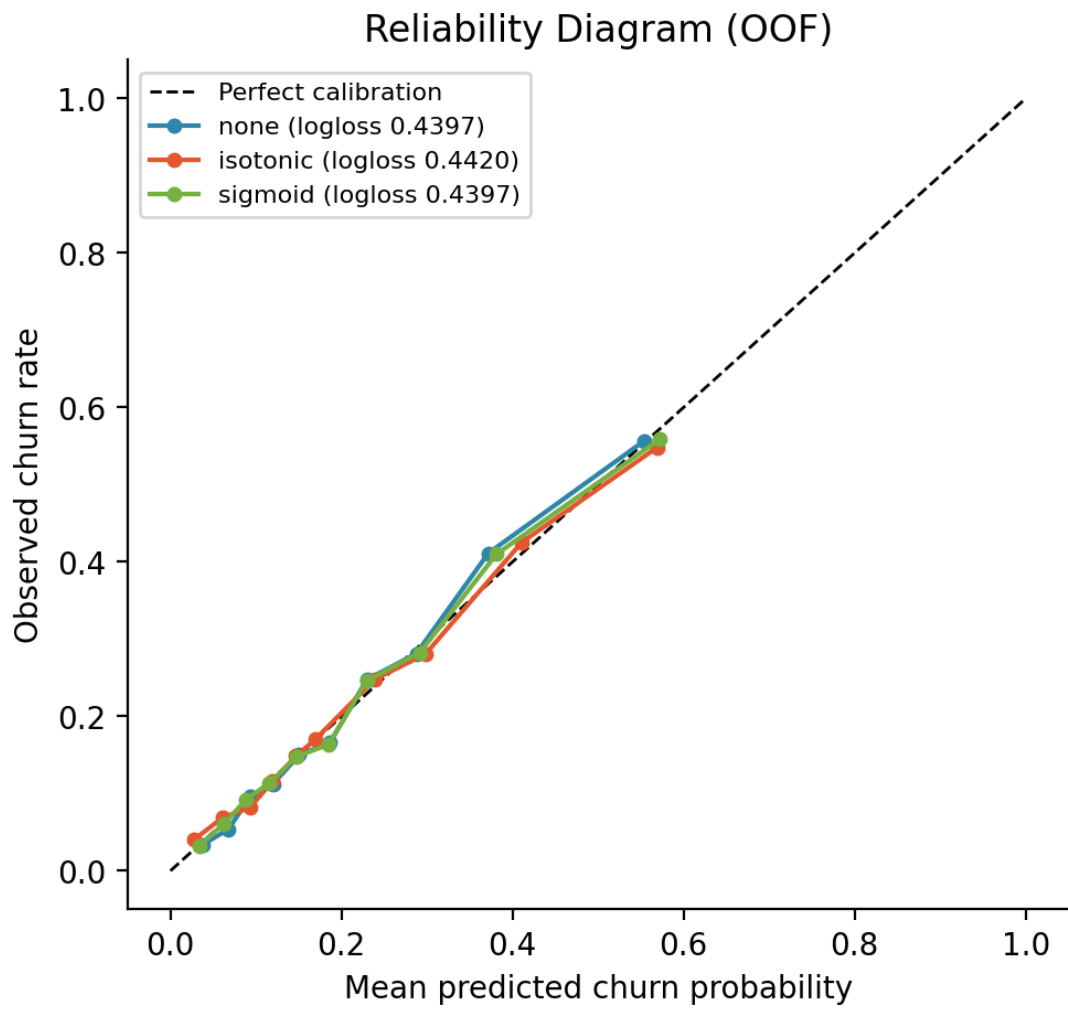
Figure F.11. ROC Curve (OOF)



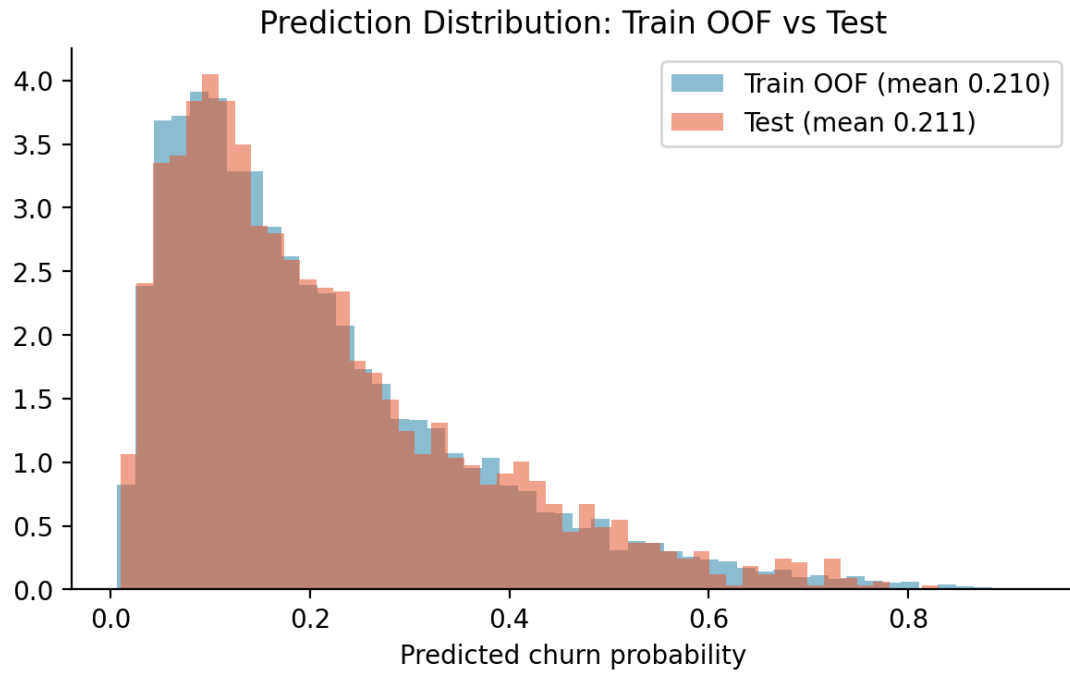
**Figure F.12.** Confusion Matrix — threshold 0.5



**Figure F.13.** Precision / Recall / F1 vs Threshold (OOF)



**Figure F.14.** Reliability Diagram (OOF)



**Figure F.15.** Prediction Distribution: Train OOF vs Test

**Table F.1.** Highest-risk segments and the most loyal segment.

| Segment (decision-tree rule)                                                                                                                   | Customers | Churn rate | Lift |
|------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|------|
| claims per year of tenure > 0.41 AND premium change at last renewal (%) > 3.85 AND payment method ≠ Auto-debit AND claim rejection rate > 0.12 | 266       | 55.3%      | 2.6× |
| claims per year of tenure > 0.41 AND premium change at last renewal (%) ≤ 3.85 AND claim rejection rate > 0.58                                 | 209       | 38.3%      | 1.8× |
| claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) > 6.55 AND payment method ≠ Auto-debit AND tenure (months) ≤ 26        | 590       | 37.3%      | 1.8× |
| claims per year of tenure ≤ 0.41 AND premium change at last renewal (%) ≤ 6.55 AND auto-renewal = Yes AND number of ANS products ≥ 2           | 885       | 6.1%       | 0.3× |

**Table F.2.** Full model comparison: AUC with DeLong CI, PR-AUC, Brier, ECE, KS and EMPC.

| Model              | Roc auc | Auc ci low | Auc ci high | Pr auc | Brier | Ece   | Ks    | Empe per customer | Delong p holm |
|--------------------|---------|------------|-------------|--------|-------|-------|-------|-------------------|---------------|
| FINAL: hill_climb  | 0.760   | 0.747      | 0.773       | 0.475  | 0.140 | 0.012 | 0.395 | 115,005           |               |
| xgb fs             | 0.758   | 0.745      | 0.771       | 0.469  | 0.141 | 0.013 | 0.389 | 114,750           | 0.025         |
| logreg             | 0.757   | 0.744      | 0.769       | 0.470  | 0.141 | 0.012 | 0.382 | 114,505           | 0.013         |
| xgb multiseed      | 0.755   | 0.742      | 0.768       | 0.465  | 0.142 | 0.014 | 0.386 | 114,294           | 3.1e-05       |
| catboost multiseed | 0.755   | 0.742      | 0.768       | 0.469  | 0.141 | 0.015 | 0.382 | 114,178           | 0.002         |
| xgb tuned          | 0.755   | 0.742      | 0.768       | 0.466  | 0.142 | 0.015 | 0.382 | 114,236           | 6.9e-05       |
| catboost tuned     | 0.753   | 0.740      | 0.766       | 0.468  | 0.142 | 0.014 | 0.383 | 113,713           | 3.6e-04       |
| catboost           | 0.752   | 0.739      | 0.765       | 0.466  | 0.142 | 0.012 | 0.380 | 113,417           | 6.9e-05       |
| lgbm tuned         | 0.751   | 0.738      | 0.764       | 0.462  | 0.142 | 0.012 | 0.379 | 113,073           | 2.1e-09       |
| ebm                | 0.748   | 0.735      | 0.761       | 0.461  | 0.142 | 0.004 | 0.365 | 112,655           | 7.7e-08       |
| xgb                | 0.747   | 0.734      | 0.760       | 0.454  | 0.143 | 0.014 | 0.377 | 112,397           | 2.3e-08       |
| lgbm               | 0.745   | 0.732      | 0.758       | 0.450  | 0.144 | 0.014 | 0.374 | 111,854           | 7.2e-10       |
| mlp                | 0.737   | 0.724      | 0.750       | 0.436  | 0.146 | 0.025 | 0.355 | 111,843           | 4.8e-12       |
| hgb                | 0.734   | 0.721      | 0.748       | 0.435  | 0.146 | 0.022 | 0.352 | 110,558           | 1.4e-15       |
| rf                 | 0.726   | 0.713      | 0.740       | 0.426  | 0.147 | 0.023 | 0.338 | 108,382           | 6.7e-19       |
| et                 | 0.724   | 0.710      | 0.737       | 0.418  | 0.148 | 0.019 | 0.338 | 108,529           | 8.1e-20       |

**Table F.3.** Ablation study by feature group (DeLong tests, Holm-corrected) (first 10 columns; full table in report\_tables.xlsx).

| Ablation         | N features removed | Features                                                                                                                        | Auc full | Auc without | Auc drop | Ci low | Ci high | Delong p | Holm p  |
|------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------|----------|-------------|----------|--------|---------|----------|---------|
| – premium_change | 3                  | premium change at last renewal (%), premium increased at renewal, premium increase above 10%                                    | 0.741    | 0.708       | 0.033    | 0.024  | 0.042   | 1.2e-12  | 1.1e-11 |
| – tenure         | 6                  | tenure (months), tenure (years), new customer (<12 months), policy start date (days since), policy start date year, tenure band | 0.741    | 0.728       | 0.013    | 0.006  | 0.020   | 1.7e-04  | 0.001   |
| – payment method | 1                  | payment method                                                                                                                  | 0.741    | 0.729       | 0.012    | 0.005  | 0.019   | 0.001    | 0.007   |
| – auto-renewal   | 1                  | auto-renewal                                                                                                                    | 0.741    | 0.730       | 0.011    | 0.005  | 0.017   | 7.6e-04  | 0.005   |
| – complaints     | 2                  | complaints in the last 12 months, has complained                                                                                | 0.741    | 0.731       | 0.010    | 0.004  | 0.016   | 0.001    | 0.006   |
| – engagement     | 1                  | digital engagement score                                                                                                        | 0.741    | 0.738       | 0.003    | -0.002 | 0.009   | 0.226    | 0.714   |
| – products       | 2                  | number of ANS                                                                                                                   | 0.741    | 0.739       | 0.002    | -0.004 | 0.008   | 0.489    | 0.978   |

| Ablation                               | N features removed | Features                                                                                                                                                                            | Auc full | Auc without | Auc drop | Ci low | Ci high | Delong p | Holm p |
|----------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|----------|--------|---------|----------|--------|
|                                        |                    | products, holds multiple products                                                                                                                                                   |          |             |          |        |         |          |        |
| – average claim settlement time (days) | 1                  | average claim settlement time (days)                                                                                                                                                | 0.741    | 0.740       | 0.001    | -0.004 | 0.007   | 0.607    | 0.978  |
| – all engineered domain features       | 18                 | tenure (years), new customer (<12 months), premium-to-income ratio, sum insured to premium ratio, has made a claim, claims per year of tenure, claim rejection rate, has complained | 0.741    | 0.745       | -0.004   | -0.009 | 0.002   | 0.179    | 0.714  |

**Table F.4.** Ensemble methods compared.

| Method            | Oof roc auc | Oof auc | Weights                                                                                                                                                                                                     |
|-------------------|-------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| best_single       | 0.758       | 0.758   | {'xgb_fs': 1.0}                                                                                                                                                                                             |
| mean_top3         | 0.759       | 0.759   | {'xgb_fs': 0.333, 'logreg': 0.333, 'xgb_multiseed': 0.333}                                                                                                                                                  |
| rank_mean_top3    | 0.759       | 0.759   | {'xgb_fs': 0.333, 'logreg': 0.333, 'xgb_multiseed': 0.333}                                                                                                                                                  |
| hill_climb        | 0.760       | 0.760   | {'xgb_fs': 0.444, 'logreg': 0.333, 'catboost_multiseed': 0.111, 'catboost': 0.111}                                                                                                                          |
| optimized_weights | 0.760       | 0.760   | {'xgb_fs': 0.448, 'logreg': 0.359, 'catboost_multiseed': 0.034, 'catboost_tuned': 0.123, 'catboost': 0.036}                                                                                                 |
| stacking_logreg   | 0.760       | 0.760   | {'xgb_fs': 0.965, 'logreg': 0.429, 'xgb_multiseed': -0.897, 'catboost_multiseed': 0.161, 'xgb_tuned': 0.332, 'catboost_tuned': 0.185, 'catboost': 0.23, 'lgbm_tuned': -0.286, 'ebm': -0.082, 'xgb': -0.023} |

**Table F.5.** Bootstrap 95% confidence intervals for the final model.

| Metric  | Estimate | Ci low | Ci high |
|---------|----------|--------|---------|
| roc auc | 0.760    | 0.748  | 0.772   |

| Metric  | Estimate | Ci low | Ci high |
|---------|----------|--------|---------|
| pr_auc  | 0.475    | 0.451  | 0.501   |
| logloss | 0.440    | 0.428  | 0.452   |

**Table F.6.** Experiment log (all logged runs).

| Exp id | Stage                   | Experiment                        | Note                                                                                                            | Roc auc | Pr auc | Logloss | Time s |
|--------|-------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|---------|--------|---------|--------|
| 1      | 1 baseline model zoo    | dummy                             | Baseline (prior), default hyperparameters                                                                       | 0.500   | 0.211  | 0.515   | 0.320  |
| 2      | 1 baseline model zoo    | logreg                            | Logistic regression, default hyperparameters                                                                    | 0.757   | 0.470  | 0.442   | 2.505  |
| 3      | 1 baseline model zoo    | rf                                | Random forest, default hyperparameters                                                                          | 0.726   | 0.426  | 0.461   | 15.2   |
| 4      | 1 baseline model zoo    | et                                | Extra trees, default hyperparameters                                                                            | 0.724   | 0.418  | 0.462   | 13.9   |
| 5      | 1 baseline model zoo    | hgb                               | Hist. gradient boosting, default hyperparameters                                                                | 0.734   | 0.435  | 0.457   | 14.0   |
| 6      | 1 baseline model zoo    | lgbm                              | LightGBM, default hyperparameters                                                                               | 0.745   | 0.450  | 0.449   | 27.2   |
| 7      | 1 baseline model zoo    | xgb                               | XGBoost, default hyperparameters                                                                                | 0.747   | 0.454  | 0.448   | 11.2   |
| 8      | 1 baseline model zoo    | catboost                          | CatBoost, default hyperparameters                                                                               | 0.752   | 0.466  | 0.444   | 168.3  |
| 9      | 1 baseline model zoo    | ebm                               | Explainable boosting machine (EBM), default hyperparameters                                                     | 0.748   | 0.461  | 0.446   | 102.8  |
| 10     | 1 baseline model zoo    | mlp                               | Neural network (MLP), default hyperparameters                                                                   | 0.737   | 0.436  | 0.458   | 9.673  |
| 11     | 2 hyperparameter tuning | lgbm_tuned                        | Optuna TPE, 39 trials, 3-fold inner CV                                                                          | 0.751   | 0.462  | 0.445   | 48.2   |
| 12     | 2 hyperparameter tuning | catboost_tuned                    | Optuna TPE, 10 trials, 3-fold inner CV                                                                          | 0.753   | 0.468  | 0.444   | 92.2   |
| 13     | 2 hyperparameter tuning | xgb_tuned                         | Optuna TPE, 40 trials, 3-fold inner CV                                                                          | 0.755   | 0.466  | 0.444   | 8.278  |
| 14     | 3 multi-seed averaging  | xgb_multiseed                     | average of 3 seeds (base: xgb_tuned)                                                                            | 0.755   | 0.465  | 0.443   | 25.6   |
| 15     | 3 multi-seed averaging  | catboost_multiseed                | average of 3 seeds (base: catboost_tuned)                                                                       | 0.755   | 0.469  | 0.443   | 245.1  |
| 16     | 4 feature selection     | xgb_fs                            | drop 31 features with permutation importance $\leq 0$                                                           | 0.758   | 0.469  | 0.442   | 16.9   |
| 17     | 5 ablation study        | ablation reference (all features) | reference for ablation (same folds & params)                                                                    | 0.741   | 0.444  | 0.452   | 26.6   |
| 18     | 5 ablation study        | ablation – premium_change         | drop 3 feature(s): premium change at last renewal (%), premium increased at renewal, premium increase above 10% | 0.708   | 0.402  | 0.468   | 17.0   |
| 19     | 5 ablation study        | ablation – tenure                 | drop 6 feature(s): tenure (months), tenure (years), new customer (<12 months), policy start date (days)         | 0.728   | 0.423  | 0.459   | 19.7   |

| Exp id | Stage            | Experiment                                      | Note                                                                                                                                                               | Roc auc | Pr auc | Logloss | Time s |
|--------|------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|---------|--------|
|        |                  |                                                 | since)...                                                                                                                                                          |         |        |         |        |
| 20     | 5 ablation study | ablation – auto-renewal                         | drop 1 feature(s): auto-renewal                                                                                                                                    | 0.730   | 0.425  | 0.458   | 20.8   |
| 21     | 5 ablation study | ablation – complaints                           | drop 2 feature(s): complaints in the last 12 months, has complained                                                                                                | 0.731   | 0.419  | 0.458   | 19.2   |
| 22     | 5 ablation study | ablation – payment method                       | drop 1 feature(s): payment method                                                                                                                                  | 0.729   | 0.424  | 0.458   | 22.6   |
| 23     | 5 ablation study | ablation – products                             | drop 2 feature(s): number of ANS products, holds multiple products                                                                                                 | 0.739   | 0.438  | 0.453   | 22.7   |
| 24     | 5 ablation study | ablation – engagement                           | drop 1 feature(s): digital engagement score                                                                                                                        | 0.738   | 0.439  | 0.453   | 22.4   |
| 25     | 5 ablation study | ablation – average claim settlement time (days) | drop 1 feature(s): average claim settlement time (days)                                                                                                            | 0.740   | 0.443  | 0.452   | 22.7   |
| 26     | 5 ablation study | ablation – all engineered domain features       | drop 18 feature(s): tenure (years), new customer (<12 months), premium-to-income ratio, sum insured to premium ratio...                                            | 0.745   | 0.447  | 0.450   | 21.3   |
| 27     | 6 ensembling     | ensemble: best_single                           | xgb_fs 1.00                                                                                                                                                        | 0.758   | 0.469  | 0.442   |        |
| 28     | 6 ensembling     | ensemble: mean_top3                             | xgb_fs 0.33, logreg 0.33, xgb_multiseed 0.33                                                                                                                       | 0.759   | 0.472  | 0.441   |        |
| 29     | 6 ensembling     | ensemble: rank_mean_top3                        | xgb_fs 0.33, logreg 0.33, xgb_multiseed 0.33                                                                                                                       | 0.759   | 0.472  | 0.723   |        |
| 30     | 6 ensembling     | ensemble: hill_climb                            | xgb_fs 0.44, logreg 0.33, catboost_multiseed 0.11, catboost 0.11                                                                                                   | 0.760   | 0.475  | 0.440   |        |
| 31     | 6 ensembling     | ensemble: optimized_weights                     | xgb_fs 0.45, logreg 0.36, catboost_multiseed 0.03, catboost_tuned 0.12, catboost 0.04                                                                              | 0.760   | 0.475  | 0.440   |        |
| 32     | 6 ensembling     | ensemble: stacking_logreg                       | xgb_fs 0.96, logreg 0.43, xgb_multiseed - 0.90, catboost_multiseed 0.16, xgb_tuned 0.33, catboost_tuned 0.18, catboost 0.23, lgbm_tuned - 0.29, ebm -0.08, xgb -0. | 0.760   | 0.476  | 0.440   |        |

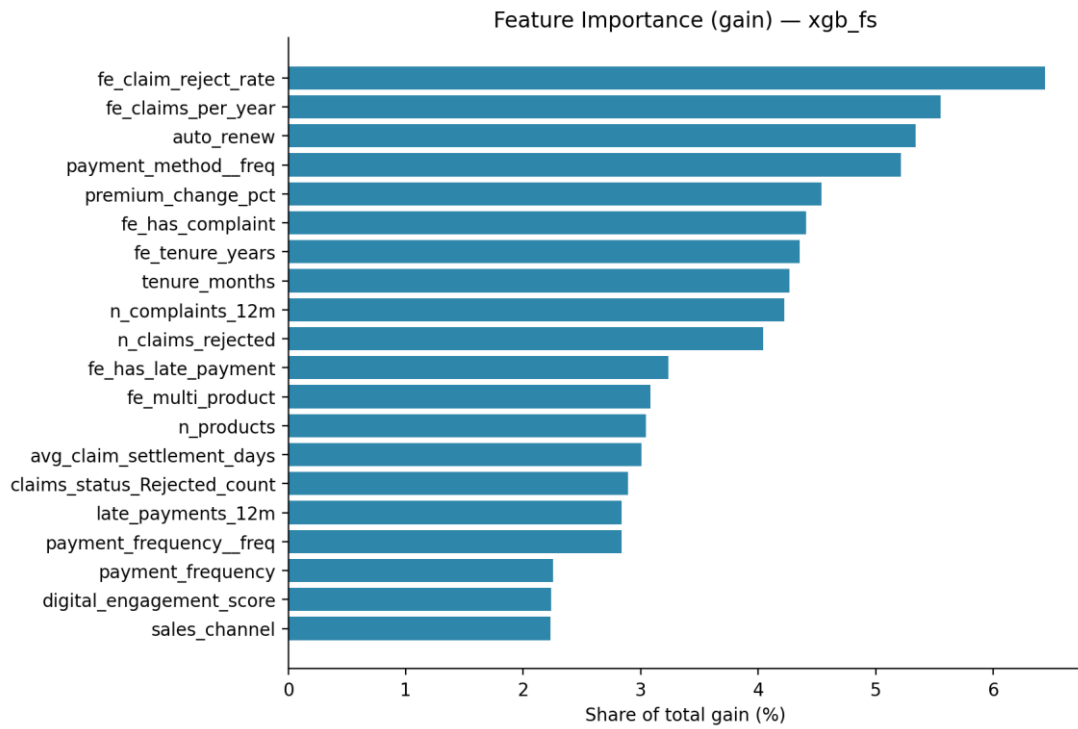
**Table F.7.** Train–test drift (PSI, KS; top 15).

| Feature      | Psi   | Ks stat | Ks p  | N unseen levels | Unseen rows % | Psi flag    |
|--------------|-------|---------|-------|-----------------|---------------|-------------|
| agent id     | 0.275 |         |       | 0               | 0             | MAJOR shift |
| claims claim | 0.062 | 0.059   | 0.503 |                 |               | stable      |

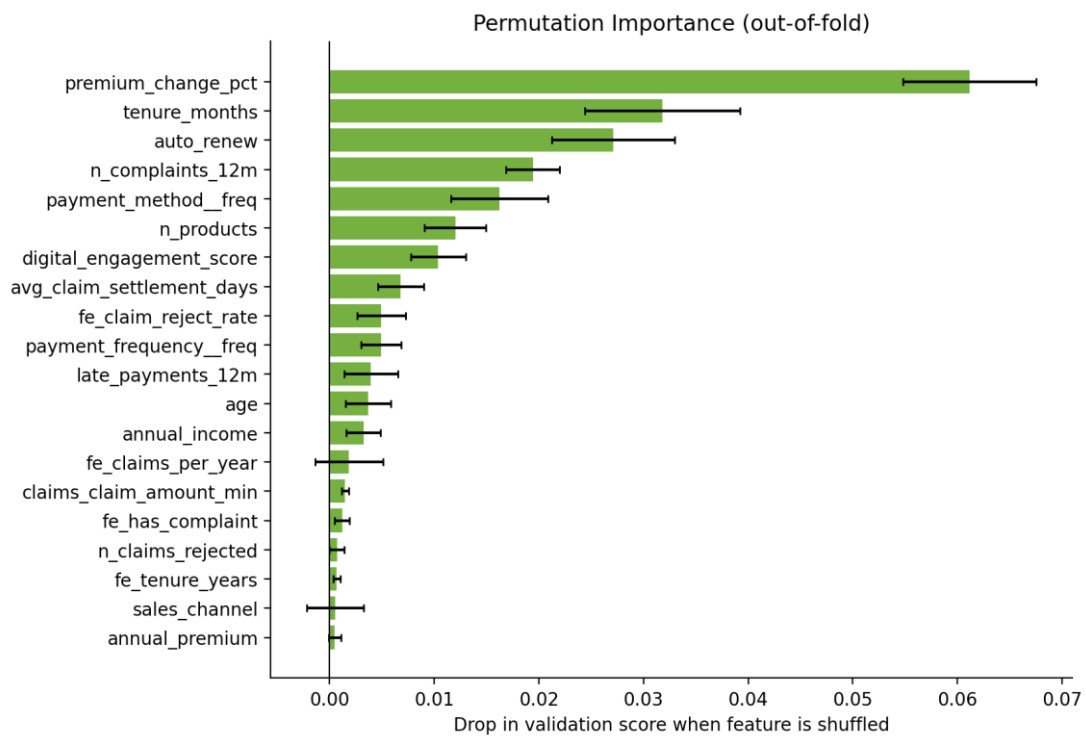
| Feature                              | Psi   | Ks stat | Ks p  | N unseen levels | Unseen rows % | Psi flag |
|--------------------------------------|-------|---------|-------|-----------------|---------------|----------|
| amount std                           |       |         |       |                 |               |          |
| annual income                        | 0.012 | 0.026   | 0.285 |                 |               | stable   |
| claims (days since) last             | 0.010 | 0.030   | 0.524 |                 |               | stable   |
| average claim settlement time (days) | 0.009 | 0.042   | 0.159 |                 |               | stable   |
| tenure (years)                       | 0.009 | 0.022   | 0.410 |                 |               | stable   |
| policy start date (days since)       | 0.009 | 0.022   | 0.410 |                 |               | stable   |
| tenure (months)                      | 0.009 | 0.022   | 0.410 |                 |               | stable   |
| claims claim amount mean             | 0.009 | 0.035   | 0.345 |                 |               | stable   |
| claims claim amount max              | 0.009 | 0.023   | 0.820 |                 |               | stable   |
| claims (days since) first            | 0.008 | 0.032   | 0.437 |                 |               | stable   |
| complaints in the last 12 months     | 0.007 | 0.028   | 0.161 |                 |               | stable   |
| annual premium                       | 0.007 | 0.019   | 0.573 |                 |               | stable   |
| policy start date month              | 0.007 | 0.017   | 0.732 |                 |               | stable   |
| policy start date year               | 0.006 | 0.020   | 0.565 |                 |               | stable   |

## Appendix G. Explainability

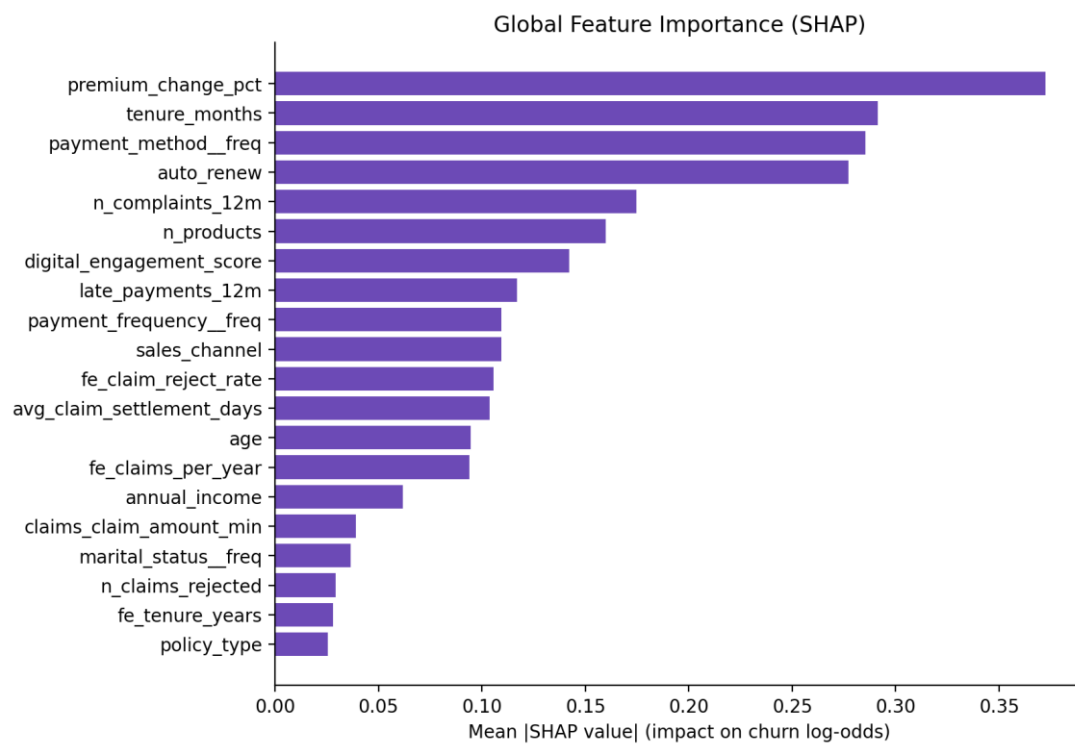
Global and local explanations: gain, permutation and SHAP importance, dependence and partial dependence, individual explanations and segment heterogeneity.



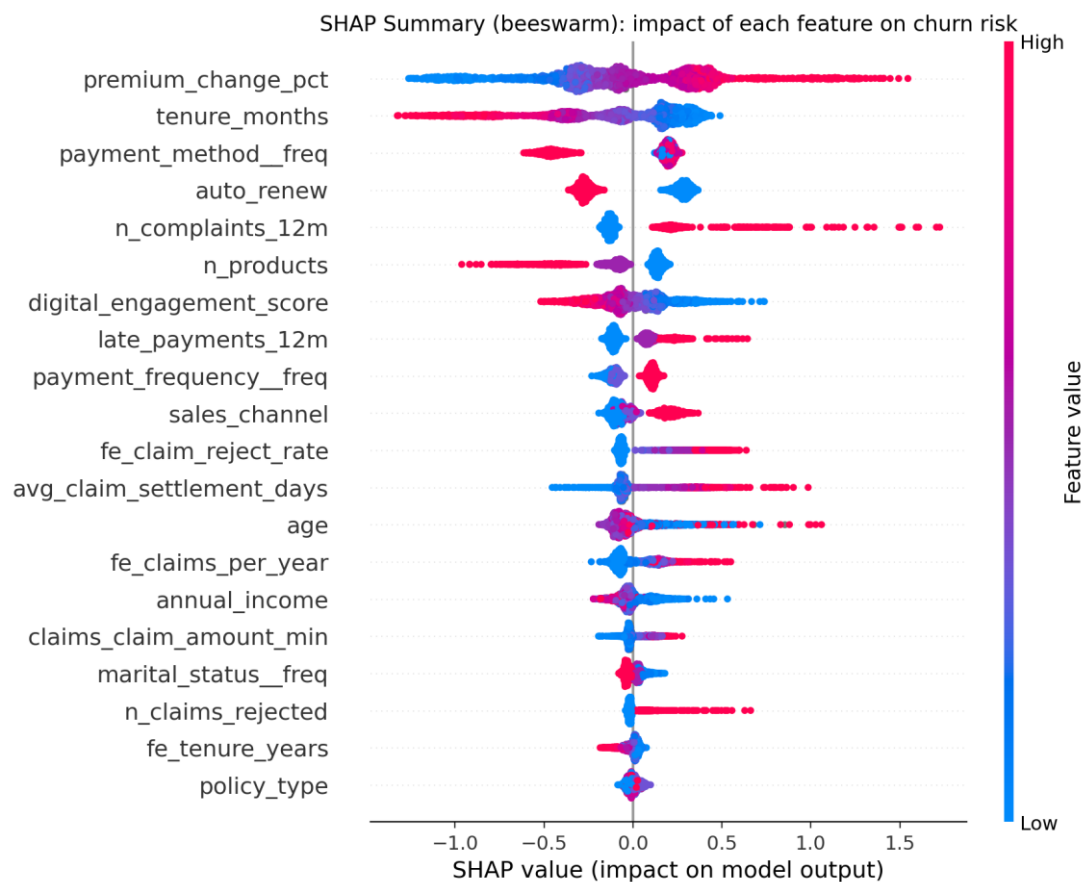
**Figure G.1.** Feature Importance (gain) — xgb\_fs



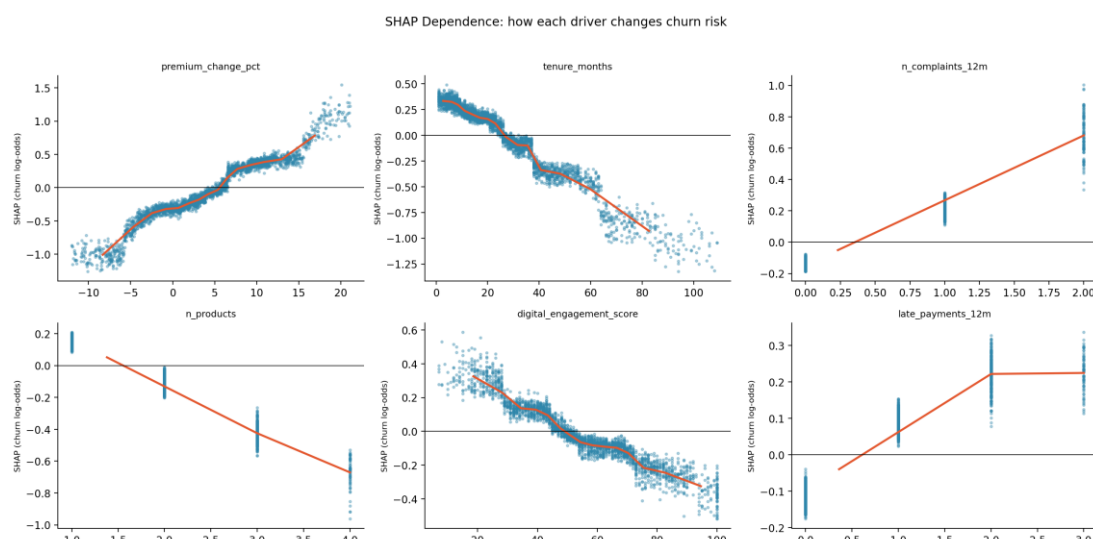
**Figure G.2.** Permutation Importance (out-of-fold)



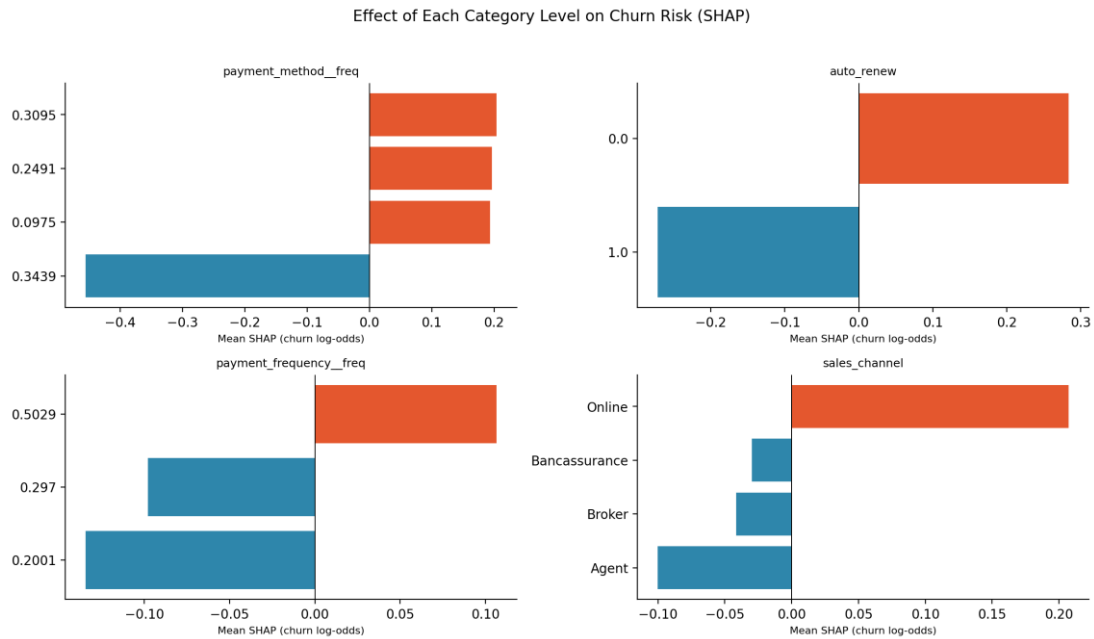
**Figure G.3.** Global Feature Importance (SHAP)



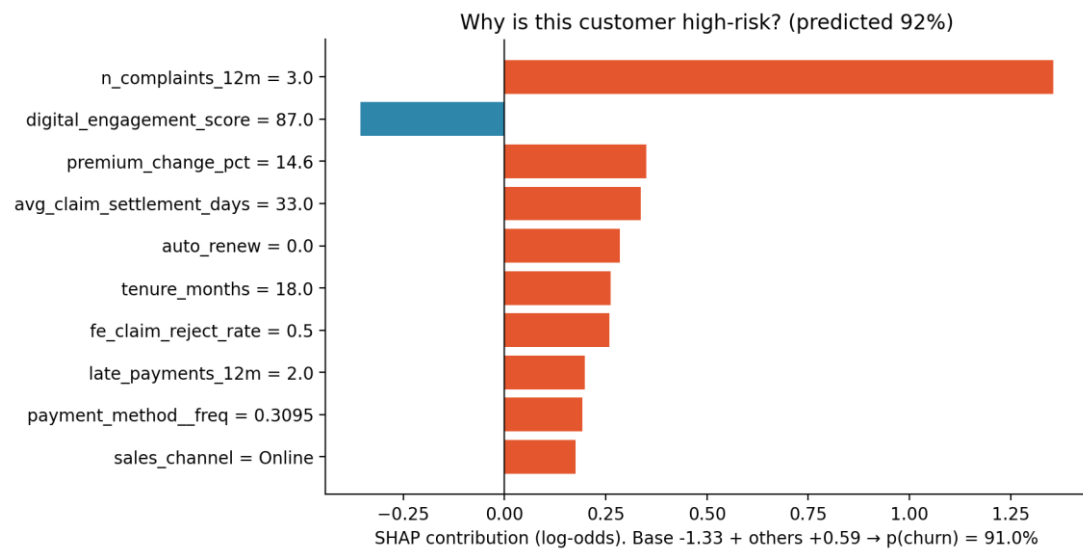
**Figure G.4.** SHAP summary: each dot is a customer; positive values push churn risk up (colour = feature value).



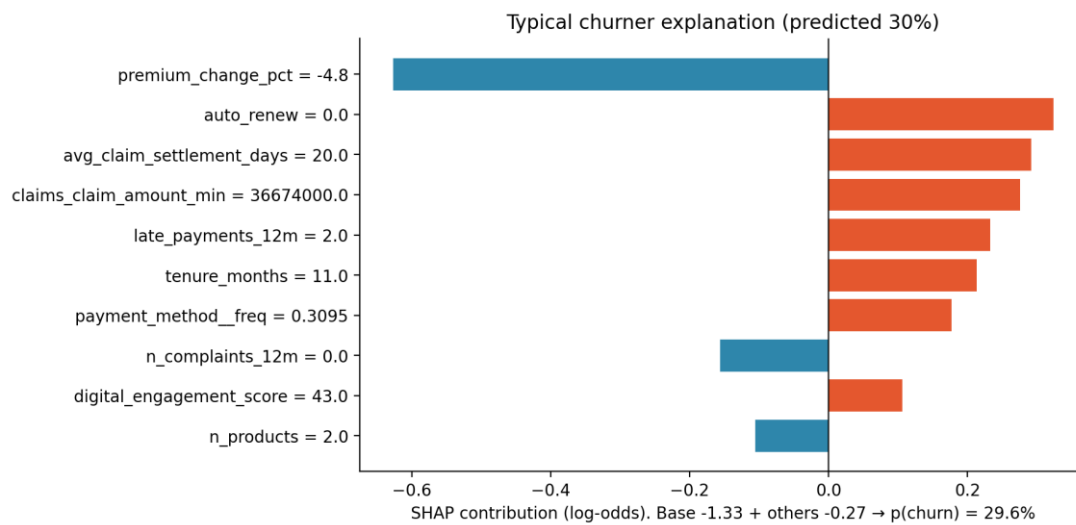
**Figure G.5.** SHAP Dependence: how each driver changes churn risk



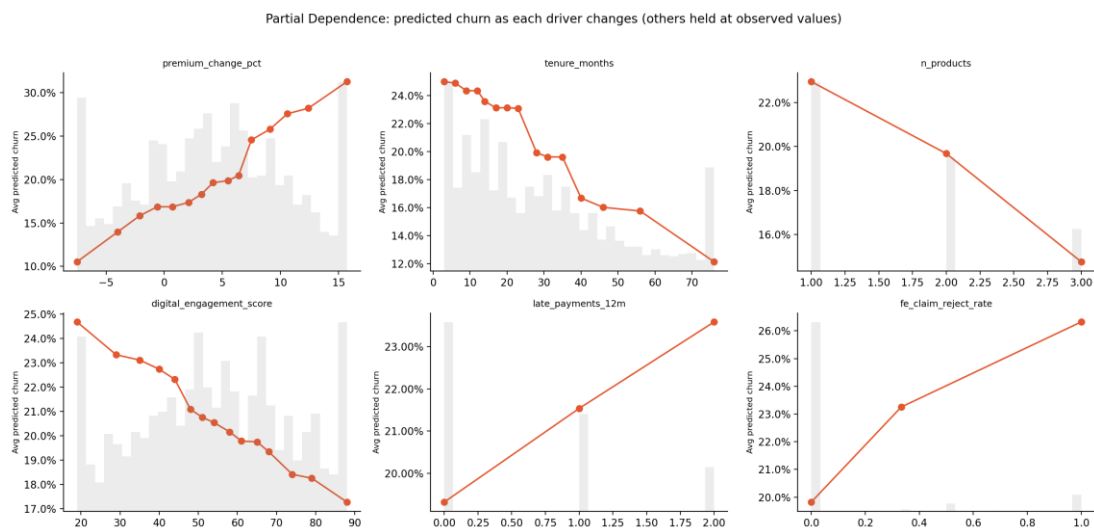
**Figure G.6.** Effect of Each Category Level on Churn Risk (SHAP)



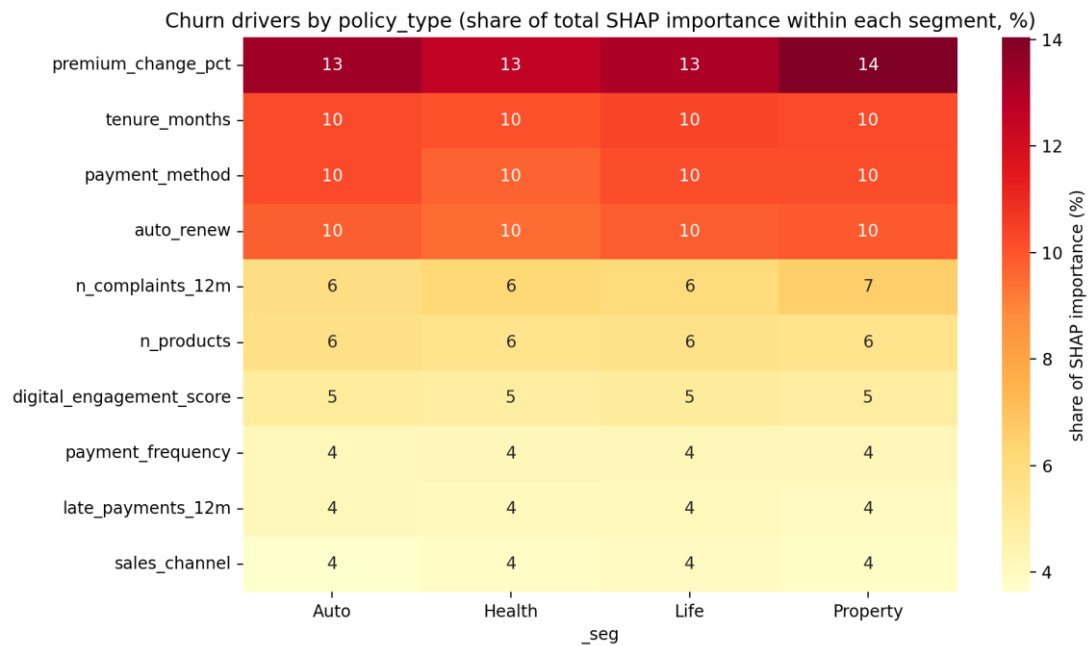
**Figure G.7.** Why is this customer high-risk? (predicted 92%)



**Figure G.8.** Typical churner explanation (predicted 30%)



**Figure G.9.** Partial Dependence: predicted churn as each driver changes (others held at observed values)



**Figure G.10.** Churn drivers by policy type (share of total SHAP importance within each segment, %)

**Table G.1.** SHAP importance and direction of effect (top 20).

| Feature                              | Mean abs shap | Share % | Direction                                           |
|--------------------------------------|---------------|---------|-----------------------------------------------------|
| premium change at last renewal (%)   | 0.372         | 13.1    | higher value → higher churn risk                    |
| tenure (months)                      | 0.291         | 10.2    | higher value → lower churn risk                     |
| payment_method_freq                  | 0.285         | 10.0    | higher value → lower churn risk                     |
| auto-renewal                         | 0.277         | 9.715   | riskiest: '0.0' (+0.28); safest: '1.0' (-0.27)      |
| complaints in the last 12 months     | 0.175         | 6.127   | higher value → higher churn risk                    |
| number of ANS products               | 0.160         | 5.604   | higher value → lower churn risk                     |
| digital engagement score             | 0.142         | 4.982   | higher value → lower churn risk                     |
| late payments in the last 12 months  | 0.117         | 4.098   | higher value → higher churn risk                    |
| payment_frequency_freq               | 0.110         | 3.839   | higher value → higher churn risk                    |
| sales channel                        | 0.109         | 3.831   | riskiest: 'Online' (+0.21); safest: 'Agent' (-0.10) |
| claim rejection rate                 | 0.106         | 3.698   | higher value → higher churn risk                    |
| average claim settlement time (days) | 0.104         | 3.631   | higher value → higher churn risk                    |
| age                                  | 0.095         | 3.318   | higher value → lower churn risk                     |
| claims per year of tenure            | 0.094         | 3.292   | higher value → higher churn risk                    |
| annual income                        | 0.062         | 2.164   | higher value → lower churn risk                     |
| claims claim amount min              | 0.039         | 1.370   | higher value → higher                               |

| Feature              | Mean abs shap | Share % | Direction                                          |
|----------------------|---------------|---------|----------------------------------------------------|
|                      |               |         | churn risk                                         |
| marital_status__freq | 0.037         | 1.285   | higher value → lower churn risk                    |
| rejected claims      | 0.029         | 1.022   | higher value → higher churn risk                   |
| tenure (years)       | 0.028         | 0.984   | higher value → lower churn risk                    |
| policy type          | 0.026         | 0.898   | riskiest: 'Health' (+0.03); safest: 'Auto' (-0.03) |

**Table G.2.** Consolidated driver evidence matrix (first 10 columns; full table in report\_tables.xlsx).

| Feature                              | Iv    | Effect size | Q value BH | Highest risk level | Highest risk rate | Odds ratio | Hazard ratio | Shap rank | Evidence score |
|--------------------------------------|-------|-------------|------------|--------------------|-------------------|------------|--------------|-----------|----------------|
| premium change at last renewal (%)   | 0.159 | 0.231       | 1.3e-44    | 13.2–30.6          | 0.335             | 1.638      | 1.385        | 1.000     | 4              |
| payment method                       | 0.098 | 0.123       | 1.3e-25    | Bank Transfer      | 0.255             | 2.330      | 1.827        | 3.000     | 4              |
| auto-renewal                         | 0.084 | 0.118       | 1.2e-25    | 0                  | 0.264             | 0.514      | 0.614        | 4.000     | 4              |
| complaints in the last 12 months     | 0.110 | 0.140       | 3.1e-30    | ≥2                 | 0.443             | 1.329      | 1.235        | 5.000     | 4              |
| tenure (months)                      | 0.148 | -0.201      | 2.9e-36    | 1–6                | 0.286             | 0.629      |              | 2.000     | 3              |
| number of ANS products               | 0.055 | -0.106      | 1.6e-13    | 1                  | 0.238             | 0.757      |              | 6.000     | 3              |
| claim rejection rate                 | 0.117 | 0.122       | 5.1e-38    | 1                  | 0.371             | 1.260      | 1.108        | 11.0      | 3              |
| average claim settlement time (days) | 0.105 | 0.118       | 1.8e-07    | 21–28              | 0.346             | 1.188      | 1.107        | 12.0      | 3              |
| digital engagement score             | 0.023 | -0.080      | 3.0e-06    | 0–27               | 0.263             |            |              | 7.000     | 2              |
| payment frequency                    | 0.024 | 0.063       | 1.9e-07    | Monthly            | 0.236             |            |              | 8.000     | 2              |
| late payments in the last 12 months  | 0.038 | 0.101       | 1.7e-12    | ≥3                 | 0.278             |            |              | 9.000     | 2              |
| sales channel                        | 0.011 | 0.044       | 0.003      | Online             | 0.241             |            |              | 10.0      | 2              |
| claims per year of tenure            | 0.134 | 0.180       | 3.6e-35    | 1.5–24             | 0.349             | 1.073      |              | 14.0      | 2              |
| claims count                         | 0.076 | 0.140       | 7.3e-23    | ≥3                 | 0.328             |            | 1.106        |           | 2              |
| annual income                        | 0.009 | -0.040      | 0.024      | 7.8M–42.1M         | 0.241             |            |              | 15.0      | 1              |
| rejected claims                      | 0.116 | 0.122       | 2.5e-38    | ≥2                 | 0.505             |            |              | 18.0      | 1              |
| tenure (years)                       | 0.148 | -0.201      | 2.9e-36    | 0.0833–0.5         | 0.286             |            |              | 19.0      | 1              |
| policy type                          | 0.009 | 0.039       | 0.011      | Health             | 0.232             |            |              | 20.0      | 1              |
| has complained                       | 0.081 | 0.120       | 1.9e-26    | 1                  | 0.294             |            |              | 27.0      | 1              |
| holds multiple                       | 0.033 | 0.074       | 7.5e-11    | 0                  | 0.238             |            |              | 28.0      | 1              |

| Feature                        | Iv    | Effect size | Q value BH | Highest risk level | Highest risk rate | Odds ratio | Hazard ratio | Shap rank | Evidence score |
|--------------------------------|-------|-------------|------------|--------------------|-------------------|------------|--------------|-----------|----------------|
| products                       |       |             |            |                    |                   |            |              |           |                |
| has late payments              | 0.033 | 0.074       | 7.3e-11    | 1                  | 0.245             |            |              | 29.0      | 1              |
| claims status Rejected count   | 0.116 | 0.122       | 2.5e-38    | $\geq 2$           | 0.505             |            |              | 31.0      | 1              |
| policy start date (days since) | 0.148 | -0.201      | 2.9e-36    | 31–183             | 0.286             |            |              |           | 1              |
| tenure band                    | 0.137 | 0.142       | 3.7e-33    | 0–6m               | 0.286             |            |              |           | 1              |
| policy start date year         | 0.131 | 0.191       | 3.4e-34    | 2,024–2,025        | 0.291             |            |              |           | 1              |

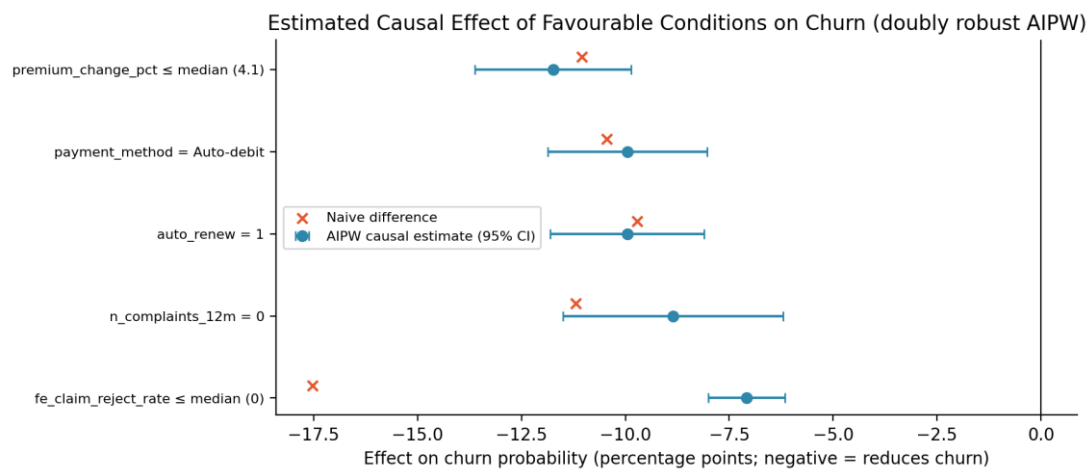
## Appendix H. Causal Inference and Uplift

Doubly robust effect estimates, identification assumptions (DAG), overlap, sensitivity (E-values), placebo tests and heterogeneous effects.

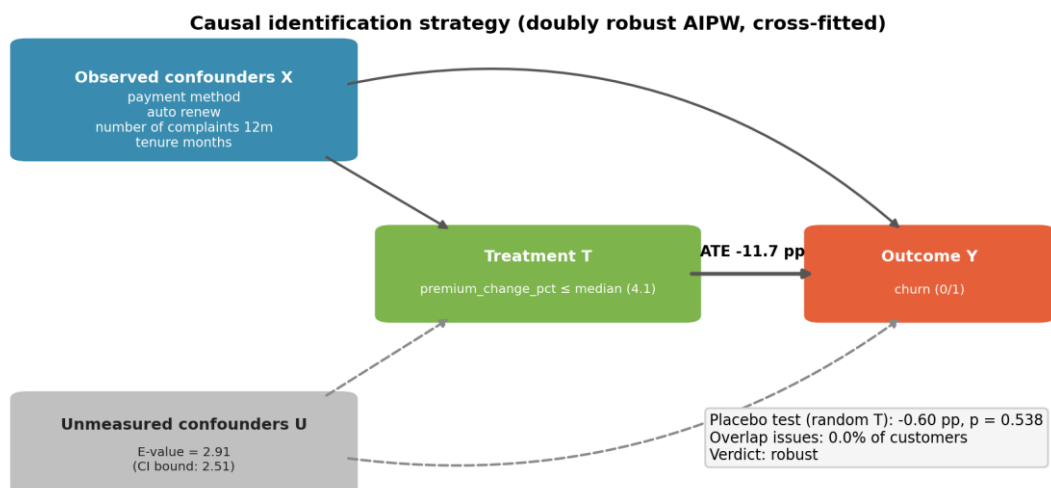
Identification. For each controllable lever a binary 'favourable condition' T is defined. Under no unmeasured confounding and overlap, the augmented inverse probability weighting estimator combines a propensity model  $e(X)$  and outcome models  $\mu_1(X)$ ,  $\mu_0(X)$  and stays consistent if either is correct (Robins et al., 1994); all nuisance models are cross-fitted with five folds to avoid own-observation bias (Chernozhukov et al., 2018). Propensity scores are clipped at 0.02/0.98 and the share of customers outside [0.05, 0.95] is reported.

Robustness. A placebo test re-estimates the effect with a randomly permuted treatment (should be  $\approx 0$ ). E-values quantify how strong an unmeasured confounder would have to be, on the risk-ratio scale, to explain away the estimate (VanderWeele & Ding, 2017). Only effects that pass all checks are labelled 'robust'.

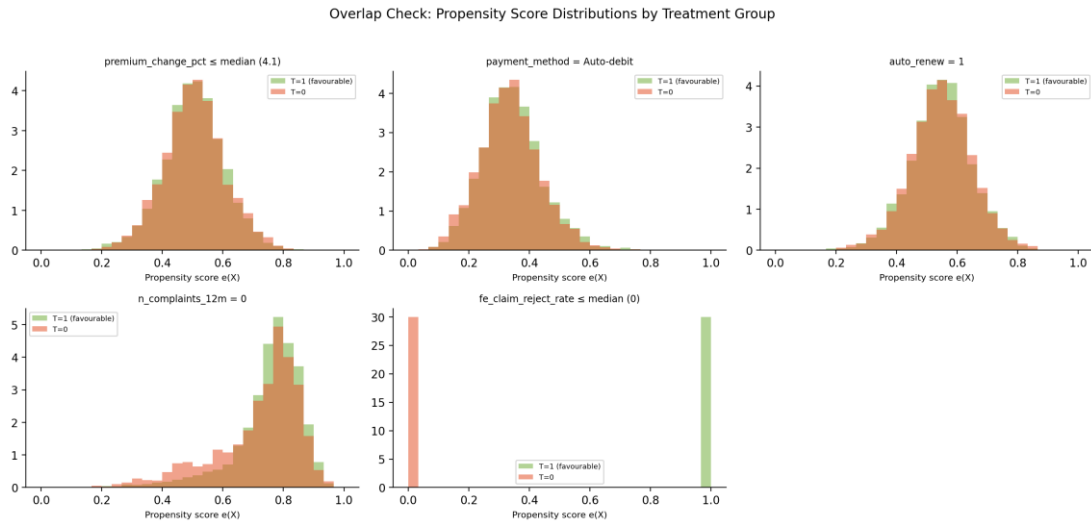
Heterogeneity and uplift. The doubly robust pseudo-outcome is regressed on customer features with cross-fitting (DR-learner; Kennedy, 2023), a meta-learner in the family studied by Künzel et al. (2019). Customers are sorted into quintiles of predicted uplift and the average pseudo-outcome per quintile (with standard errors) tests whether effects differ. Uplift-based targeting is compared with risk-based targeting, following the uplift literature for insurance retention (Guelman et al., 2015; Devriendt et al., 2021).



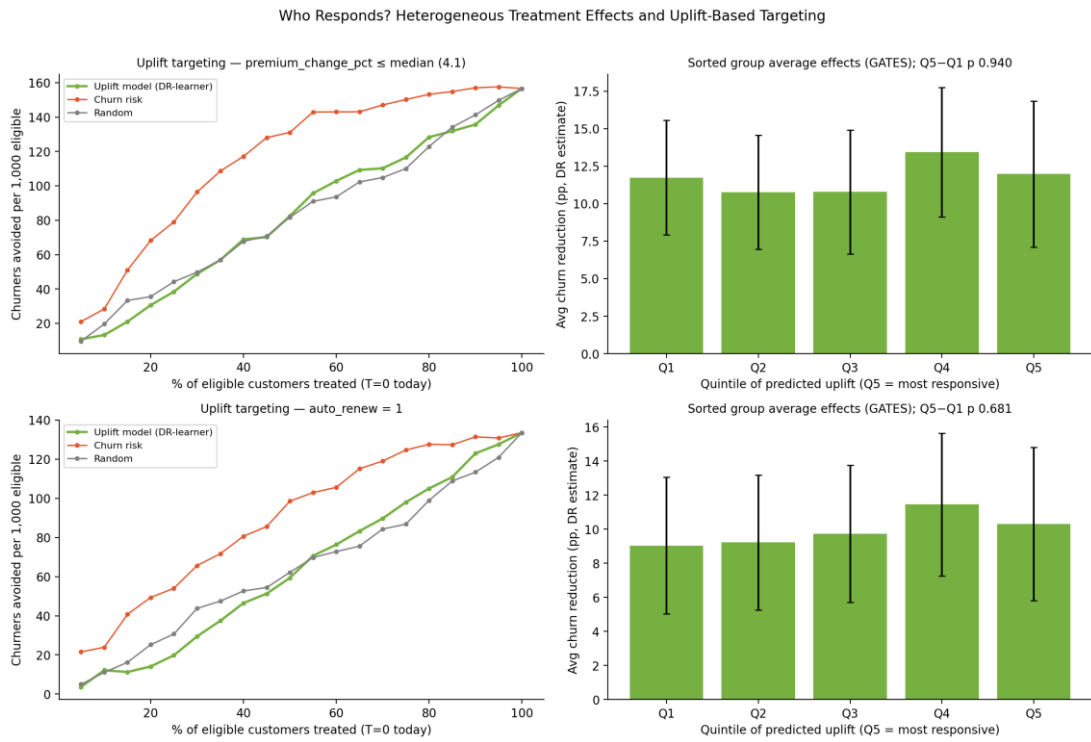
**Figure H.1.** Estimated Causal Effect of Favourable Conditions on Churn (doubly robust AIPW)



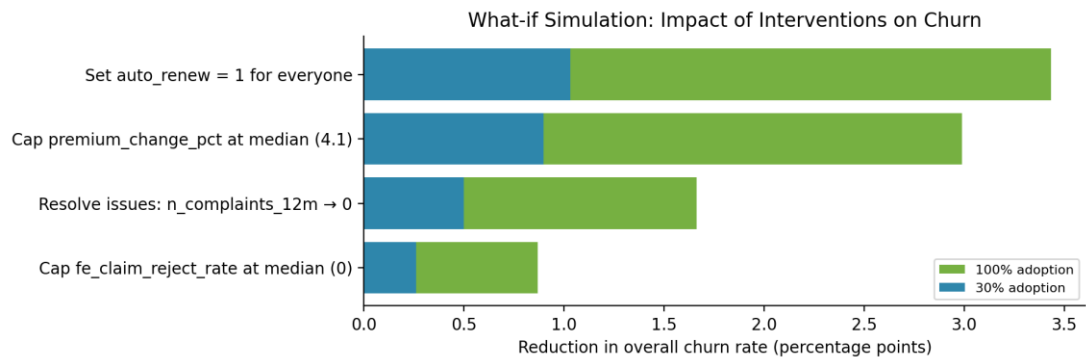
**Figure H.2.** Assumed causal diagram (DAG) behind the doubly robust effect estimates: lever, confounders and churn.



**Figure H.3.** Overlap Check: Propensity Score Distributions by Treatment Group



**Figure H.4.** Who Responds? Heterogeneous Treatment Effects and Uplift-Based Targeting



**Figure H.5.** What-if Simulation: Impact of Interventions on Churn

**Table H.1.** AIPW average treatment effects.

| Feature                            | Favorable condition (T=1)                         | Treated % | Naive diff pp | Ate pp | Ci low pp | Ci high pp | P value | Overlap issue % |
|------------------------------------|---------------------------------------------------|-----------|---------------|--------|-----------|------------|---------|-----------------|
| premium change at last renewal (%) | premium change at last renewal (%) ≤ median (4.1) | 50.4      | -11.1         | -11.7  | -13.6     | -9.863     | 0       | 0               |
| payment method                     | payment method = Auto-debit                       | 34.4      | -10.4         | -9.949 | -11.9     | -8.036     | 0       | 0.025           |
| auto-renewal                       | auto-renewal = 1                                  | 54.6      | -9.709        | -9.954 | -11.8     | -8.102     | 0       | 0               |
| complaints in the last 12 months   | complaints in the last 12 months = 0              | 73.9      | -11.2         | -8.852 | -11.5     | -6.207     | 5.4e-11 | 0.137           |
| claim rejection rate               | claim rejection rate ≤ median (0)                 | 86.7      | -17.5         | -7.082 | -8.005    | -6.159     | 0       | 100.0           |

**Table H.2.** Robustness of causal estimates (first 10 columns; full table in report\_tables.xlsx).

| Treatment (T=1)                                   | Feature                            | Ate pp | P value | Mu1 churn if T1 | Mu0 churn if T0 | Risk ratio | E value point | E value ci | Placebo ate pp |
|---------------------------------------------------|------------------------------------|--------|---------|-----------------|-----------------|------------|---------------|------------|----------------|
| premium change at last renewal (%) ≤ median (4.1) | premium change at last renewal (%) | -11.7  | 0       | 0.155           | 0.272           | 0.569      | 2.911         | 2.511      | -0.595         |
| payment method = Auto-debit                       | payment method                     | -9.949 | 0       | 0.145           | 0.245           | 0.594      | 2.758         | 2.341      | 0.699          |
| auto-renewal = 1                                  | auto-renewal                       | -9.954 | 0       | 0.166           | 0.266           | 0.626      | 2.576         | 2.232      | 1.178          |
| complaints in the last 12 months = 0              | complaints in the last 12 months   | -8.852 | 5.4e-11 | 0.188           | 0.276           | 0.679      | 2.306         | 1.902      | -2.134         |
| claim rejection                                   | claim rejection                    | -7.082 | 0       | 0.192           | 0.263           | 0.731      | 2.078         | 1.937      | -2.571         |

| Treatment (T=1)        | Feature | Ate pp | P value | Mu1 churn if T1 | Mu0 churn if T0 | Risk ratio | E value point | E value ci | Placebo ate pp |
|------------------------|---------|--------|---------|-----------------|-----------------|------------|---------------|------------|----------------|
| rate $\leq$ median (0) | rate    |        |         |                 |                 |            |               |            |                |

**Table H.3.** Uplift (DR-learner) summary.

| Treatment                                              | Feature                            | Eligible customers | Ate pp | Gates q5 pp | Gates q1 pp | Heterogeneity p | Avoided per1000 top20 uplift | Avoided per1000 top20 risk | Avoided per1000 top20 random |
|--------------------------------------------------------|------------------------------------|--------------------|--------|-------------|-------------|-----------------|------------------------------|----------------------------|------------------------------|
| premium change at last renewal (%) $\leq$ median (4.1) | premium change at last renewal (%) | 3,809              | -11.7  | 12.0        | 11.7        | 0.940           | 30.6                         | 68.2                       | 35.6                         |
| auto-renewal = 1                                       | auto-renewal                       | 3,630              | -9.954 | 10.3        | 9.041       | 0.681           | 14.1                         | 49.4                       | 25.2                         |

**Table H.4.** Sorted group average treatment effects (GATES).

| Treatment                                              | Uplift quintile | Avg churn reduction pp | Ci low pp | Ci high pp | N     |
|--------------------------------------------------------|-----------------|------------------------|-----------|------------|-------|
| premium change at last renewal (%) $\leq$ median (4.1) | Q1              | 11.7                   | 7.918     | 15.6       | 1,535 |
| premium change at last renewal (%) $\leq$ median (4.1) | Q2              | 10.8                   | 6.973     | 14.6       | 1,535 |
| premium change at last renewal (%) $\leq$ median (4.1) | Q3              | 10.8                   | 6.663     | 14.9       | 1,535 |
| premium change at last renewal (%) $\leq$ median (4.1) | Q4              | 13.4                   | 9.130     | 17.7       | 1,535 |
| premium change at last renewal (%) $\leq$ median (4.1) | Q5              | 12.0                   | 7.115     | 16.8       | 1,535 |
| auto-renewal = 1                                       | Q1              | 9.041                  | 5.026     | 13.1       | 1,600 |
| auto-renewal = 1                                       | Q2              | 9.232                  | 5.266     | 13.2       | 1,600 |
| auto-renewal = 1                                       | Q3              | 9.742                  | 5.723     | 13.8       | 1,600 |
| auto-renewal = 1                                       | Q4              | 11.5                   | 7.263     | 15.6       | 1,600 |
| auto-renewal = 1                                       | Q5              | 10.3                   | 5.807     | 14.8       | 1,600 |

**Table H.5.** Model-based what-if simulation (associational).

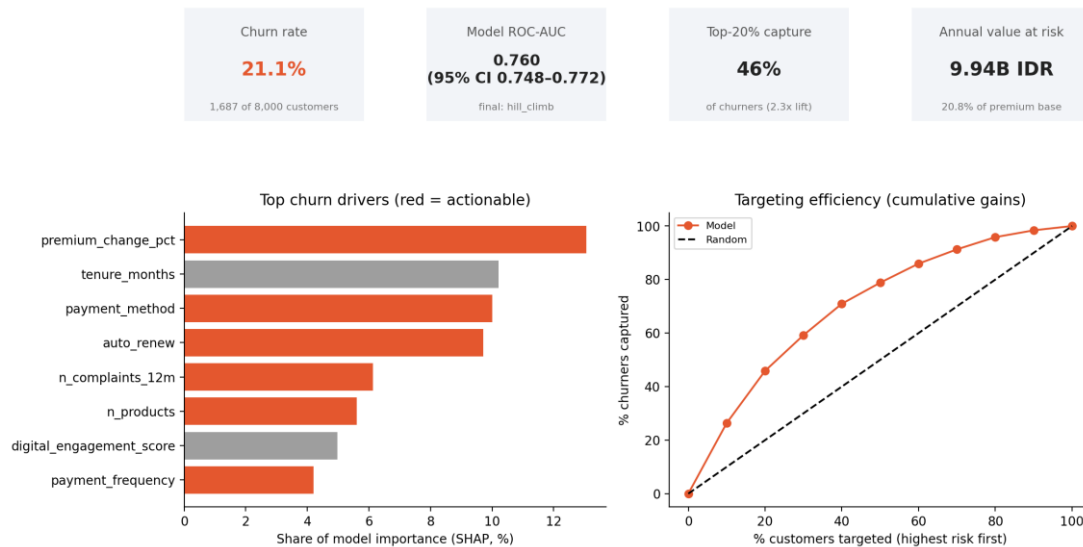
| Scenario  | Feature      | Affected % | Churn affected before | Churn affected after | Overall reduction pp full | Overall reduction pp 30pct adoption | Churners avoided per 1000 |
|-----------|--------------|------------|-----------------------|----------------------|---------------------------|-------------------------------------|---------------------------|
| Set auto- | auto-renewal | 45.4       | 0.245                 | 0.169                | 3.435                     | 1.030                               | 10.3                      |

| Scenario                                               | Feature                            | Affected % | Churn affected before | Churn affected after | Overall reduction pp full | Overall reduction pp 30pct adoption | Churners avoided per 1000 |
|--------------------------------------------------------|------------------------------------|------------|-----------------------|----------------------|---------------------------|-------------------------------------|---------------------------|
| renewal = 1 for everyone                               |                                    |            |                       |                      |                           |                                     |                           |
| Cap premium change at last renewal (%) at median (4.1) | premium change at last renewal (%) | 48.4       | 0.256                 | 0.194                | 2.989                     | 0.897                               | 8.968                     |
| Resolve issues: complaints in the last 12 months → 0   | complaints in the last 12 months   | 26.0       | 0.271                 | 0.207                | 1.662                     | 0.499                               | 4.985                     |
| Cap claim rejection rate at median (0)                 | claim rejection rate               | 12.3       | 0.347                 | 0.276                | 0.868                     | 0.260                               | 2.605                     |

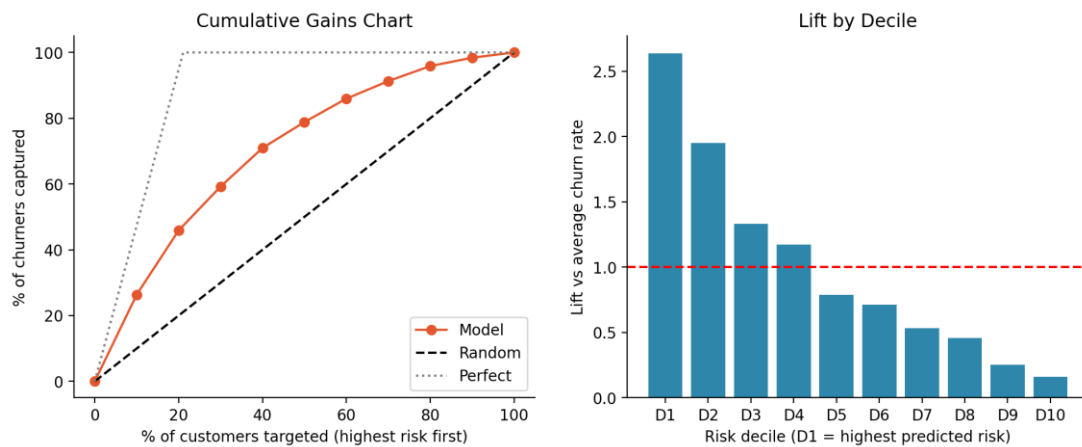
## Appendix I. Business Analytics

Lift and gains, risk tiers, risk × value prioritisation, campaign simulation and sensitivity, churn personas.

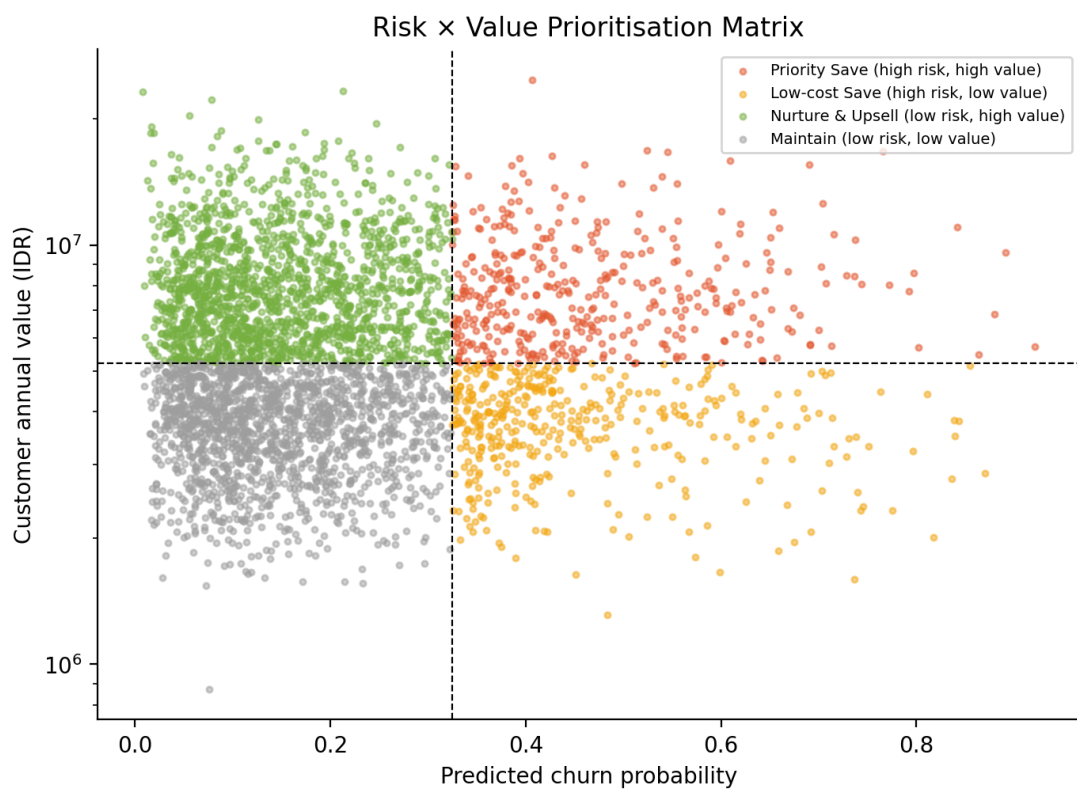
### Customer Churn — Executive Summary



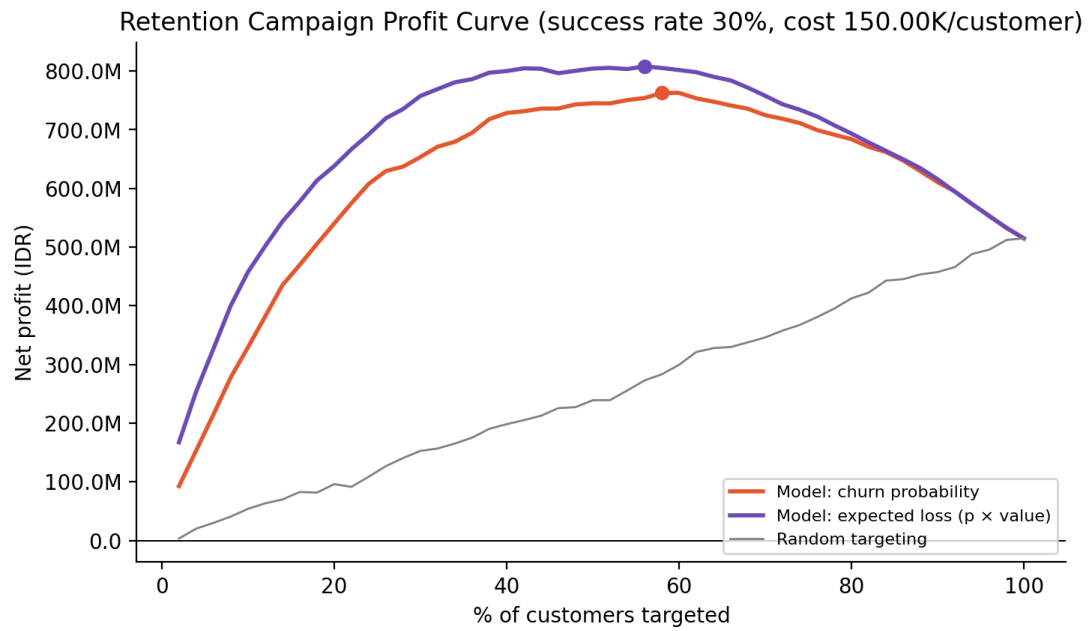
**Figure I.1.** Customer Churn — Executive Summary



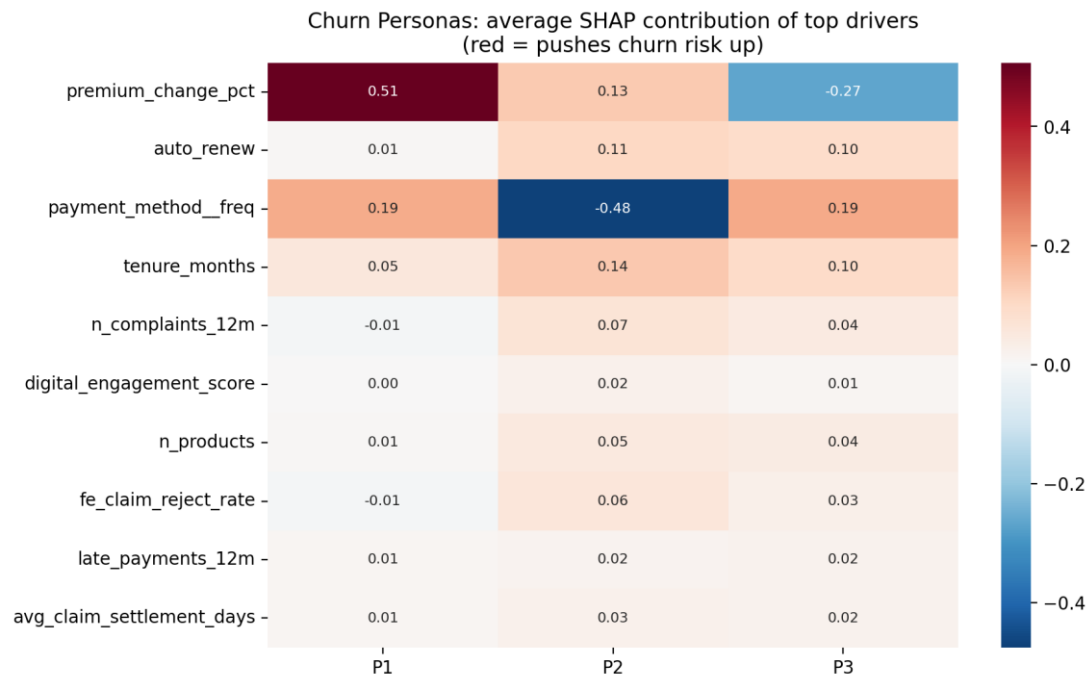
**Figure I.2.** Cumulative Gains Chart



**Figure I.3.** Risk x Value Prioritisation Matrix



**Figure I.4.** Retention Campaign Profit Curve (success rate 30%, cost 150.00K/customer)



**Figure I.5.** Churn Personas: average SHAP contribution of top drivers  
(red = pushes churn risk up)

**Table I.1.** Lift by risk decile.

| Decile | N   | Churners | Avg pred | Min pred | Churn rate | Lift  | Cum customers % | Cum churners captured % | Cum lift |
|--------|-----|----------|----------|----------|------------|-------|-----------------|-------------------------|----------|
| D1     | 800 | 445      | 0.553    | 0.427    | 0.556      | 2.638 | 10.0            | 26.4                    | 2.638    |
| D2     | 800 | 329      | 0.372    | 0.324    | 0.411      | 1.950 | 20.0            | 45.9                    | 2.294    |
| D3     | 800 | 225      | 0.288    | 0.255    | 0.281      | 1.334 | 30.0            | 59.2                    | 1.974    |
| D4     | 800 | 198      | 0.230    | 0.207    | 0.247      | 1.174 | 40.0            | 71.0                    | 1.774    |
| D5     | 800 | 133      | 0.187    | 0.168    | 0.166      | 0.788 | 50.0            | 78.8                    | 1.577    |
| D6     | 800 | 120      | 0.151    | 0.135    | 0.150      | 0.711 | 60.0            | 86.0                    | 1.433    |
| D7     | 800 | 90       | 0.120    | 0.106    | 0.113      | 0.533 | 70.0            | 91.3                    | 1.304    |
| D8     | 800 | 77       | 0.094    | 0.081    | 0.096      | 0.456 | 80.0            | 95.9                    | 1.198    |
| D9     | 800 | 43       | 0.068    | 0.055    | 0.054      | 0.255 | 90.0            | 98.4                    | 1.093    |
| D10    | 800 | 27       | 0.038    | 0.006    | 0.034      | 0.160 | 100.0           | 100.0                   | 1.000    |

**Table I.2.** Risk tiers.

| Tier   | N train | Actual churn rate | Avg predicted | Share churners % | N test |
|--------|---------|-------------------|---------------|------------------|--------|
| High   | 1,600   | 0.484             | 0.463         | 45.9             | 412    |
| Medium | 2,400   | 0.232             | 0.235         | 33.0             | 597    |
| Low    | 4,000   | 0.089             | 0.094         | 21.2             | 991    |

**Table I.3.** Risk × value quadrants.

| Quadrant                                | N     | Actual churn rate | Avg pred | Total value | Value at risk | Share value at risk % |
|-----------------------------------------|-------|-------------------|----------|-------------|---------------|-----------------------|
| Low-cost Save (high risk, low value)    | 821   | 0.481             | 0.463    | 3.02B       | 1.39B         | 14.0                  |
| Maintain (low risk, low value)          | 3,179 | 0.142             | 0.146    | 11.77B      | 1.71B         | 17.2                  |
| Nurture & Upsell (low risk, high value) | 3,221 | 0.143             | 0.148    | 26.73B      | 3.92B         | 39.5                  |
| Priority Save (high risk, high value)   | 779   | 0.487             | 0.462    | 6.31B       | 2.91B         | 29.3                  |

**Table I.4.** Optimal campaign size by strategy.

| Strategy                         | Target % | N targeted | Churners reached | Benefit | Cost    | Net profit | Roi   |
|----------------------------------|----------|------------|------------------|---------|---------|------------|-------|
| Model: churn probability         | 58.0     | 4,640      | 1,426            | 1.46B   | 696.00M | 762.60M    | 1.096 |
| Model: expected loss (p × value) | 56.0     | 4,480      | 1,343            | 1.48B   | 672.00M | 807.70M    | 1.202 |
| Random targeting                 | 100.0    | 8,000      | 1,687            | 1.71B   | 1.20B   | 514.76M    | 0.429 |

**Table I.5.** Campaign sensitivity analysis.

| Success rate | Cost per contact | Optimal target % | Max net profit | Roi   |
|--------------|------------------|------------------|----------------|-------|
| 0.150        | 75,000           | 55.1             | 403.44M        | 1.221 |
| 0.150        | 150,000          | 26.5             | 202.00M        | 0.635 |
| 0.150        | 300,000          | 6.083            | 56.03M         | 0.384 |
| 0.300        | 75,000           | 67.3             | 1.18B          | 2.922 |

| Success rate | Cost per contact | Optimal target % | Max net profit | Roi   |
|--------------|------------------|------------------|----------------|-------|
| 0.300        | 150,000          | 55.1             | 806.88M        | 1.221 |
| 0.300        | 300,000          | 26.5             | 404.01M        | 0.635 |
| 0.450        | 75,000           | 87.8             | 2.01B          | 3.814 |
| 0.450        | 150,000          | 67.3             | 1.57B          | 1.941 |
| 0.450        | 300,000          | 34.7             | 972.24M        | 1.169 |

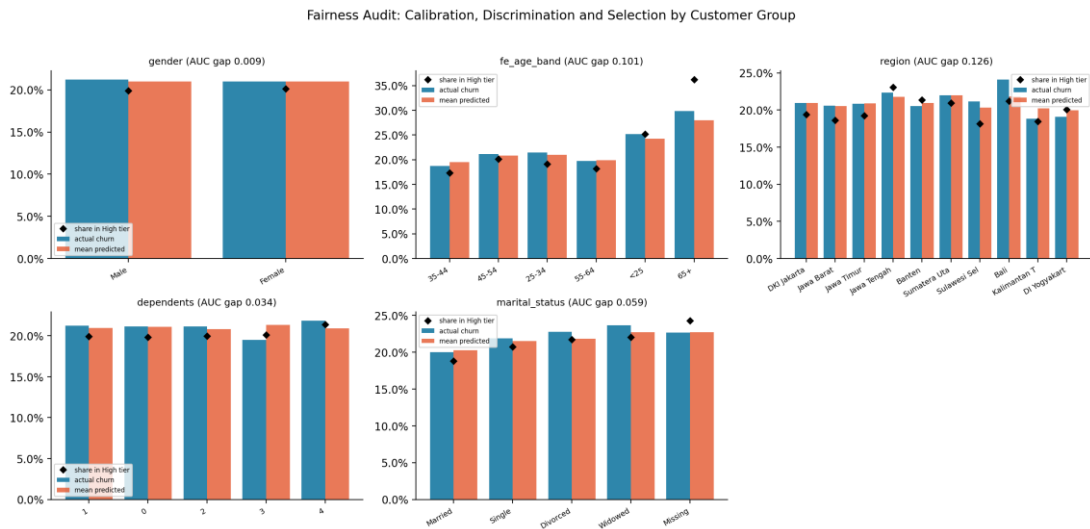
**Table I.6.** Churn personas (first 10 columns; full table in report\_tables.xlsx).

| Persona | N in sample | Share of at risk % | Avg pred churn | Actual churn rate | Top drivers                                                                                 | Premium change at last renewal (%) | Payment method | Auto-renewal | Complaints in the last 12 months |
|---------|-------------|--------------------|----------------|-------------------|---------------------------------------------------------------------------------------------|------------------------------------|----------------|--------------|----------------------------------|
| P3      | 643         | 41.8               | 0.306          | 0.333             | payment method (+0.19); tenure (months) (+0.10); auto-renewal (+0.10)                       | 0.910                              | Bank Transfer  | 0.350        | 0.460                            |
| P1      | 597         | 38.8               | 0.377          | 0.382             | premium change at last renewal (%) (+0.51); payment method (+0.19); tenure (months) (+0.05) | 12.0                               | Bank Transfer  | 0.500        | 0.350                            |
| P2      | 298         | 19.4               | 0.285          | 0.315             | tenure (months) (+0.14); premium change at last renewal (%) (+0.13); auto-renewal (+0.11)   | 7.290                              | Auto-debit     | 0.310        | 0.540                            |

## Appendix J. Governance, Deployment and Implementation

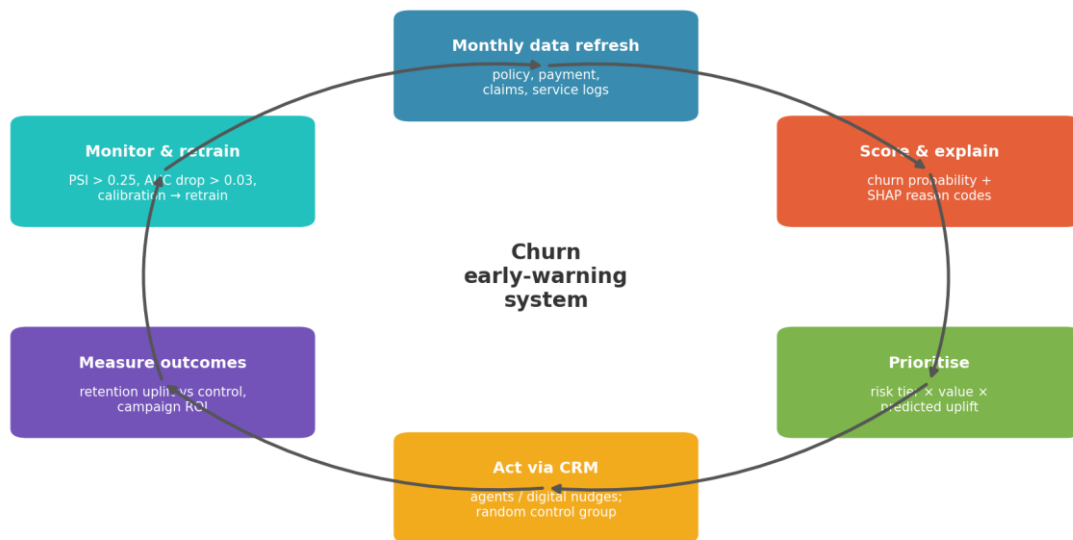
Fairness audit, model card, deployment loop, pilot design and implementation roadmap.

The model card follows Mitchell et al. (2019). The fairness audit compares calibration (mean predicted vs actual churn), discrimination (AUC) and selection into the High-risk tier across customer groups; differences in selection that mirror genuine differences in churn are expected, whereas calibration gaps would indicate bias. The score is restricted to retention outreach.

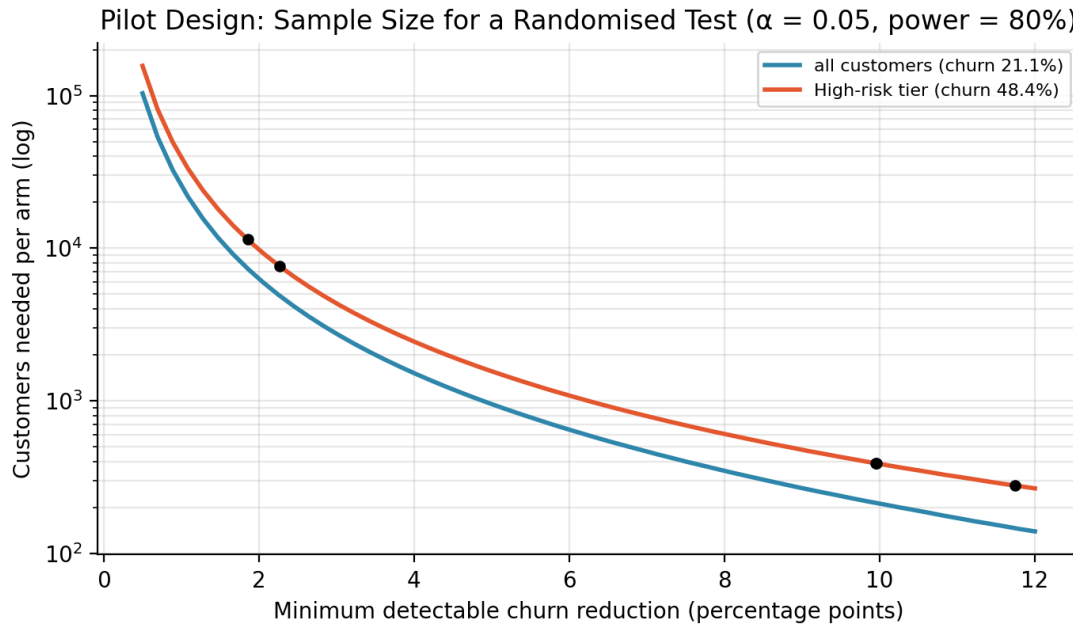


**Figure J.1.** Fairness Audit: Calibration, Discrimination and Selection by Customer Group

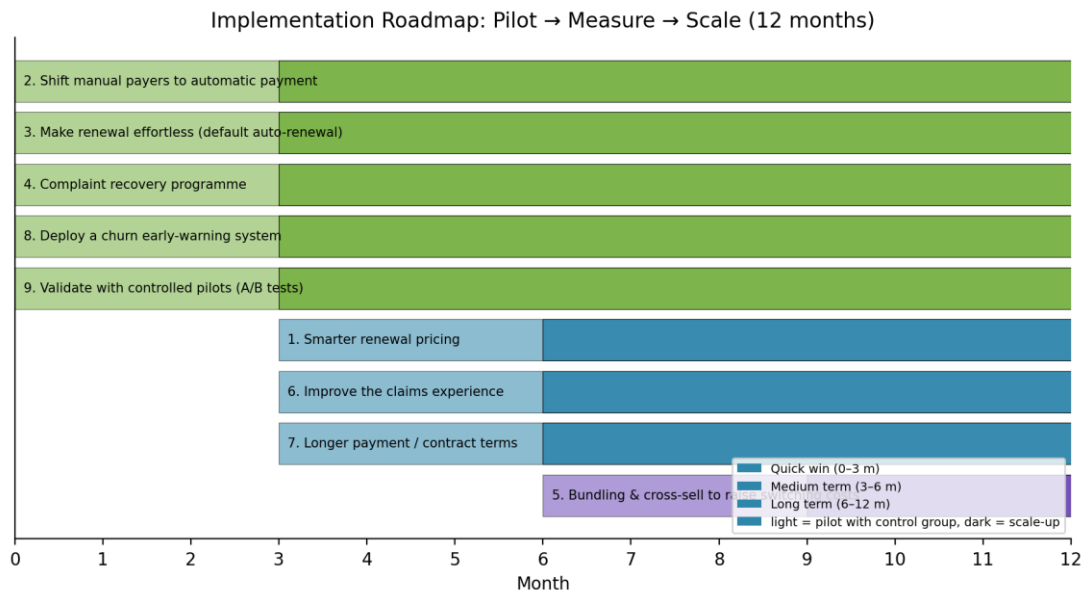
### Deployment & Continuous-Learning Loop (MLOps)



**Figure J.2.** Deployment & Continuous-Learning Loop (MLOps)



**Figure J.3.** Pilot Design: Sample Size for a Randomised Test ( $\alpha = 0.05$ , power = 80%)



**Figure J.4.** Twelve-month implementation roadmap: each initiative is piloted against a control group before scale-up.

**Table J.1.** Model card (Mitchell et al., 2019).

| Item             | Description                                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------|
| Model            | hill_climb (xgb_fs 0.44, logreg 0.33, catboost_multiseed 0.11, catboost 0.11)                                    |
| Intended use     | Rank active policyholders by 12-month churn risk to prioritise retention outreach and allocate retention budget. |
| Out-of-scope use | Not for premium pricing, underwriting, claims decisions or                                                       |

| Item               | Description                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                    | denial of service; not validated for other products/markets.                                                                                |
| Training data      | 8,000 customers, 53 input features; target 'churn' (churn = Yes); churn rate 21.1%.                                                         |
| Excluded variables | cancellation reason (post-event leakage)                                                                                                    |
| Validation         | Stratified 5-fold CV (out-of-fold), target encoding inside folds, shuffled-label & adversarial checks, bootstrap & DeLong CIs.              |
| Performance        | ROC-AUC 0.760 (95% CI 0.748–0.772); PR-AUC 0.475; Brier 0.140; ECE 0.012; EMPC 115.01K IDR/customer.                                        |
| Calibration        | none (selected by cross-validated log-loss).                                                                                                |
| Decision rule      | Risk tiers at the 50%/80% quantiles; campaign size chosen by expected profit (see Section 4).                                               |
| Explainability     | Global: SHAP, permutation importance, PDP, EBM shape functions (if used). Local: SHAP waterfall per customer for agents.                    |
| Fairness           | Audited on gender, age band, region, dependents, marital status; max AUC gap 0.126.                                                         |
| Monitoring         | Monthly: PSI per feature (alert > 0.25), score distribution, realised churn by tier, calibration; retrain quarterly or if AUC drops > 0.03. |
| Limitations        | Observational snapshot data; causal effects assume no unmeasured confounding; ROI depends on stated cost/acceptance assumptions.            |

**Table J.2.** Fairness summary by attribute.

| Attribute      | Min sel | Max sel | Min auc | Max auc | Min cal | Max cal | Selection ratio | Auc gap |
|----------------|---------|---------|---------|---------|---------|---------|-----------------|---------|
| dependents     | 0.198   | 0.214   | 0.733   | 0.767   | 0.957   | 1.093   | 0.929           | 0.034   |
| age band       | 0.174   | 0.362   | 0.740   | 0.841   | 0.939   | 1.041   | 0.480           | 0.101   |
| gender         | 0.199   | 0.201   | 0.756   | 0.765   | 0.989   | 1.002   | 0.988           | 0.009   |
| marital status | 0.188   | 0.243   | 0.721   | 0.781   | 0.958   | 1.014   | 0.774           | 0.059   |
| region         | 0.181   | 0.231   | 0.708   | 0.835   | 0.907   | 1.075   | 0.786           | 0.126   |

**Table J.3.** Pilot design: sample size per arm.

| Intervention                                                     | Effect source                 | Baseline churn target segment | Expected reduction pp | N per arm | Total pilot size | Share of high tier % |
|------------------------------------------------------------------|-------------------------------|-------------------------------|-----------------------|-----------|------------------|----------------------|
| Shift to: premium change at last renewal (%) $\leq$ median (4.1) | AIPW causal estimate          | 0.484                         | 11.7                  | 278       | 556              | 135.0                |
| Shift to: auto-renewal = 1                                       | AIPW causal estimate          | 0.484                         | 9.954                 | 388       | 776              | 188.3                |
| Shift to: payment method = Auto-debit                            | AIPW causal estimate          | 0.484                         | 9.949                 | 389       | 778              | 188.8                |
| Set auto-renewal = 1 for everyone                                | what-if $\times$ 30% adoption | 0.484                         | 2.271                 | 7,587     | 15,174           | 3,683                |
| Cap premium change at last renewal (%) at median (4.1)           | what-if $\times$ 30% adoption | 0.484                         | 1.854                 | 11,389    | 22,778           | 5,529                |

**Table J.4.** Implementation roadmap.

| Priority | Initiative                                     | Phase               | Start month | Pilot months | Scale months | Kpi                                                           |
|----------|------------------------------------------------|---------------------|-------------|--------------|--------------|---------------------------------------------------------------|
| 2        | Shift manual payers to automatic payment       | Quick win (0–3 m)   | 0           | 3            | 3            | % of policies on auto-pay; lapse rate of converted vs control |
| 3        | Make renewal effortless (default auto-renewal) | Quick win (0–3 m)   | 0           | 3            | 3            | Auto-renew adoption; renewal rate                             |
| 4        | Complaint recovery programme                   | Quick win (0–3 m)   | 0           | 3            | 3            | Complaint resolution time; churn rate of complainants         |
| 8        | Deploy a churn early-warning system            | Quick win (0–3 m)   | 0           | 3            | 3            | Precision/recall of alerts; retained revenue vs control group |
| 9        | Validate with controlled pilots (A/B tests)    | Quick win (0–3 m)   | 0           | 3            | 3            | Uplift in retention rate (treated – control)                  |
| 1        | Smarter renewal pricing                        | Medium term (3–6 m) | 3           | 3            | 3            | Churn rate among customers with increases; price elasticity   |
| 6        | Improve the claims experience                  | Medium term (3–6 m) | 3           | 3            | 3            | Claim settlement days; post-claim churn rate; claim NPS       |
| 7        | Longer payment / contract terms                | Medium term (3–6 m) | 3           | 3            | 3            | Share of annual payers; churn by payment frequency            |
| 5        | Bundling & cross-sell to raise switching costs | Long term (6–12 m)  | 6           | 3            | 6            | Products per customer; churn of bundled vs single             |

**Table J.5.** Full recommendation list with evidence and KPIs.

| Priority | Recommendation                           | Action                                                                                                                                                         | Evidence                                                                                                                 | Estimated impact                                                             | Kpi                                                           |
|----------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1        | Smarter renewal pricing                  | Cap or phase in premium increases for at-risk loyal customers, explain the value behind increases, and offer coverage/deductible adjustments instead of lapse. | '13.2–30.6' churns at 33.5% (1.6x avg); IV 0.16; OR 1.64; HR 1.39; SHAP rank #1; causal (AIPW) -11.7 pp [-13.6, -9.9]    | -2.99 pp churn at full adoption (≈9 churners avoided /1,000 at 30% adoption) | Churn rate among customers with increases; price elasticity   |
| 2        | Shift manual payers to automatic payment | Offer a small incentive (e.g. premium discount/cashback) and one-click enrolment for auto-debit/recurring card payment; send pre-due reminders to              | 'Bank Transfer' churns at 25.5% (1.2x avg); IV 0.10; OR 2.33; HR 1.83; SHAP rank #3; causal (AIPW) -9.9 pp [-11.9, -8.0] |                                                                              | % of policies on auto-pay; lapse rate of converted vs control |

| Priority | Recommendation                                 | Action                                                                                                                                                                               | Evidence                                                                                                      | Estimated impact                                                              | Kpi                                                           |
|----------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------|
|          |                                                | remaining manual payers.                                                                                                                                                             |                                                                                                               |                                                                               |                                                               |
| 3        | Make renewal effortless (default auto-renewal) | Default new and renewing policies to auto-renewal with clear opt-out; 30/14/7-day renewal reminders with a one-click renew link.                                                     | '0' churns at 26.4% (1.3x avg); IV 0.08; OR 0.51; HR 0.61; SHAP rank #4; causal (AIPW) -10.0 pp [-11.8, -8.1] | -3.43 pp churn at full adoption (≈10 churners avoided /1,000 at 30% adoption) | Auto-renew adoption; renewal rate                             |
| 4        | Complaint recovery programme                   | Trigger a proactive call within 48h of any complaint, enforce resolution SLAs, offer service-recovery gestures to high-value complainants, and fix top root causes.                  | '≥2' churns at 44.3% (2.1x avg); IV 0.11; OR 1.33; HR 1.23; SHAP rank #5; causal (AIPW) -8.9 pp [-11.5, -6.2] | -1.66 pp churn at full adoption (≈5 churners avoided /1,000 at 30% adoption)  | Complaint resolution time; churn rate of complainants         |
| 5        | Bundling & cross-sell to raise switching costs | Offer multi-policy discounts and relevant riders to single-product customers with good risk profiles.                                                                                | '1' churns at 23.8% (1.1x avg); IV 0.05; OR 0.76; SHAP rank #6                                                |                                                                               | Products per customer; churn of bundled vs single             |
| 6        | Improve the claims experience                  | Fast-track settlement, explain rejections transparently with an appeal path, and assign a claims concierge for high-value / high-risk customers after a claim event.                 | '1' churns at 37.1% (1.8x avg); IV 0.12; OR 1.26; HR 1.11; SHAP rank #11; causal (AIPW) -7.1 pp [-8.0, -6.2]  | -0.87 pp churn at full adoption (≈3 churners avoided /1,000 at 30% adoption)  | Claim settlement days; post-claim churn rate; claim NPS       |
| 7        | Longer payment / contract terms                | Encourage annual/quarterly payment or longer contracts with a discount; monthly payers get extra reminders.                                                                          | 'Monthly' churns at 23.6% (1.1x avg); IV 0.02; SHAP rank #8                                                   |                                                                               | Share of annual payers; churn by payment frequency            |
| 8        | Deploy a churn early-warning system            | Score all active policies monthly with this model; route High-risk/Priority-Save customers to retention teams, lower tiers to automated digital nudges; refresh the model quarterly. | Model ROC-AUC 0.760; top-20% captures 46% of churners                                                         | optimal campaign: target top 56% → net 807.70M                                | Precision/recall of alerts; retained revenue vs control group |
| 9        | Validate with controlled pilots (A/B tests)    | For each intervention keep a random control group (e.g. 20% of targeted customers) to measure true                                                                                   | What-if results are associational                                                                             |                                                                               | Uplift in retention rate (treated – control)                  |

| Priority | Recommendation | Action                 | Evidence | Estimated impact | Kpi |
|----------|----------------|------------------------|----------|------------------|-----|
|          |                | uplift before scaling. |          |                  |     |

## Appendix K. Reproducibility

All results were produced by one notebook executed top-to-bottom (Restart & Run All) with random seed 42, 5-fold stratified cross-validation and the settings in Appendix B. Software: Python 3.11.9, pandas 2.3.3, scikit-learn 1.6.1, LightGBM 4.7.0, XGBoost 3.2.0, CatBoost 1.2.10, SHAP 0.51.0, statsmodels 0.15.0, lifelines 0.30.3, Optuna 4.9.0; platform Windows 10. The submission package contains the executed notebook with all outputs, every figure and table (CSV and Excel), the model card, the per-customer action list and an interactive dashboard.

## References for the Appendices

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